HSF Knockdown and Nab-Paclitaxel Effects on PDCL5 Kinome Signaling_STK

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Table of contents

1 Introduction **3**
1.1 Background . . . 3

2 Run Design **5**

3 Results **6**
3.1 Image Analysis . . . 6
3.2 Reading Data . . . 6

4 Objective **7**

5 Dataset overview **8**

6 Custom cycle-level quality control **9**

7 Exposure-time behavior at the final cycle **11**
7.1 Median raw signal versus exposure time . . . 11
7.2 Number of measurable peptides versus exposure time . . . 12
7.3 Peptides losing measurable signal from 50 to 100 ms . . . 12

8 Standard KRSA QC in the two exposure branches **14**

9 Final peptide-list comparison **15**

10 Conclusion **16**

11 Detailed output files **17**
11.1 QC Initial Steps and Groups Assignments . . . 17
11.2 QC Steps and Model Fitting . . . 17
11.3 Global Signal Intensity . . . 17
11.3.1 Global CV Plots . . . 17
11.3.2 Global Violin Plots . . . 18
11.3.3 Global Heatmaps . . . 18
11.4 Group Comparison . . . 18

12 Session Info **78**

1 Introduction

1.1 Background

The Pamstation12 instrument provides a profiling of kinase activity of cell or tissue samples. The device is loaded with either serine/threonine or tyrosine microarray chips. Each chip has 4 wells, so four samples can be loaded on a single chip, and the Pamstation12 can accommodate 3 chips per run. The microarray represents 144 (STK chip) or 196 (PTK chip) reporter peptides that can be phosphorylated by serine/threonine or tyrosine kinases. The device measures the degree of the phosphorylation in real time by detecting fluorescent labeled antibodies at different exposure times. The list of peptides present in each microarray can be viewed here: STK chip, PTK chip

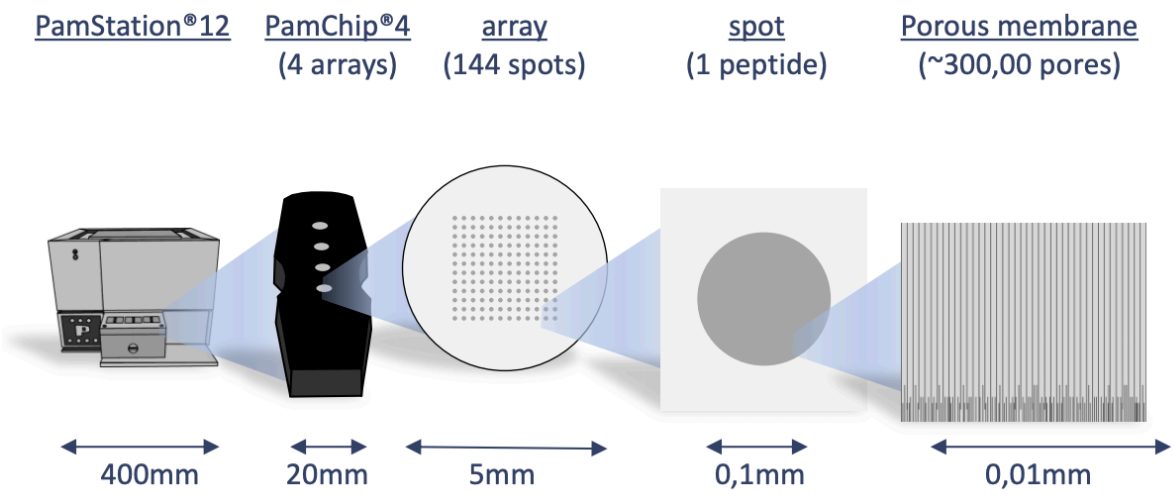


Figure 1.1: PamGene Kinase Activity Platform

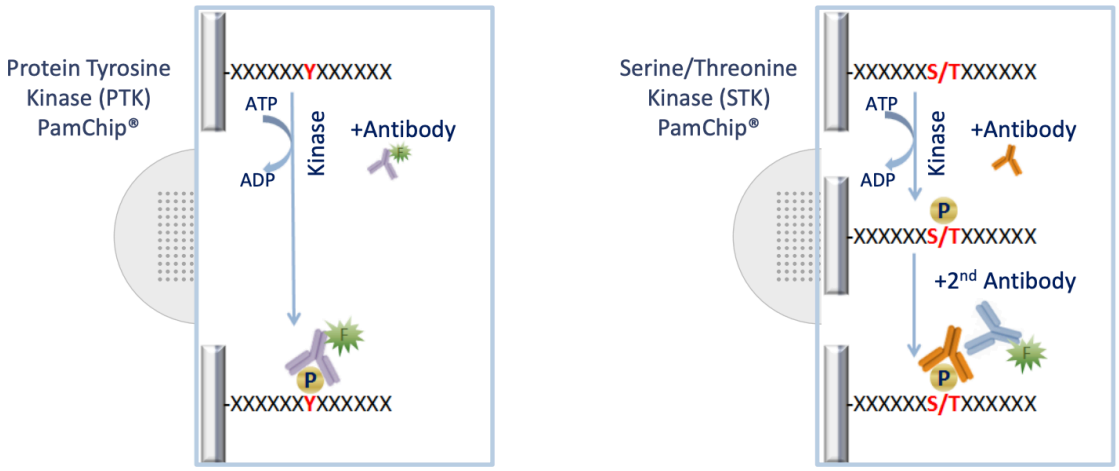
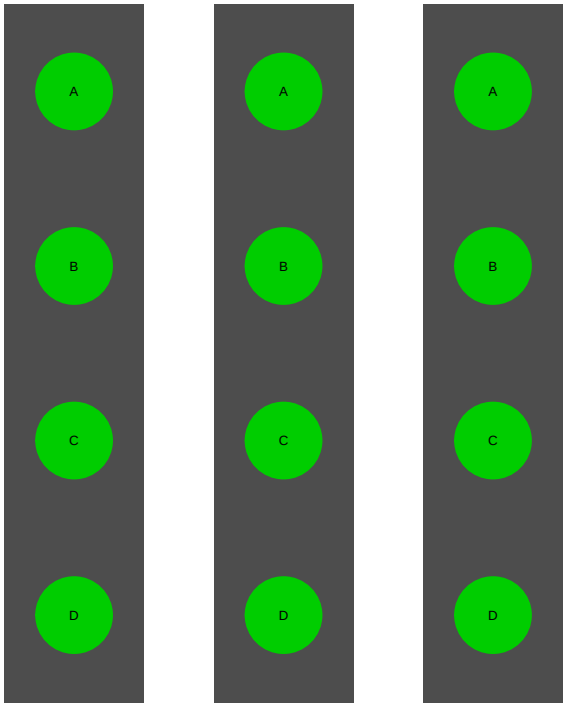


Figure 1.2: PamGene Platform Detection

2 Run Design

Designing the placement of the samples on the chips and arrays is important to consider due to the variability across different chips and batches. During the run some wells are subject to fail and their data cannot be analyzed and shown below as red.



3 Results

3.1 Image Analysis

The first step of analyzing the run is to convert the images taken by the PamStation of each array at different exposure times to numerical values. This is done by the Bionavigator software developed by PamGene. The software recognizes the grid of the array with the aid of the searching algorithm (Pamgrid) to correctly identify each spot on the array. The numbers produced by this software represent the median value of the foreground pixels minus the median value of the background pixels to produce the median signal minus background (Median_SigmBg).

3.2 Reading Data

The first step will be reading the crosstab view bionavigator files (Median_SigmBg and Signal_Saturation) and defining the PamChip type (STK or PTK). The raw data is read and then transformed to be in tidy format for an easier analysis, modeling, and visualizing.

4 Objective

This analysis evaluates whether inclusion of the 100-ms exposure affects raw signal behavior, exposure-response linearity, and peptide retention through the standard KRSA quality-control pipeline. A corrected branch excluding 100 ms is compared with the original branch that includes exposures through 100 ms.

```
#>
#> === INITIAL QC DATA ===
#> Rows: 59939
#> Columns: Barcode, Row, Col, Filter, ExposureTime, Cycle, Image, Temperature, Instrument ty
#>
#> === ORIGINAL SAMPLE NAMES ===
#>  [1] "ShSr-C-1"      "ShSr-NPTX-1"   "ShHSFi-C-1"    "ShHSFi-NPTX-1"
#>  [5] "ShSr-C-2"      "ShSr-NPTX-2"   "ShHSFi-C-2"    "ShHSFi-NPTX-2"
#>  [9] "ShSr-C-3"      "ShSr-NPTX-3"   "ShHSFi-NPTX-3"
#>
#> === DETECTED GROUPS ===
#> [1] "ShSr-C"      "ShSr-NPTX"   "ShHSFi-C"    "ShHSFi-NPTX"
#>
#> === NUMERIC CONVERSION DIAGNOSTICS ===
#> # A tibble: 1 x 5
#>   total_rows valid_exposure valid_cycle valid_signal valid_all
#>        <int>          <int>       <int>        <int>     <int>
#> 1      55105          55105       55105        55105     55105
#>
#> === RETAINED GROUPS ===
#> # A tibble: 4 x 2
#>   Group       n_samples
#>   <chr>           <int>
#> 1 ShHSFi-C            2
#> 2 ShHSFi-NPTX         3
#> 3 ShSr-C              3
#> 4 ShSr-NPTX           3
#>
#> === AVAILABLE CYCLES ===
#> [1]  92  97 102 107 112 117 122 124
```

5 Dataset overview

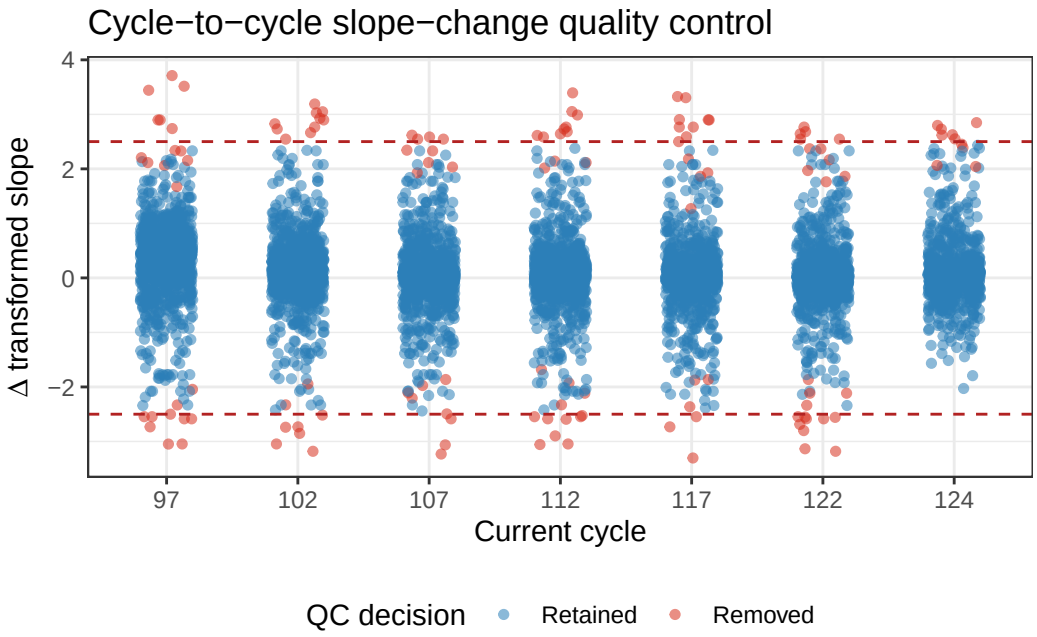
Dataset characteristic	Value
Chip type	STK
Samples	11
Barcodes	3
Analytical peptides	141
Available exposure times (ms)	1, 5, 10, 25, 50, 100
Cycle used for exposure plots	124
Signal threshold	3
Linearity threshold	0.4

6 Custom cycle-level quality control

Because the 100-ms images are suspected to be overexposed, the preliminary high-signal screen uses exposures through 50 ms. This screen selects measurable peptides for cycle-level anomaly detection; it is not the final KRSA peptide filter.

For each sample, barcode, peptide, and cycle, the model (Signal = slope ×ExposureTime) is fitted through the origin. Consecutive cycles are compared using the change in transformed slope. A transition fails when its absolute delta exceeds 2.5, or when its peptide-specific absolute z-score exceeds 3 and either adjacent model has (R^2 < 0.8).

Cycle-level QC metric	Result
Evaluated cycle transitions	7623
Failed transitions	135
Failed peptide-cycle combinations	135
Measurement rows removed	560
Measurement rows retained	54545

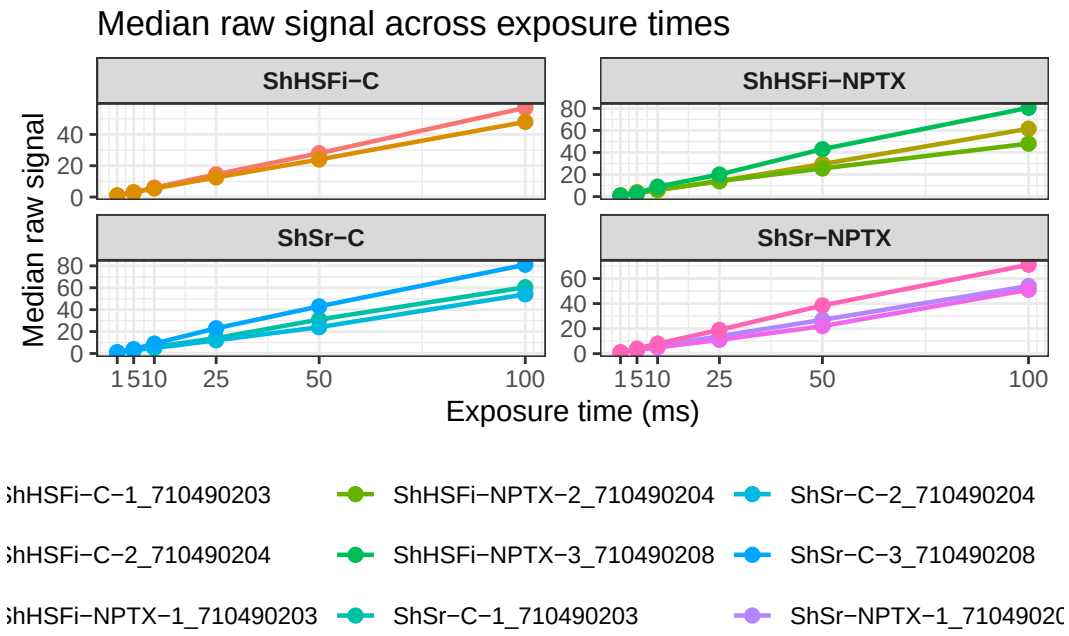


The failed peptide-cycle combinations were removed before creating either exposure-time branch, ensuring that both branches start from the same cleaned dataset.

7 Exposure-time behavior at the final cycle

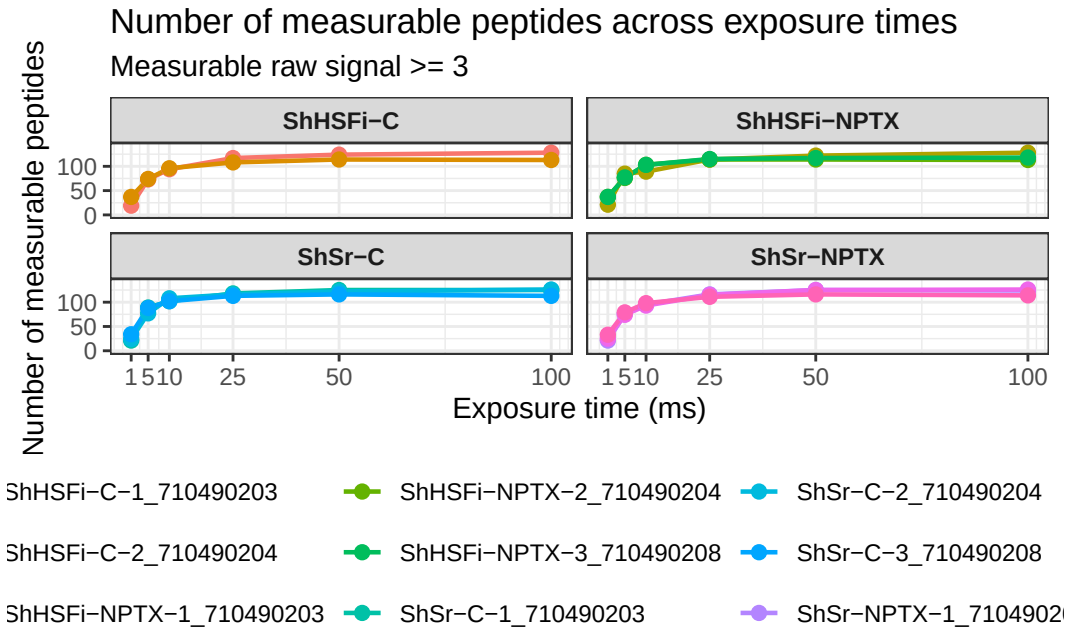
The following plots use unmodeled signal values from cycle 124 and exclude reference/control peptides. Conditions and replicates are taken from the data; no study-specific group names are hard-coded.

7.1 Median raw signal versus exposure time



This plot shows whether median signal increases consistently with exposure time or declines at the 100-ms exposure within individual samples.

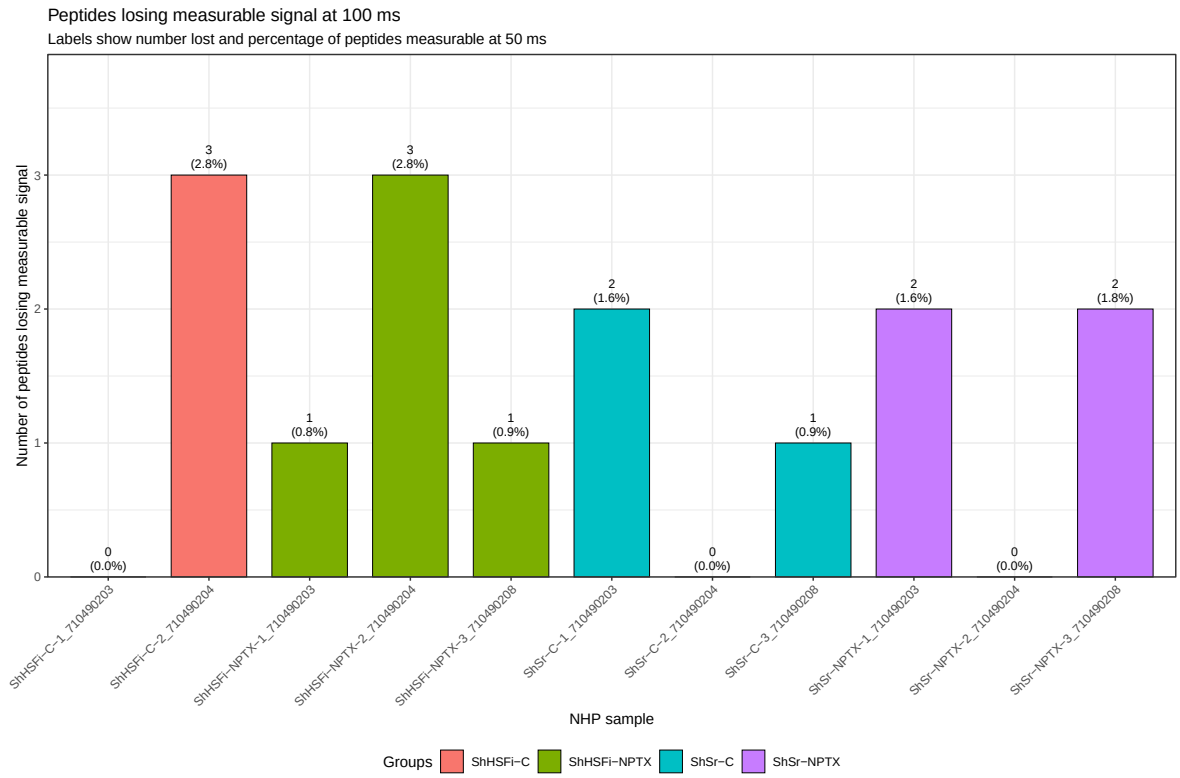
7.2 Number of measurable peptides versus exposure time



This plot evaluates whether longer exposure improves peptide detection or causes the number of measurable peptides to fall at 100 ms.

7.3 Peptides losing measurable signal from 50 to 100 ms

Only peptides measured at both exposure times enter this calculation. Signal loss is defined as signal 5 at 50 ms followed by signal <5 at 100 ms. The percentage denominator is the number of paired peptides measurable at 50 ms.


Peptides losing measurable signal at 100 ms
Labels show number lost and percentage of peptides measurable at 50 ms
Number of peptides losing measurable signal
0
1
2
3
0 (0.0%)
3 (2.8%)
1 (0.8%)
3 (2.8%)
1 (0.9%)
2 (1.6%)
0 (0.0%)
1 (0.9%)
2 (1.6%)
0 (0.0%)
2 (1.8%)
ShHSFi−C−1_710490203
ShHSFi−C−2_710490204
ShHSFi−NPTX−1_710490203
ShHSFi−NPTX−2_710490204
ShHSFi−NPTX−3_710490208
ShSr−C−1_710490203
ShSr−C−2_710490204
ShSr−C−3_710490208
ShSr−NPTX−1_710490203
ShSr−NPTX−2_710490204
ShSr−NPTX−3_710490208
NHP sample
Groups
ShHSFi−C
ShHSFi−NPTX
ShSr−C
ShSr−NPTX

8 Standard KRSA QC in the two exposure branches

The observed-100-ms branch uses the standard KRSA pipeline. In the corrected branch, exposure-response models are fitted using observed measurements through 50 ms, the fitted slopes are extrapolated to obtain predicted 100-ms signals for low-signal filtering, and linearity is evaluated using observed measurements only.

Analysis	Maximum observed exposure	Low-signal end-point	Total analytical peptides	Passed low-signal filter	Passed linearity filter	Passed all KRSA filters
Observed through 100 ms	100 ms	Observed 100 ms	141	94	94	94
Predicted 100 ms from observed exposures through 50 ms	50 ms	Predicted 100 ms	141	109	107	107

The corrected analysis, which fits exposure-response models using observed measurements through 50 ms and uses the predicted 100-ms signal for low-signal filtering, retains **107** of **141** analytical peptides after all KRSA filters, compared with **94** when 100 ms is included.

9 Final peptide-list comparison

Final peptide status	Number of peptides
Passed both	91
Passed only with predicted 100 ms	16
Passed only with observed 100 ms	3
Failed both	31

The corrected-branch-specific set contains **16** peptides that pass all KRSA filters only when the predicted 100-ms endpoint, derived from observed exposures through 50 ms, is used. Conversely, **3** peptides pass only when the observed 100-ms measurements are included. These classifications describe changes in QC retention; they do not, by themselves, prove detector saturation.

10 Conclusion

The branch comparison quantifies how exclusion of the problematic 100-ms measurements changes final KRSA peptide retention. The paired well-level analysis provides complementary evidence by identifying measurements that are detectable at 50 ms but fall below the signal threshold at 100 ms. A peptide that passes only in the corrected branch and repeatedly loses signal at 100 ms provides stronger evidence of exposure-associated measurement distortion than either result alone.

11 Detailed output files

Detailed peptide lists, fitted model tables, cycle-QC diagnostics, and paired 50-versus-100-ms measurements are saved as CSV files in:

/Users/jvergis/Documents/work/Su_STK_2/qc_output_dir

These detailed objects are intentionally not printed as raw console output in the main report.

11.1 QC Initial Steps and Groups Assignments

We will perform a couple of quality control steps to deal with negative values in the data and adjust based on signal saturation (optional). Next, we will define a new column to represent the grouping. And then, we will extract end point signal values

11.2 QC Steps and Model Fitting

We will filter out peptides with low signals. In order to combine the values from different exposure times into a single value, a simple linear regression model of the *Median_SigmBg* as a function of exposure time is fitted. The slope of of the model fit and R^2 are then used for quality control and samples comparison. The slope is also scaled by multiplying by 100 and log2 transformed (*Slope_Transformed*). We then filter out peptides with poor linear fit and references peptides.

11.3 Global Signal Intensity

For a global signal intensity across all samples/groups, few figures can be plotted based on the *Slope_Transformed* values.

11.3.1 Global CV Plots

We will plot the coefficient of variation on both the normal and normalized fits. This will help us to identify groups with high variation that could be explained by sample outliers.

11.3.2 Global Violin Plots

We will plot violin figures to examine global signal differences between groups/samples.

11.3.3 Global Heatmaps

The heatmap represent all the peptides present on the chip except the positive/internal controls and peptides that failed to pass QC. The heatmaps are scaled by row to highlight the peptide signal differences across the samples. A hierarchical unsupervised clustering is applied both on the peptides and the samples to group potentially similar signatures.

11.4 Group Comparison

To compare between samples, a two-group comparison is performed. In this case, there are three group comparisons:

- Case 1 vs Control 1

The *Slope_Transformed* ratio between each group, paired by chip, is calculated to the fold change. Based on the fold change, peptides that pass a certain fold change threshold are considered significant hits. Also, quality control steps applied in each comparison to filter out peptides that do not reach specific criteria:

- The *Median_SigmBg* at max exposure *100ms* must be above a certain value
- R^2 of the linear model fit must be above a threshold value

These *Filtering Parameters* (fold change threshold, QC criteria) can be modified to adjust the stringency of the analysis. The *Filtering Parameters* that are used for this analysis:

- The *Median_SigmBg* at max exposure *100ms* must be equal or above 5
- R^2 of the linear model fit must be above or equal 0.8
- Log fold change (LFC) cutoffs at (0.2,0.3,0.4)

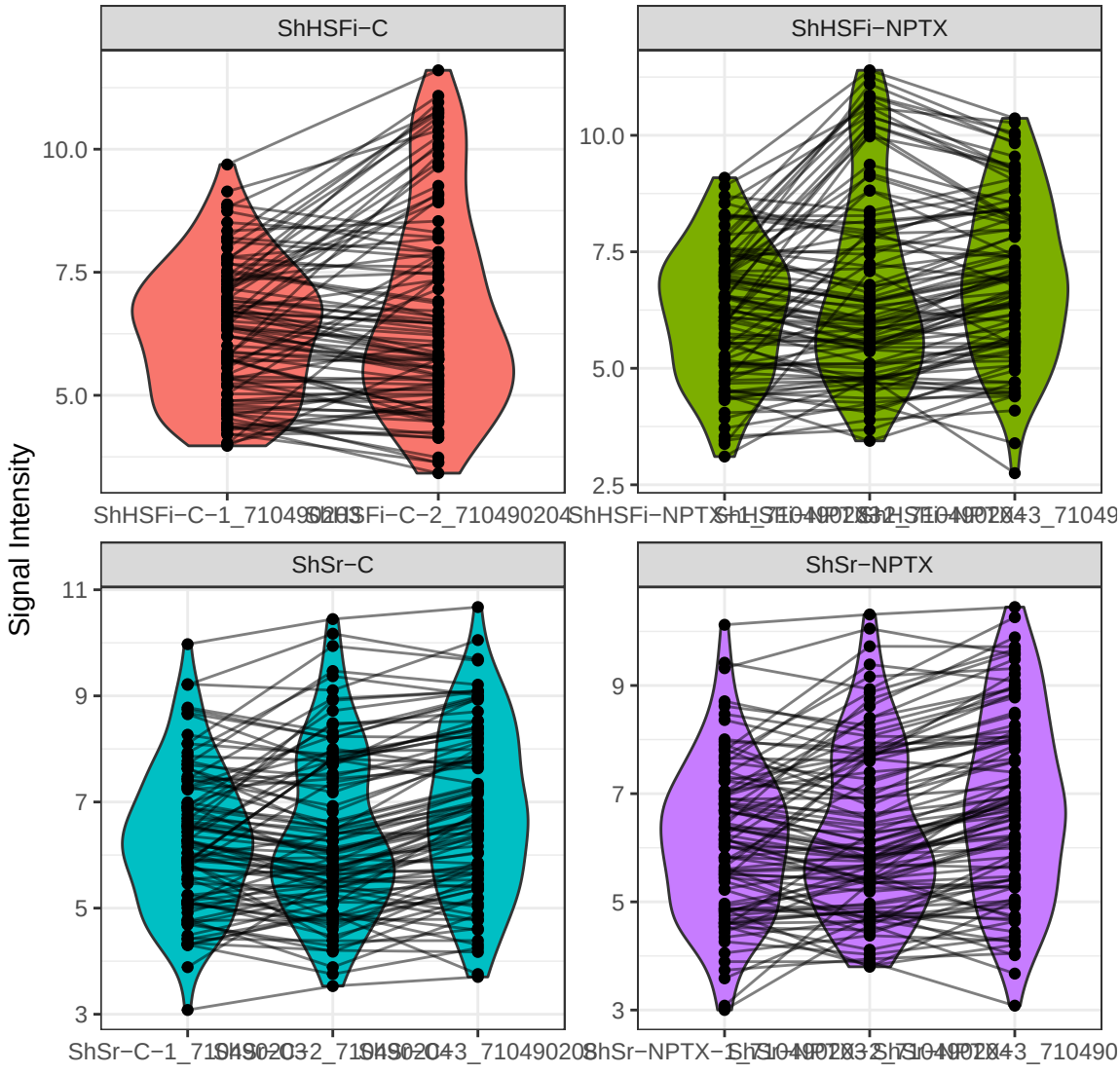


Figure 11.1: Violin Plots for signal intensity Distribution Across Groups for all replicates

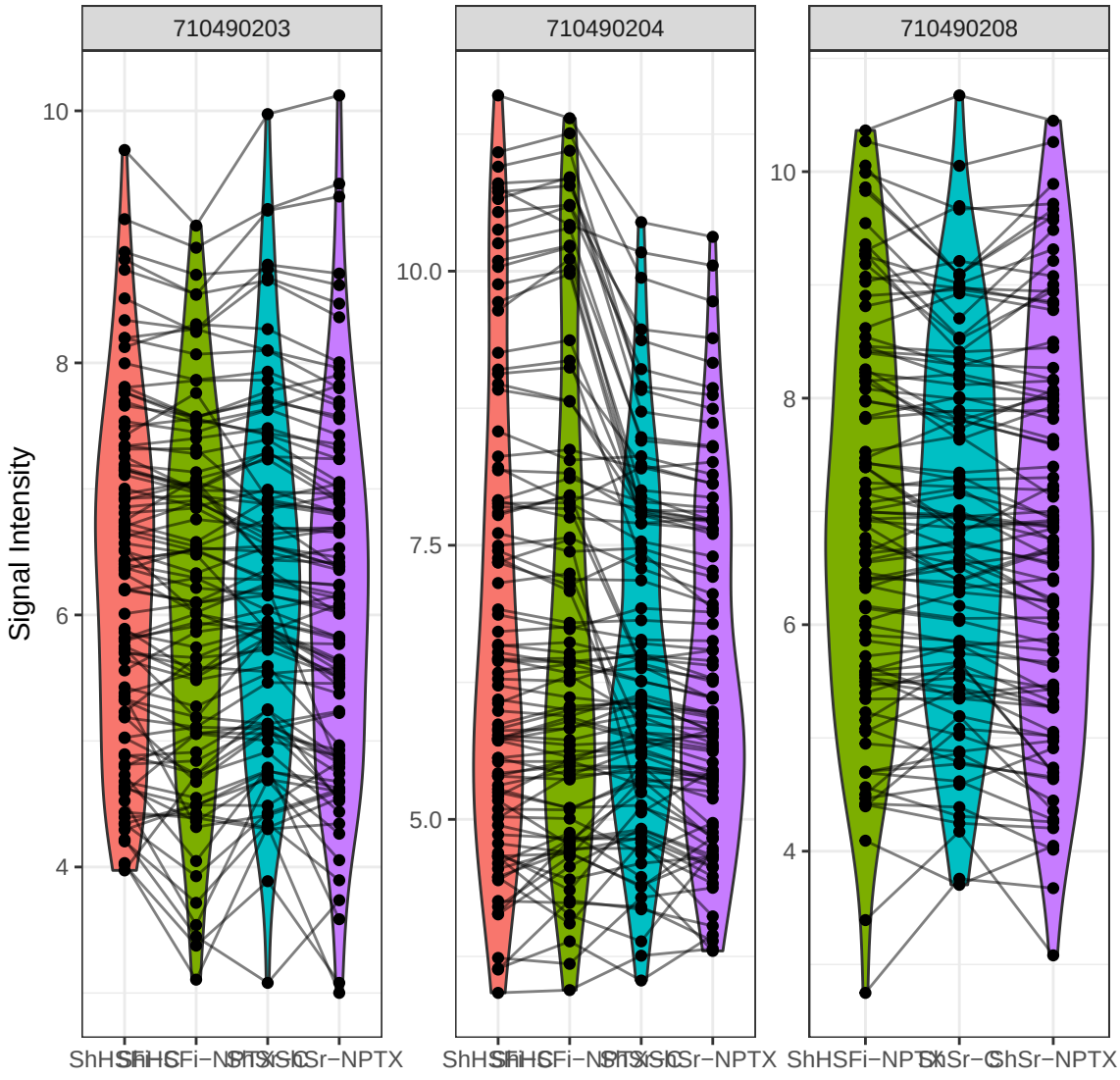


Figure 11.2: Violin Plots for signal intensity Distribution Across Chips for all replicates

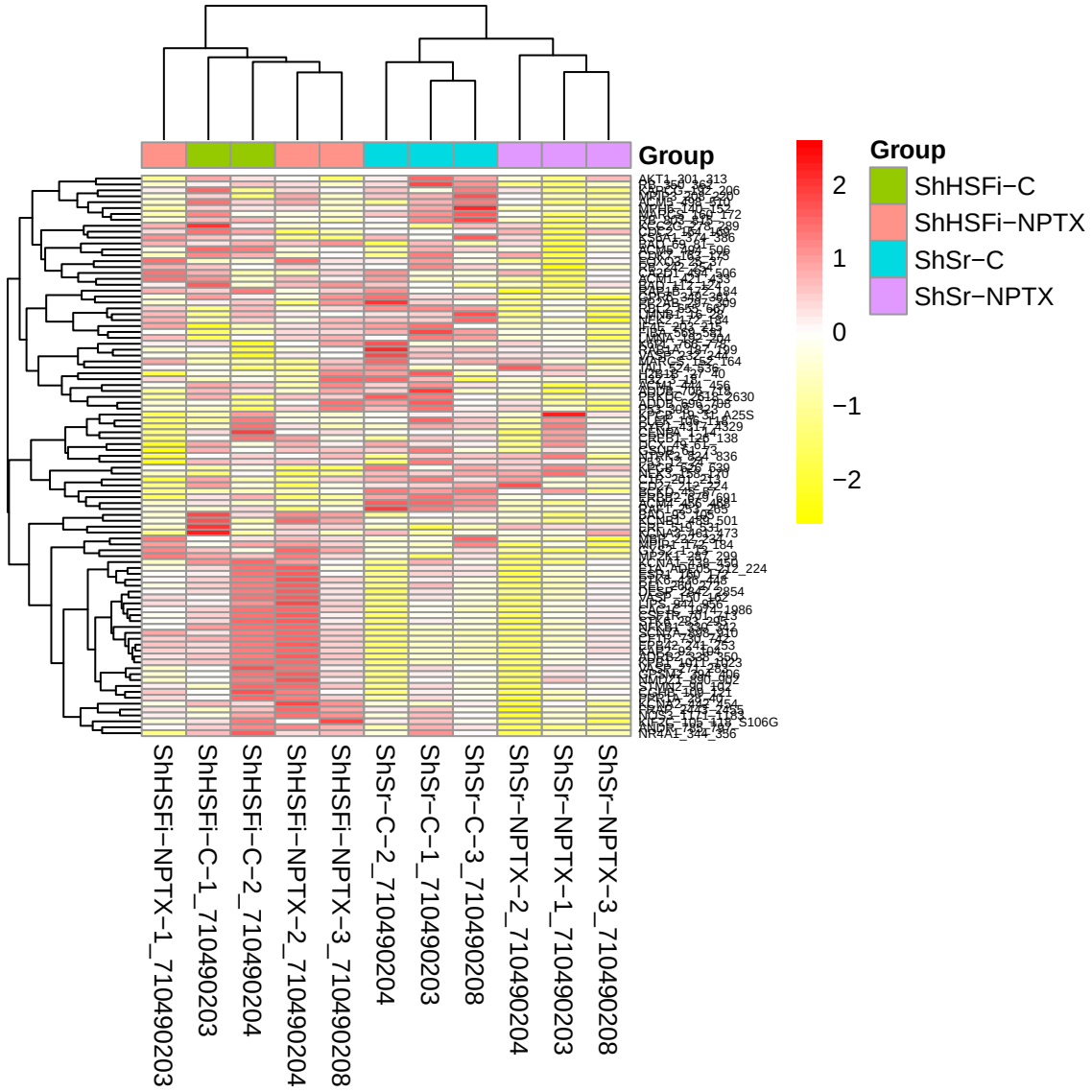


Figure 11.3: Row and chip normalized intensity values for the selected peptides

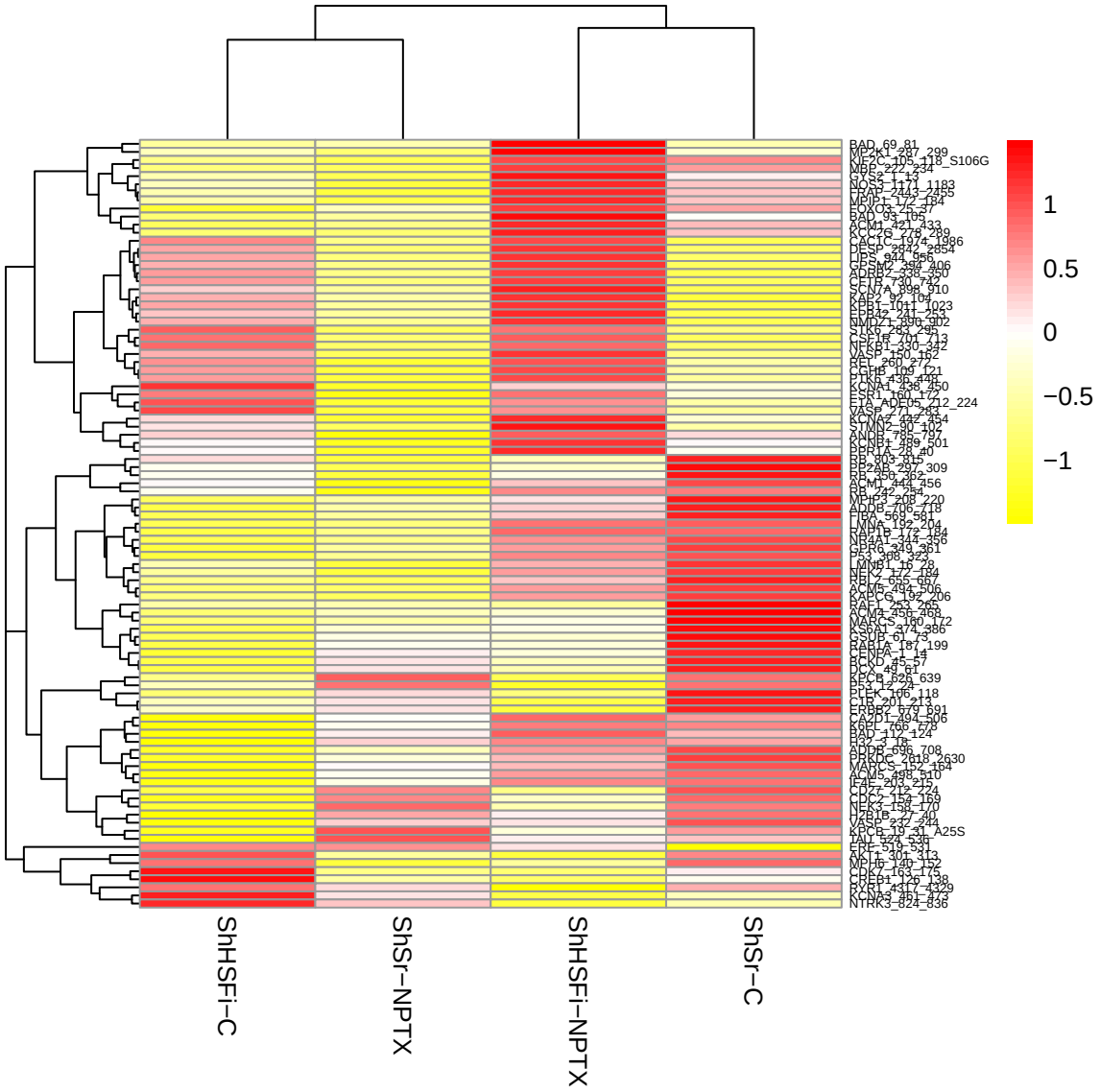
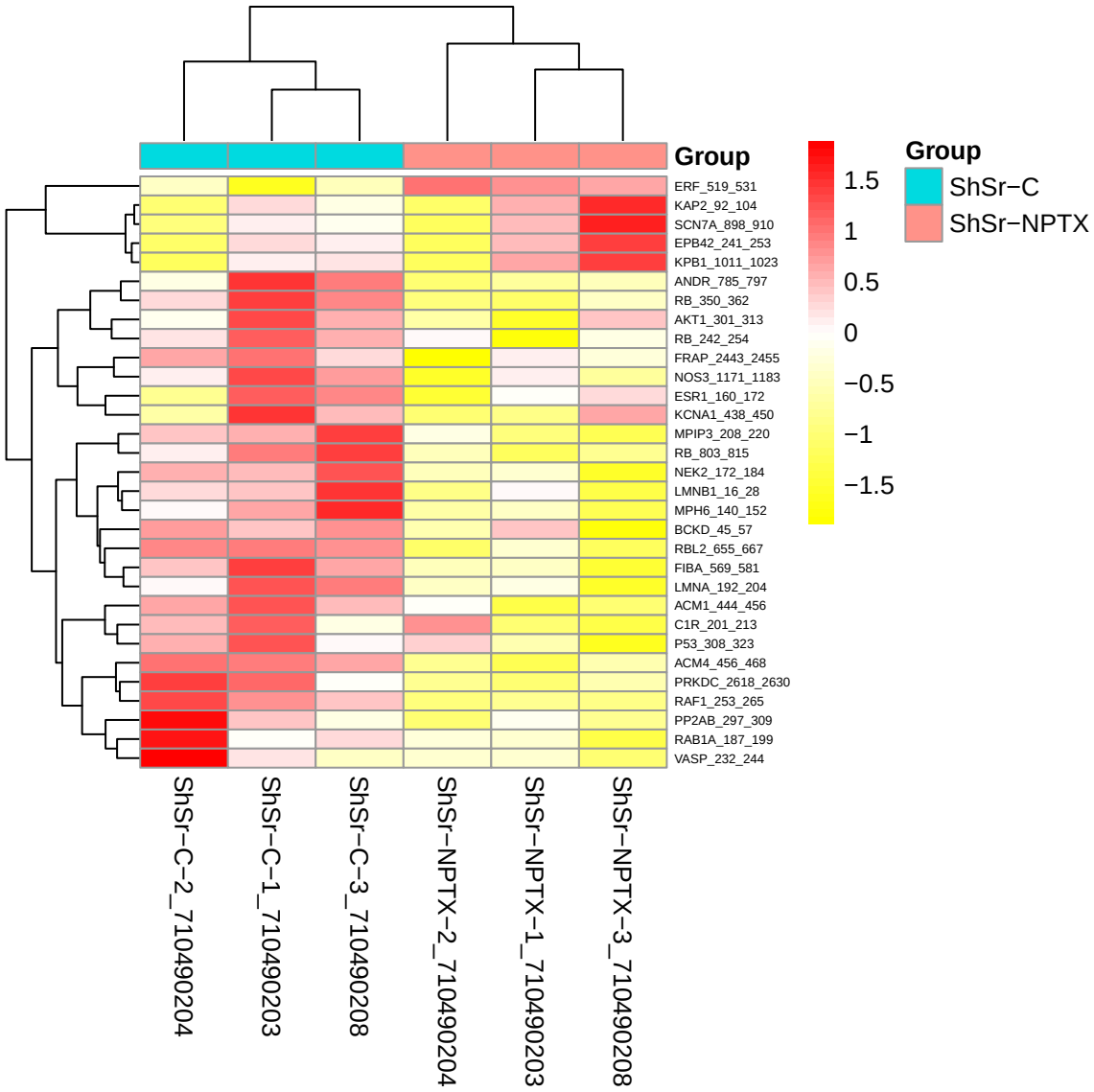


Figure 11.4: Row and group normalized intensity values for the selected peptides



ShSr-NPTX vs ShSr-C

11.4.0.1 Heatmap

After applying the *Filtering Parameters* for this group comparison, only *31*/141 peptides carried forward in the analysis (i.e. *31 hits*). Below are some figures to visualize the differences between these samples for considering these *hits*.

11.4.0.2 Violin Plot

Below, the violin plot visualizes the distribution of selected peptides for the analysis.

11.4.0.3 Waterfall Plot

This waterfall represents the log2 fold changes between the two groups at each peptide.

11.4.0.4 Upstream Kinase Analysis

The lab carefully curated and mapped the kinases that can act and phosphorylate each peptide present on the chip. This was achieved by using multiple sources including GPS 3.0, Kinexus Phosphonet, PhosphoELM and PhosphoSite Plus. Based on that association between peptides and kinases, a random sampling analysis is performed for these hits. The basic idea of *KRSA* is: For each iteration (*2000* iterations performed in this analysis), the same number of hits are randomly selected from the total 141/or 193 peptides present on the chip. Predicted kinases are then mapped to this sample list of peptides and number of kinases are determined. The kinase count from the actual hits and random sampling is then compared to determine the significance.

Kinase	AvgZ
HASPIN	1.5966667
SYK	1.1116667
NKF2	0.6958333
KIS	0.3991667
LMR	0.2966667
RCK	0.1558333
BUD32	0.0600000
CDK	0.0433333
CAMK1	-2.2425000
CK1	-2.3916667
AKT	-2.4308333
CAMKL	-2.4450000
PDK1	-2.4583333
RSK	-2.4750000
WNK	-2.5775000
DAPK	-2.7091667
PKC	-2.9775000
STE20	-3.5550000

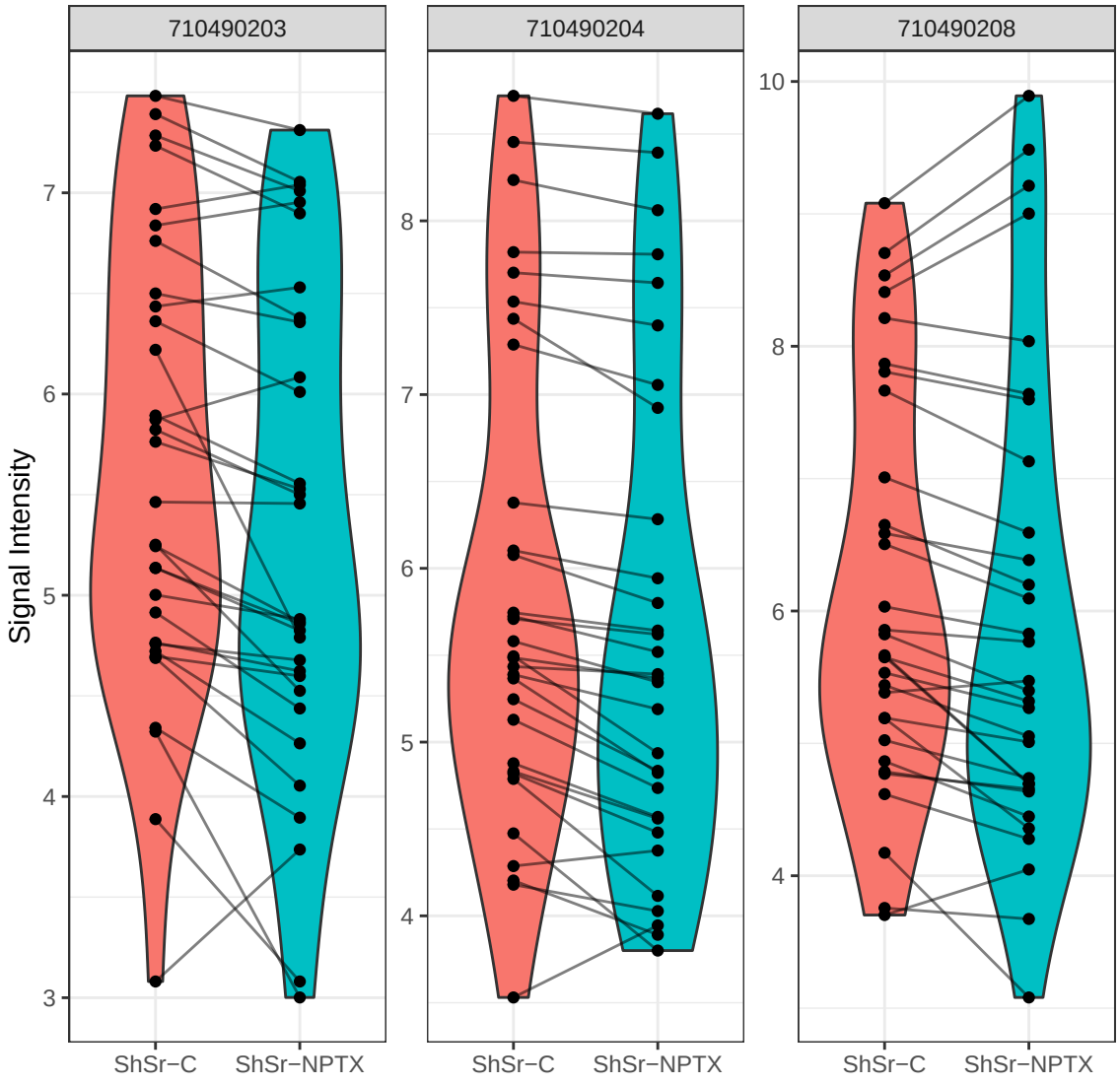


Figure 11.5: Violin plot of two groups

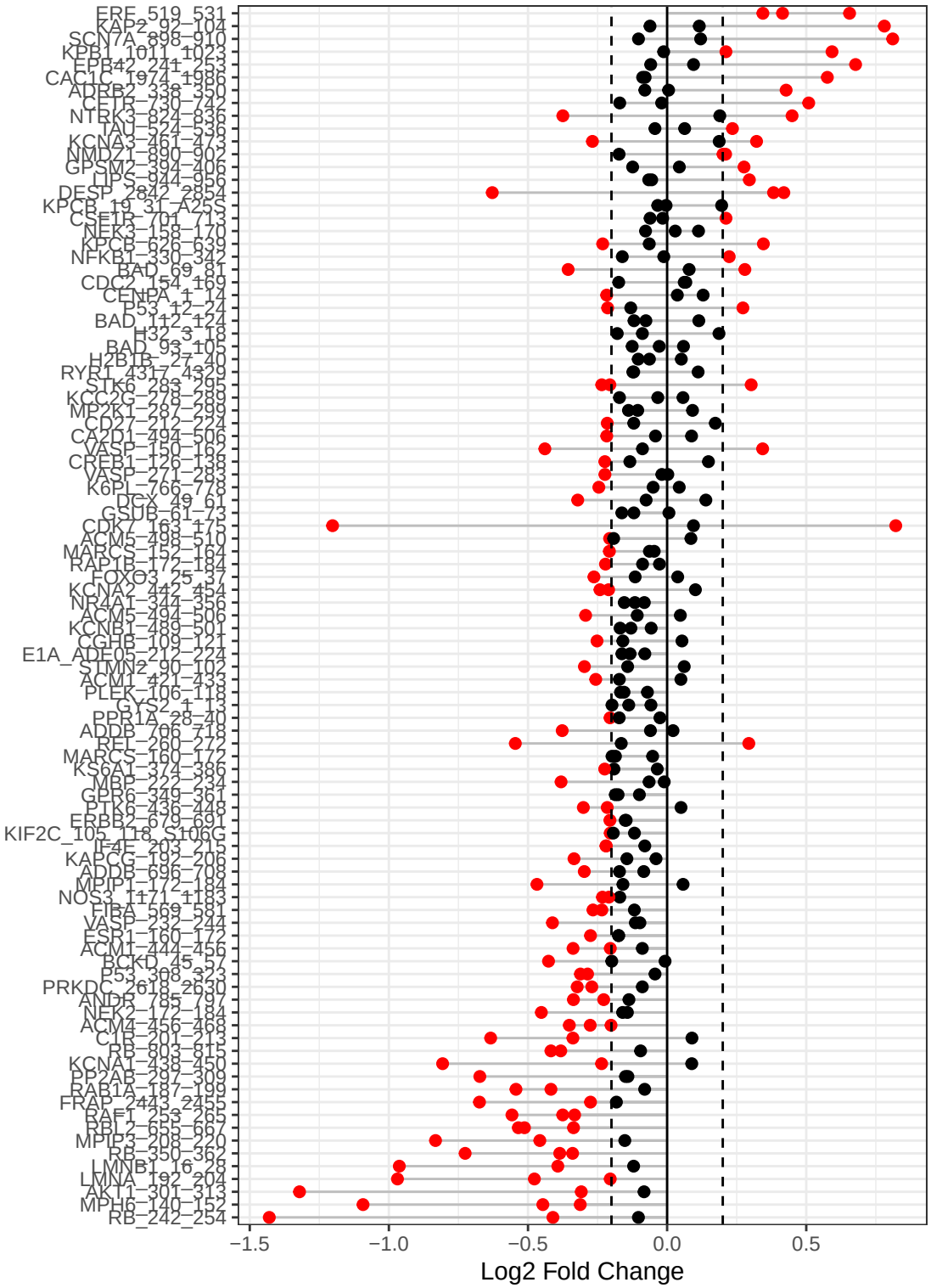


Figure 11.6: Waterfall Plot to show the distribution of change in peptide phosphorylation

Method	NumberOfPeptides
meanLFC.0.15	41
meanLFC.0.2	31
meanLFC.0.3	14
meanLFC.0.4	10
710490203.0.15	52
710490203.0.2	42
710490203.0.3	25
710490203.0.4	13
710490204.0.15	52
710490204.0.2	35
710490204.0.3	15
710490204.0.4	9
710490208.0.15	58
710490208.0.2	46
710490208.0.3	29
710490208.0.4	19

11.4.0.5 Z Scores Plot

We will plot the individual and averaged Z scores using both the across and within chip analyses.

11.4.0.6 Reverse KRSA Plot

We will use the reverse KRSA plot function, to plot the log2 fold change values for all peptides mapped to kinase hits. This will help us examine the activity of the kinase

11.4.0.7 Coverage Plot

To view the coverage of kinases across the full list of peptides on the chip, we will use the coverage plot function

11.4.0.8 Ball Model Network

We will view the ball model network function, to generate a model representing the protein-protein interactions between kinases

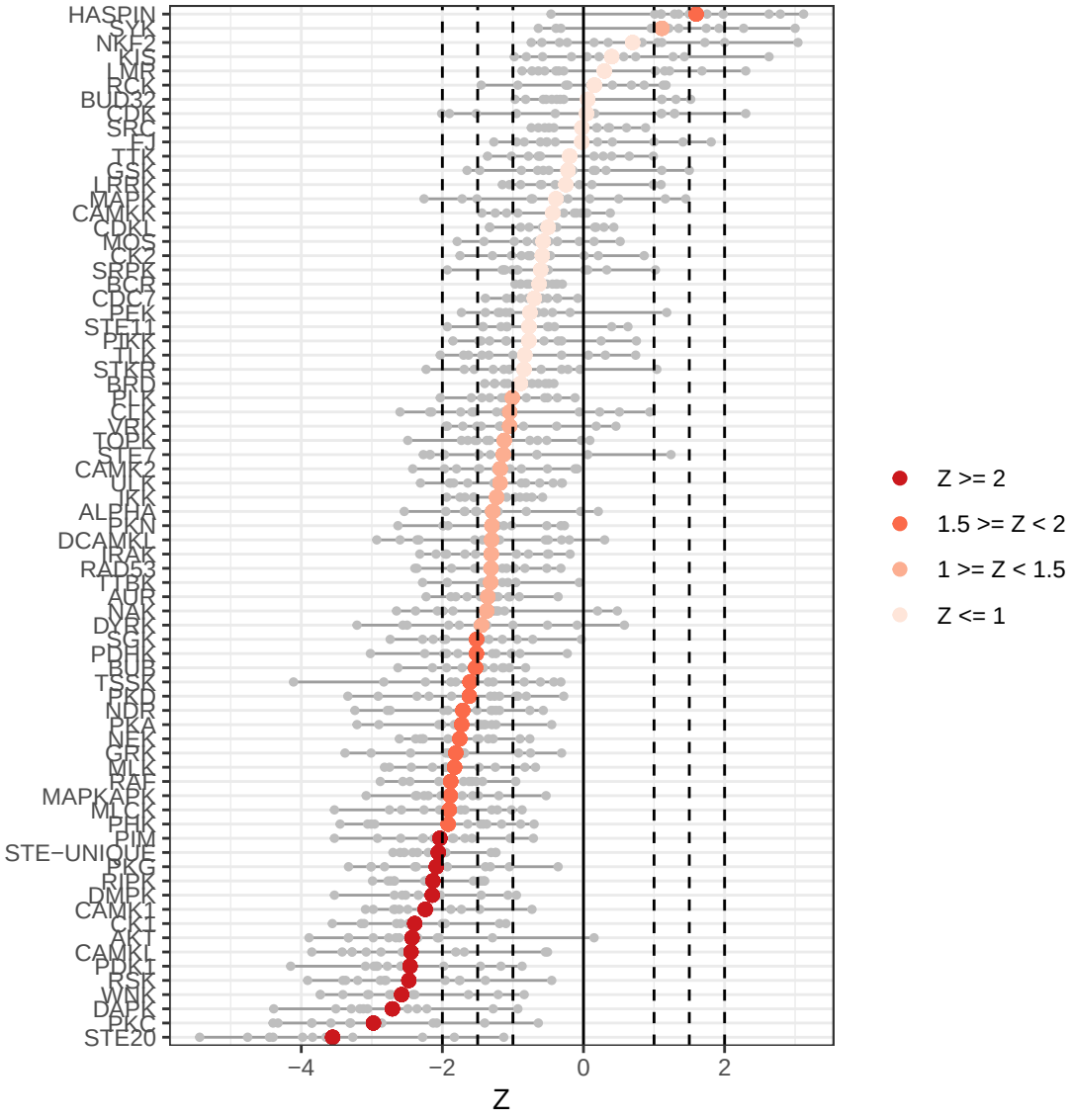


Figure 11.7: Waterfall plot of the Z Scores of each kinase family

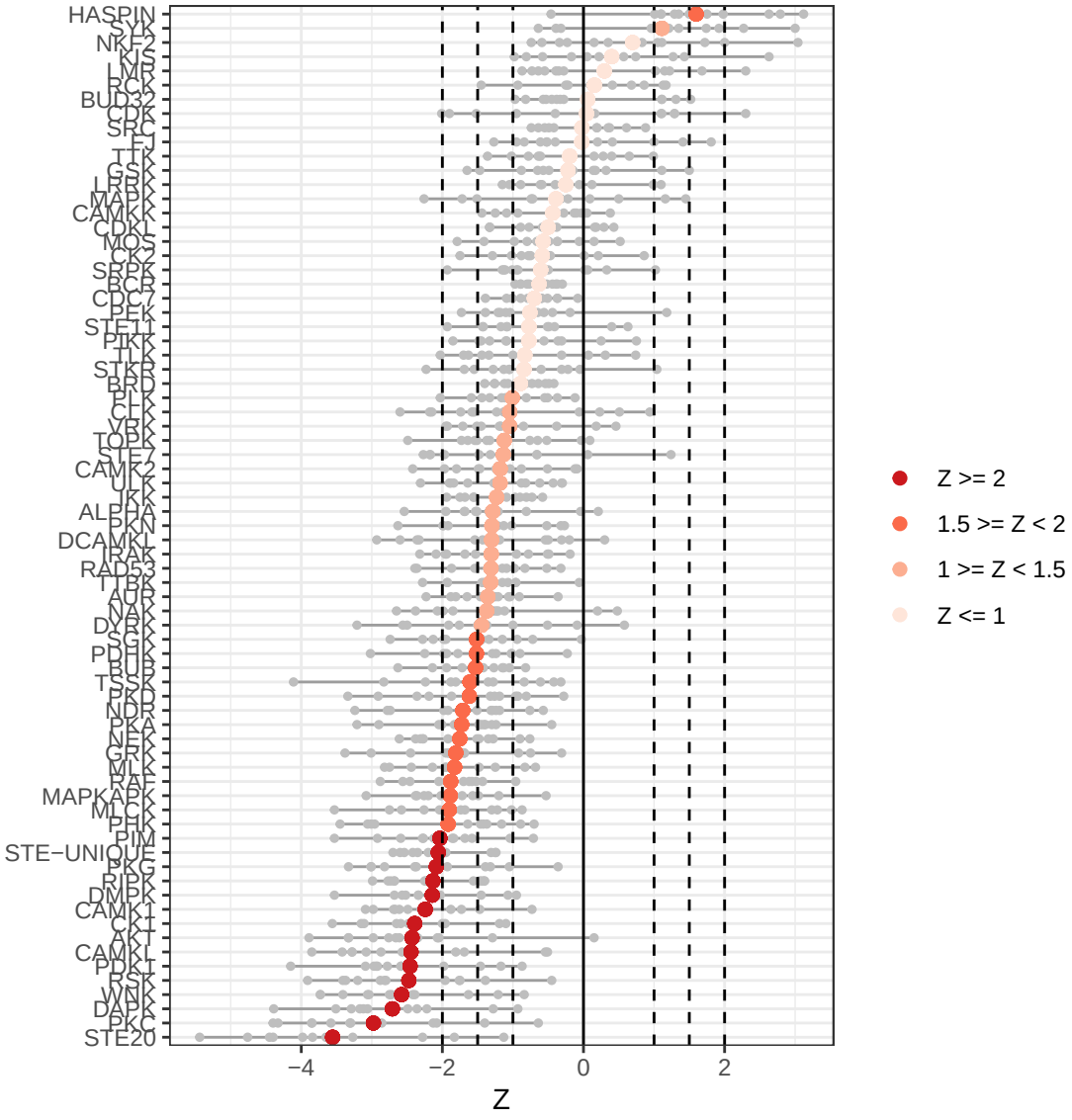


Figure 11.8: Waterfall plot of the Z Scores of each kinase family

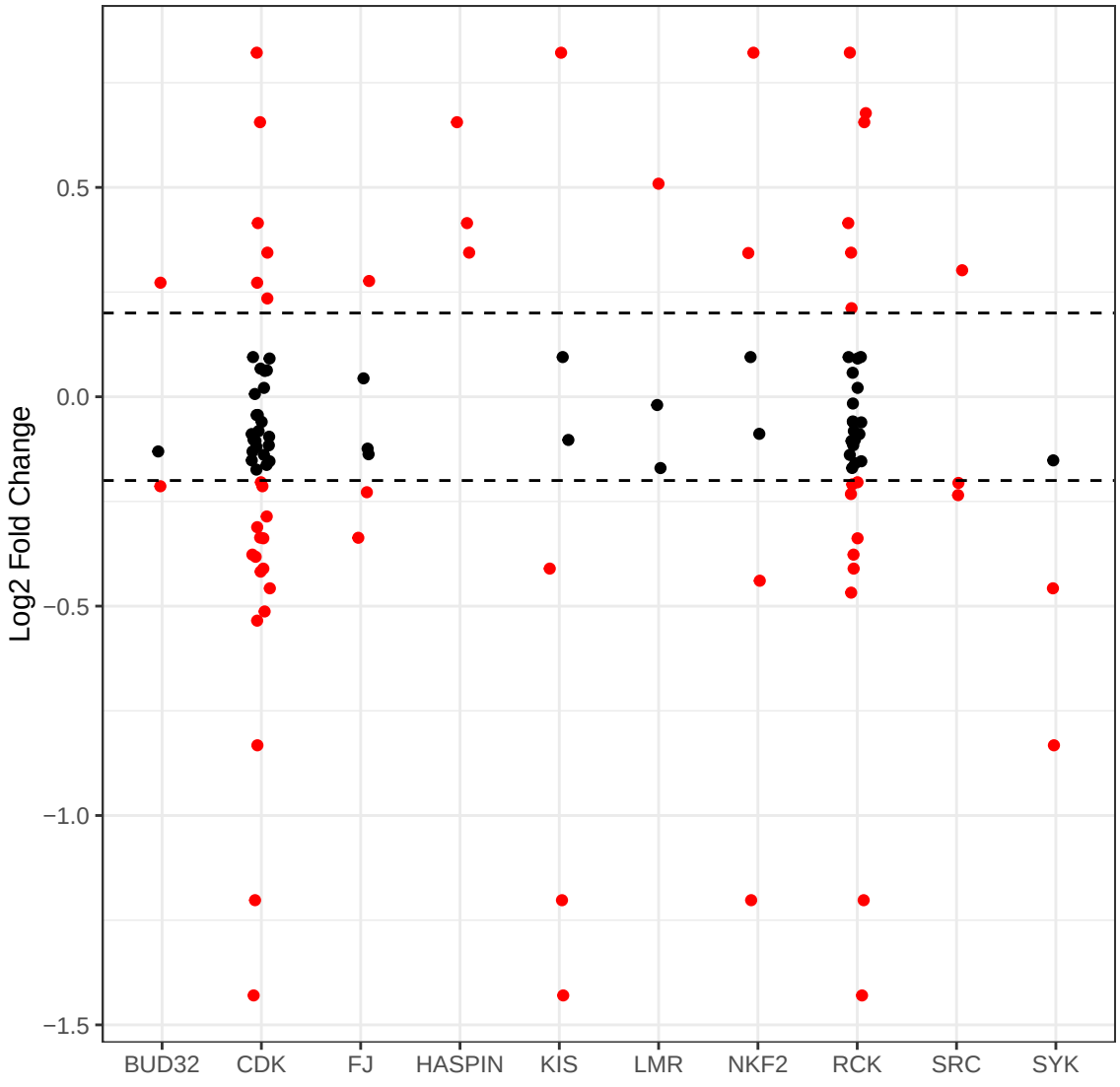


Figure 11.9: Kinase Activity summary for each kinase family based on peptide phosphorylation

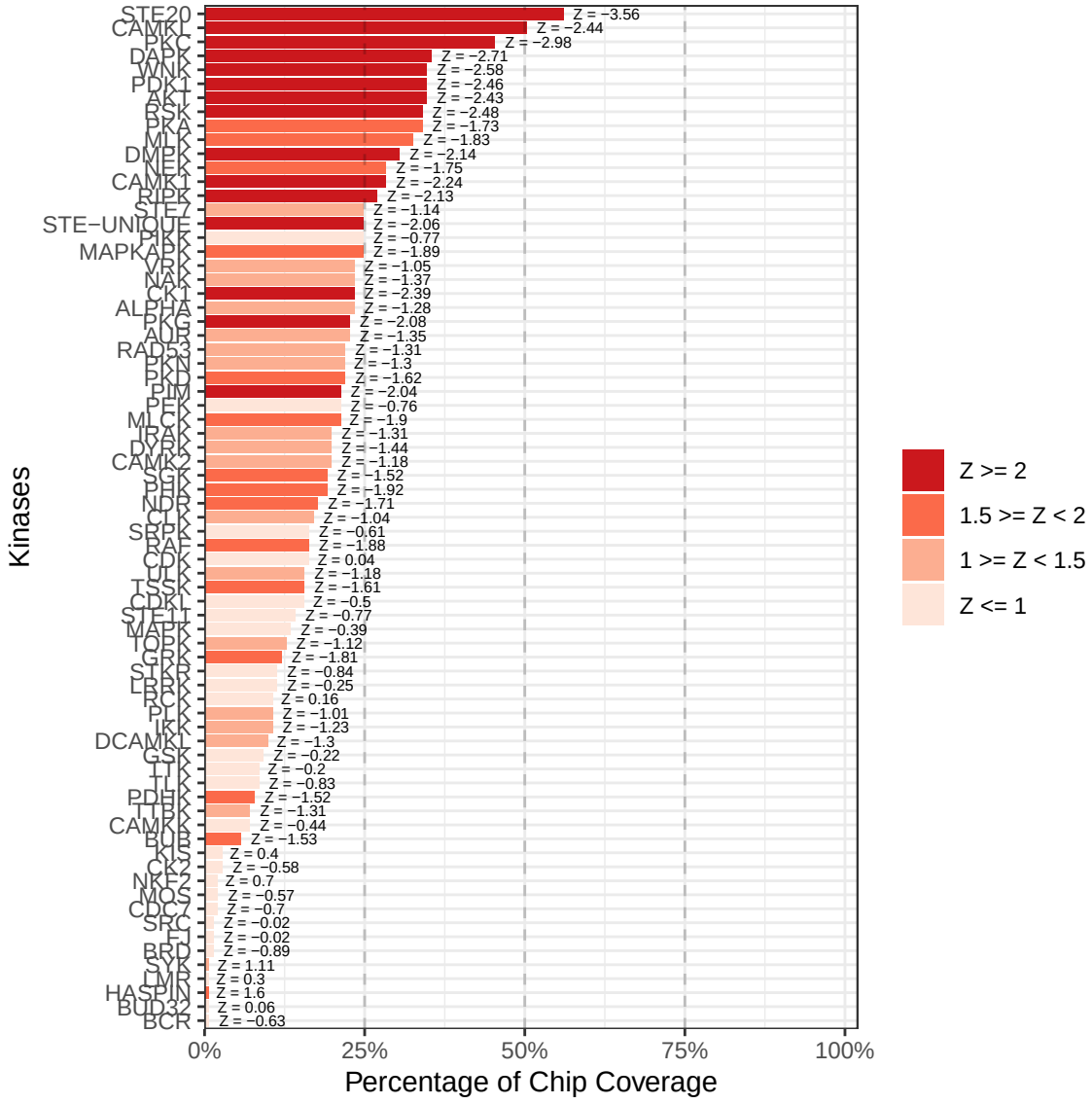


Figure 11.10: Percentage of peptides each kinase family phosphorylates

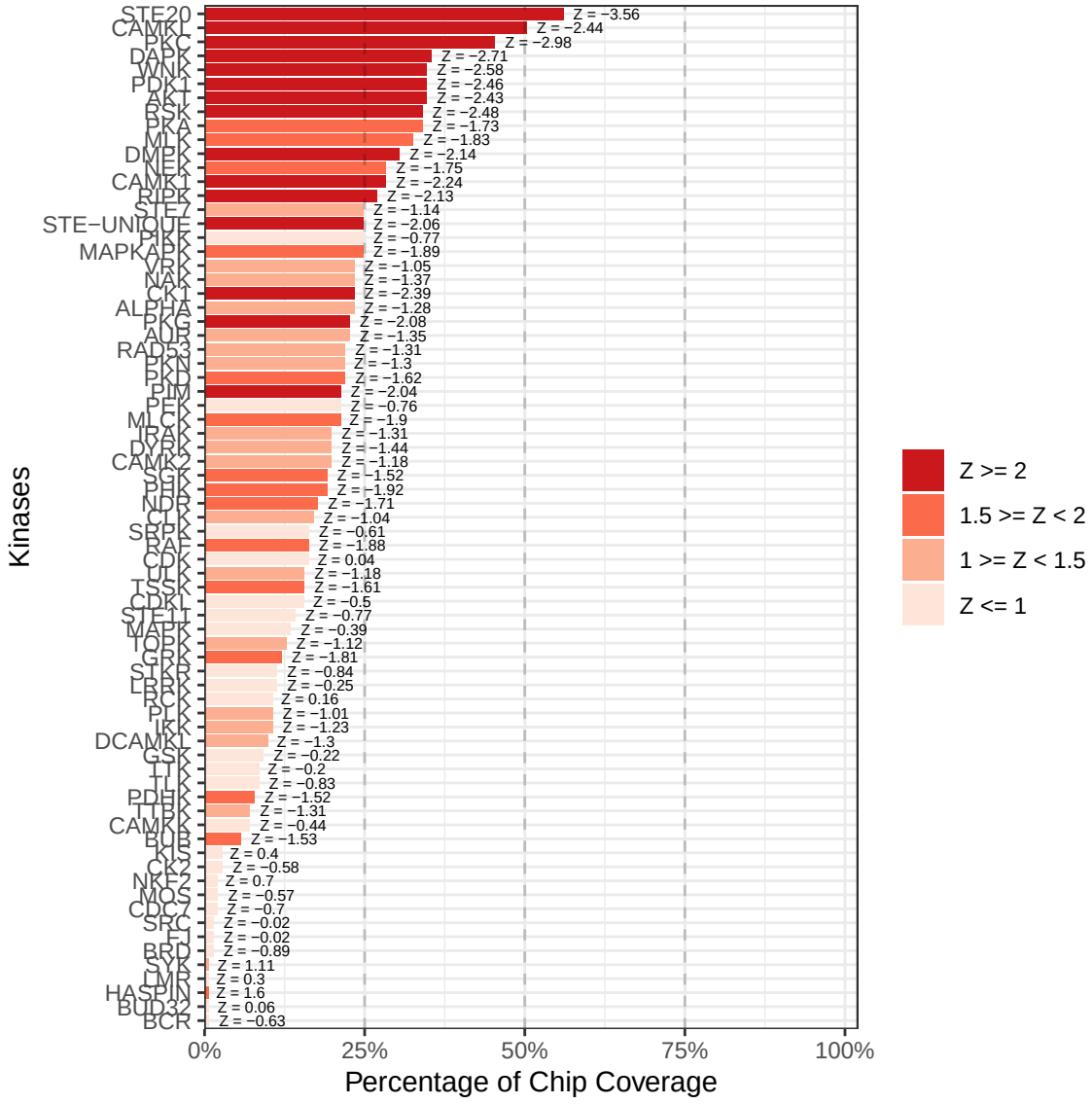
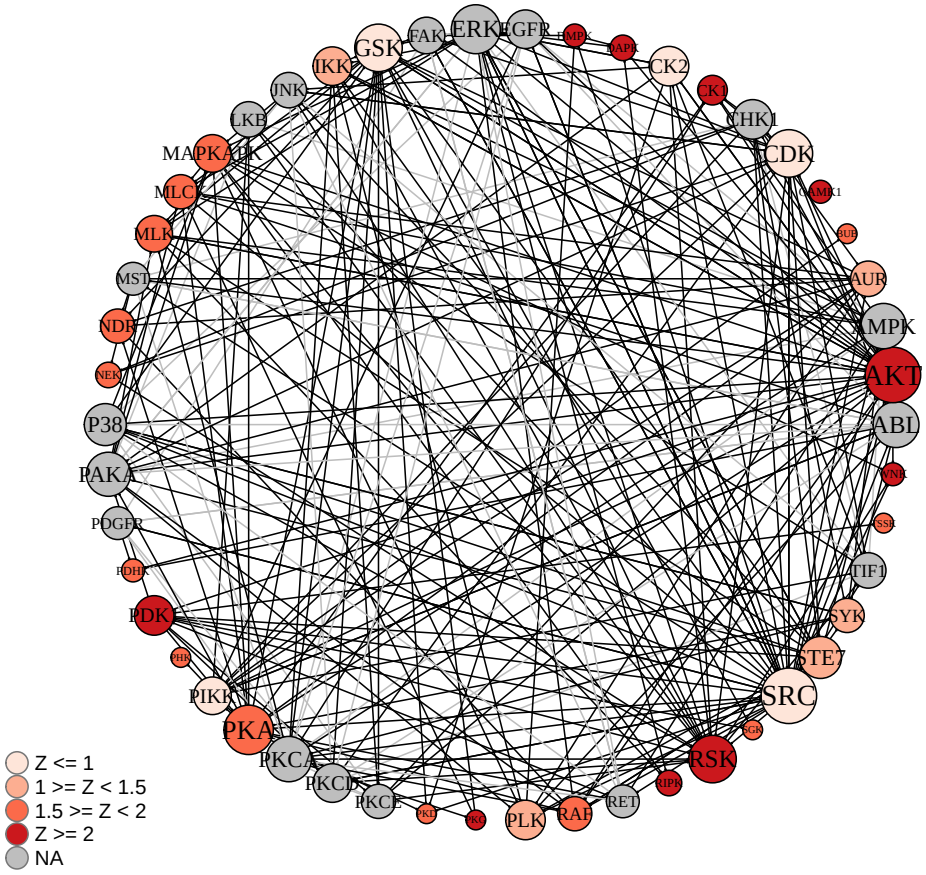
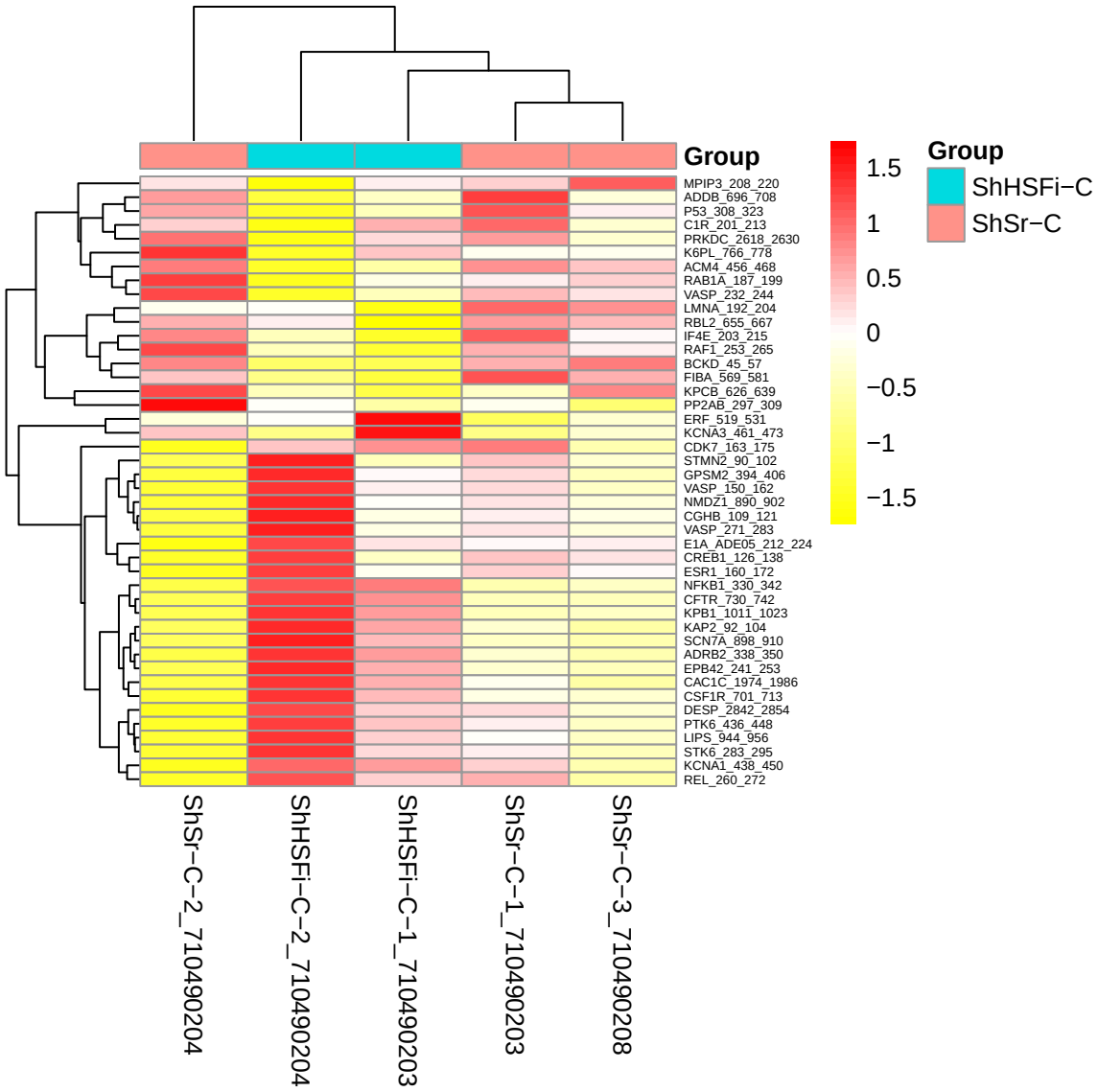


Figure 11.11: Percentage of peptides each kinase family phosphorylates


ERK
EGFR
GSK
FAK
IKKE
JNK
CK2
CHK1
CDK
LKB
MAPKAPK
MLCK
MLK
MST
NDR
AUR
AMPK
AKT
P38
ABL
PAKA
PDGFR
PDHK
PDK1
PIKK
PKA
PKCA
PKCD
PKCE
PLK
RAF
RET
RSK
SRC
STE7
SYK
TIF1
Z <= 1
1 >= Z < 1.5
1.5 >= Z < 2
Z >= 2
NA



ShHSFi-C vs ShSr-C

11.4.0.9 Heatmap

After applying the *Filtering Parameters* for this group comparison, only *44*/141 peptides carried forward in the analysis (i.e. *44 hits*). Below are some figures to visualize the differences between these samples for considering these *hits*.

11.4.0.10 Violin Plot

Below, the violin plot visualizes the distribution of selected peptides for the analysis.

11.4.0.11 Waterfall Plot

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11.4.0.12 Upstream Kinase Analysis

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Kinase	AvgZ
LMR	1.57375
CAMKK	0.56000
HASPIN	0.52750
SYK	0.34250
FJ	0.28625
PDK1	-2.16875
WNK	-2.28875
TOPK	-2.31250
DAPK	-2.36375
NEK	-2.38125
CAMKL	-2.38250
RSK	-2.50125
STE-UNIQUE	-2.51375
PKC	-2.99750
STE20	-4.27500

Method	NumberOfPeptides
meanLFC.0.15	51

Method	NumberOfPeptides
meanLFC.0.2	44
meanLFC.0.3	36
meanLFC.0.4	27
710490203.0.15	43
710490203.0.2	35
710490203.0.3	24
710490203.0.4	14
710490204.0.15	59
710490204.0.2	52
710490204.0.3	42
710490204.0.4	35

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11.4.0.16 Ball Model Network

We will view the ball model network function, to generate a model representing the protein-protein interactions between kinases

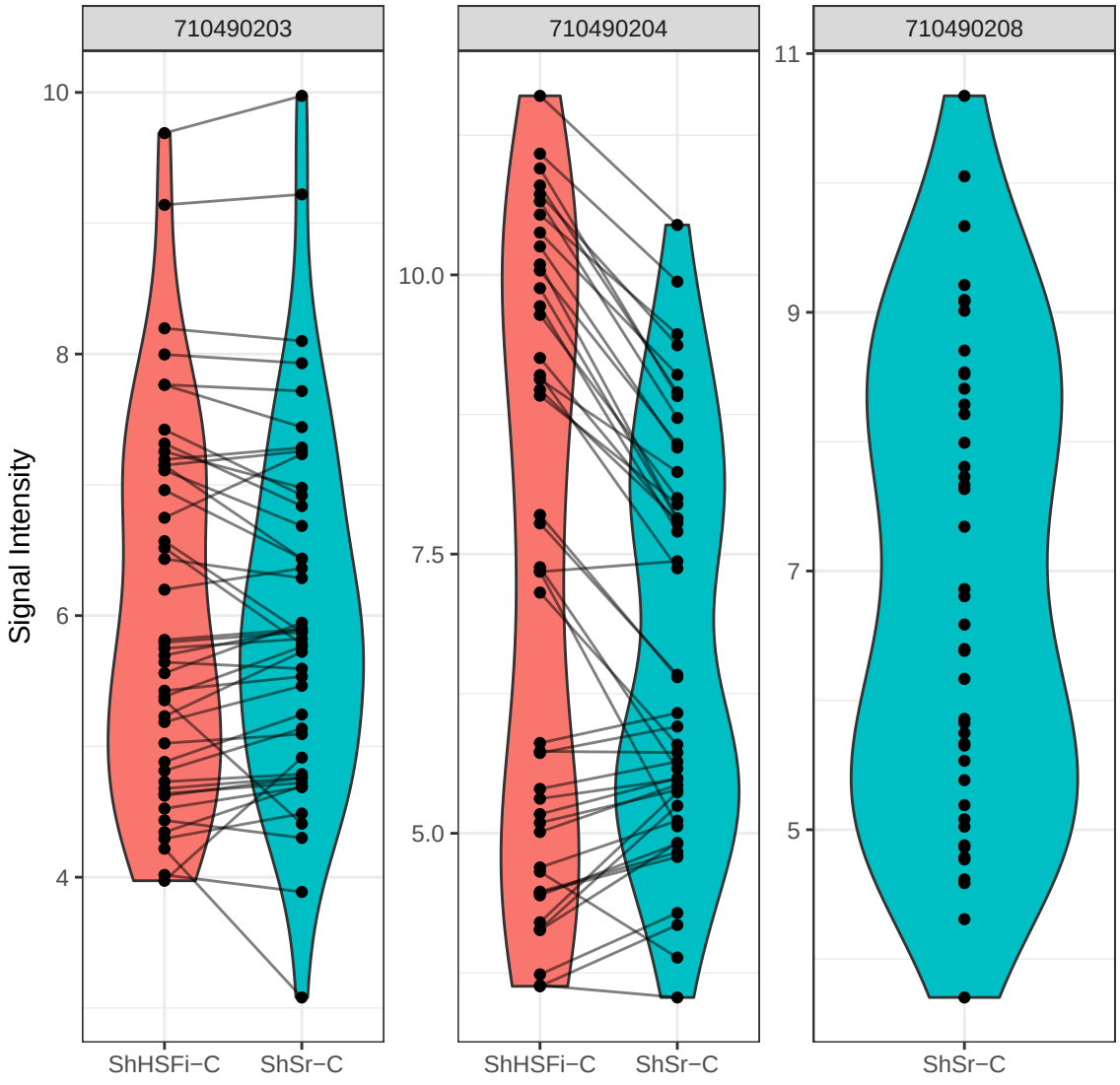


Figure 11.12: Violin plot of two groups

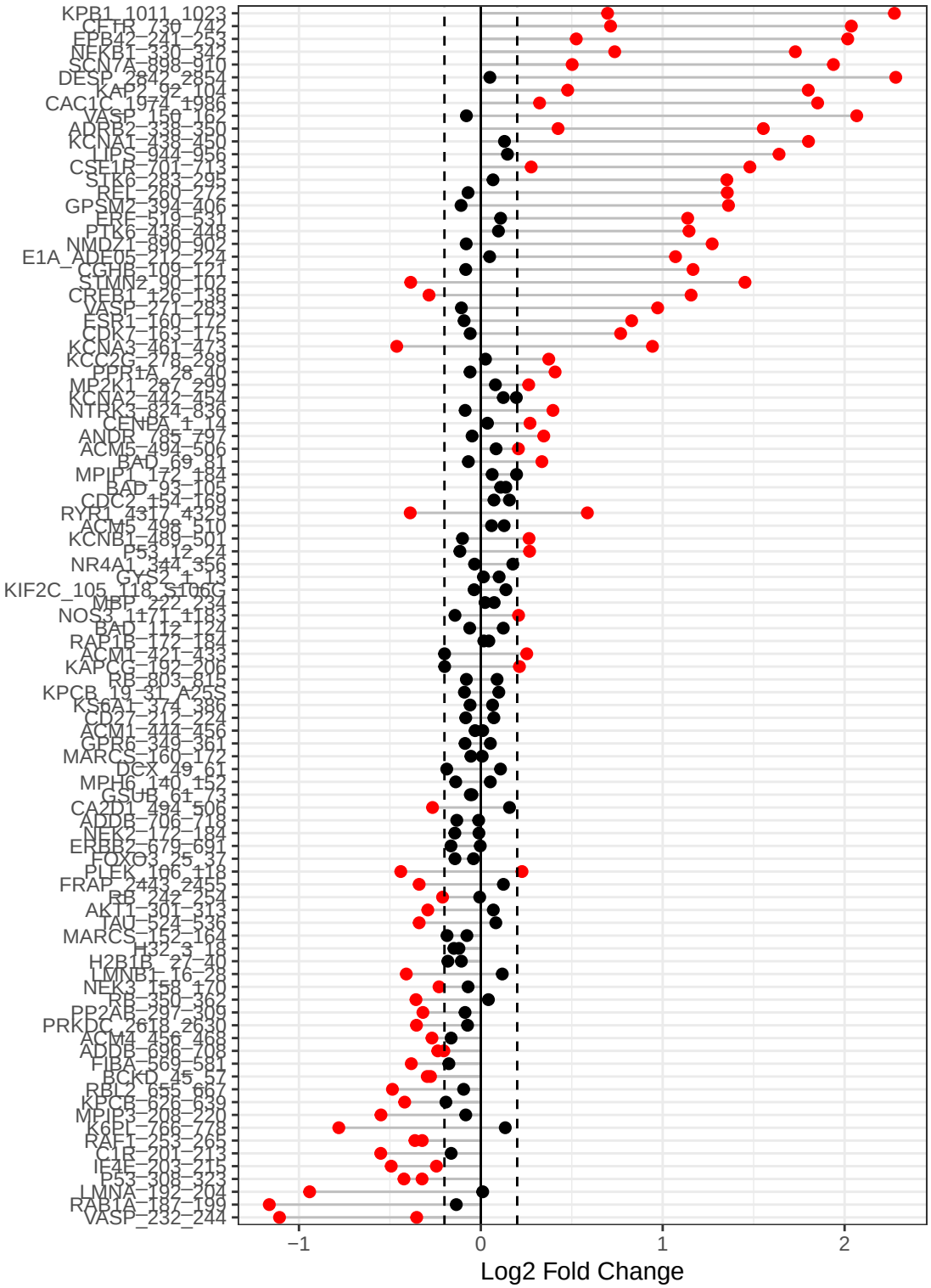


Figure 11.13: Waterfall Plot to show the distribution of change in peptide phosphorylation

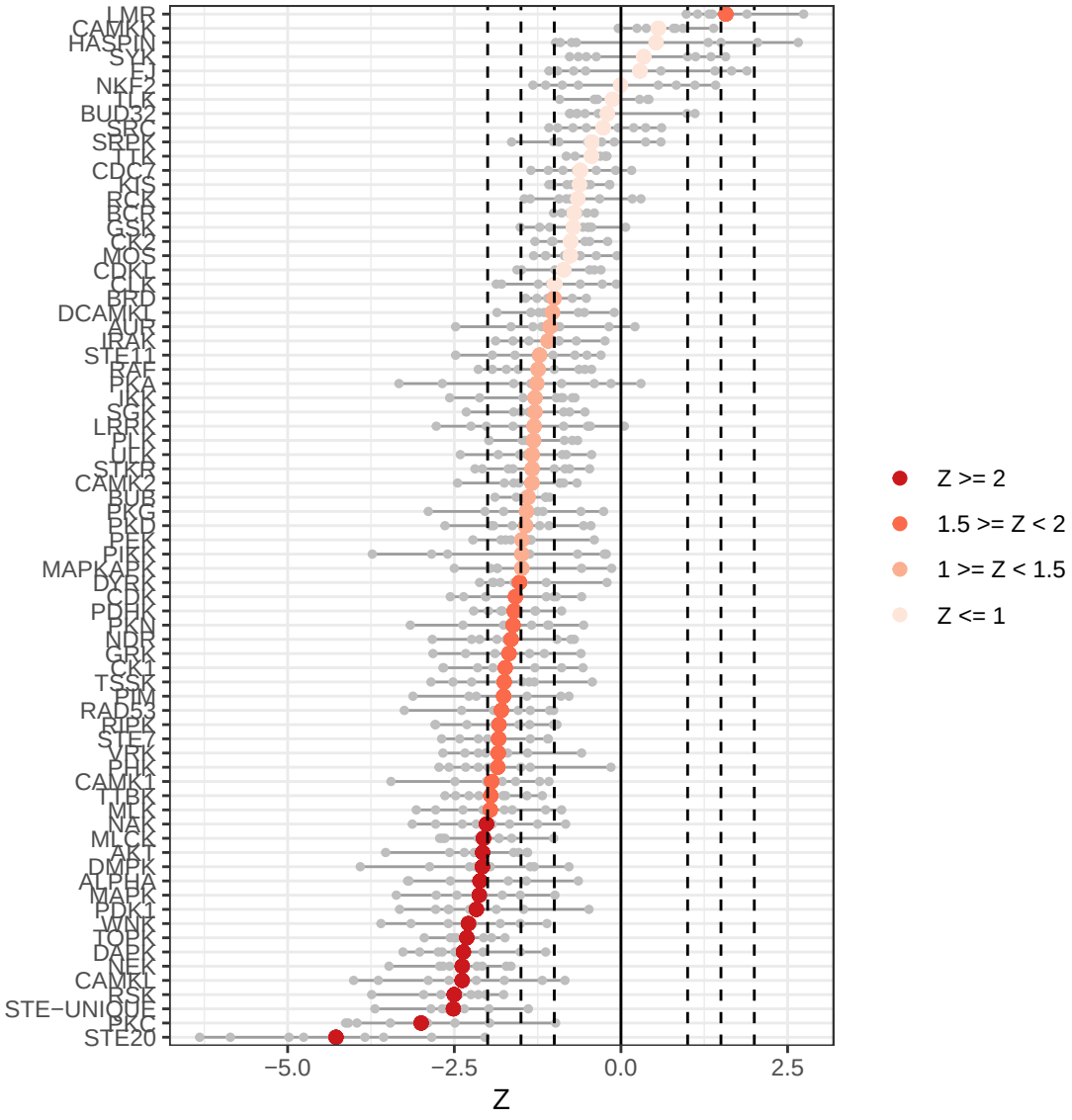


Figure 11.14: Waterfall plot of the Z Scores of each kinase family

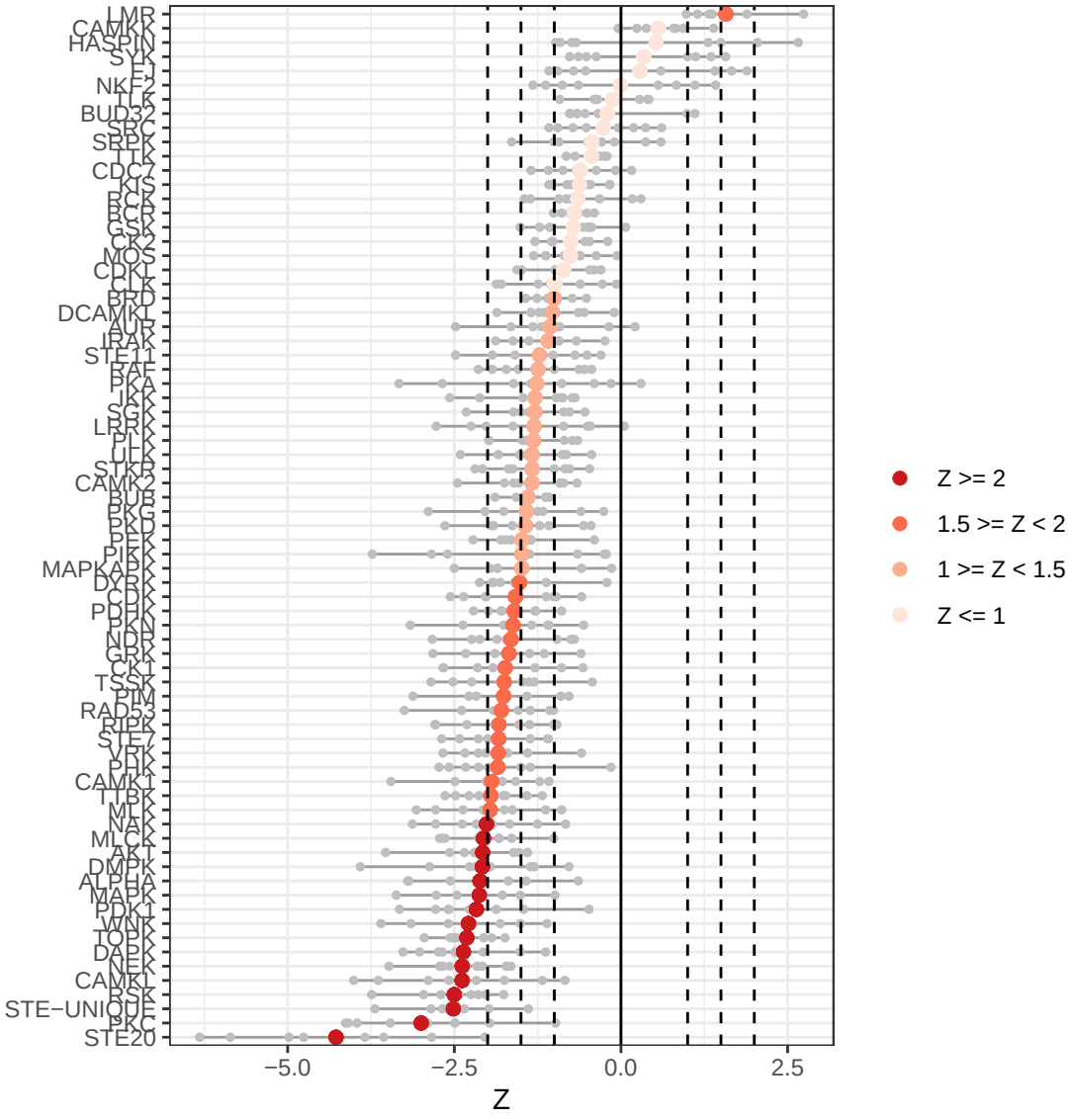


Figure 11.15: Waterfall plot of the Z Scores of each kinase family

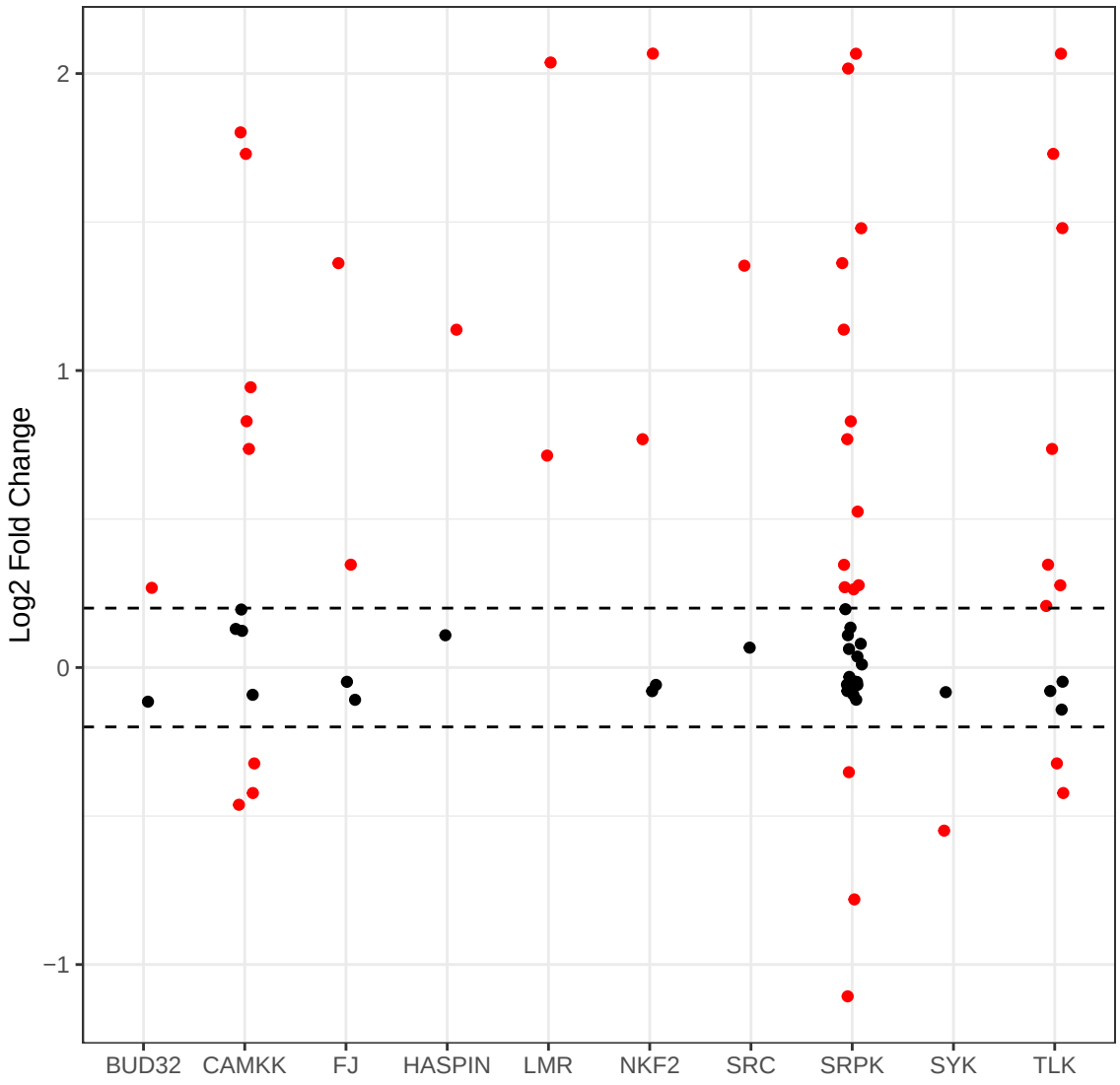


Figure 11.16: Kinase Activity summary for each kinase family based on peptide phosphorylation

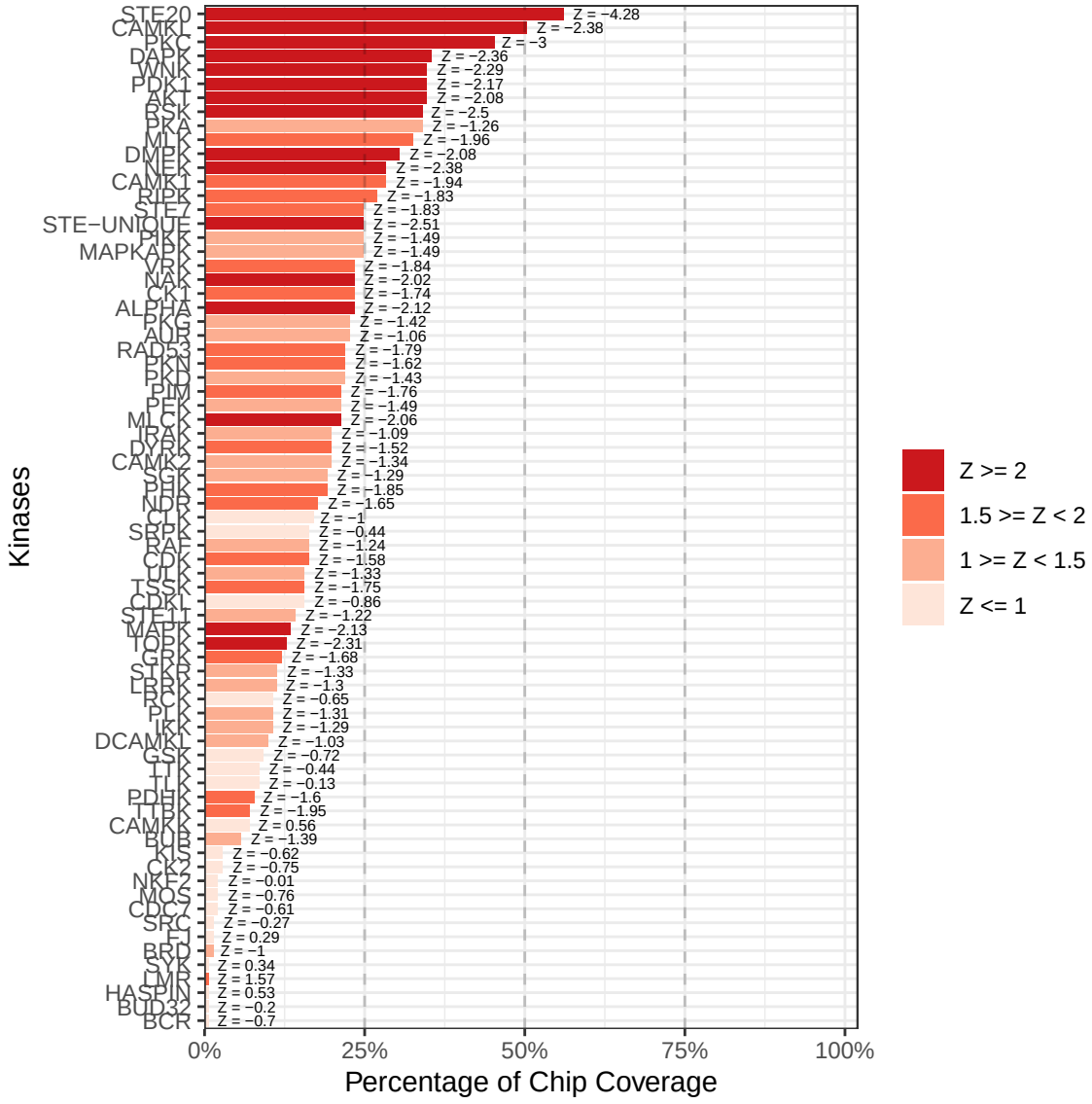


Figure 11.17: Percentage of peptides each kinase family phosphorylates

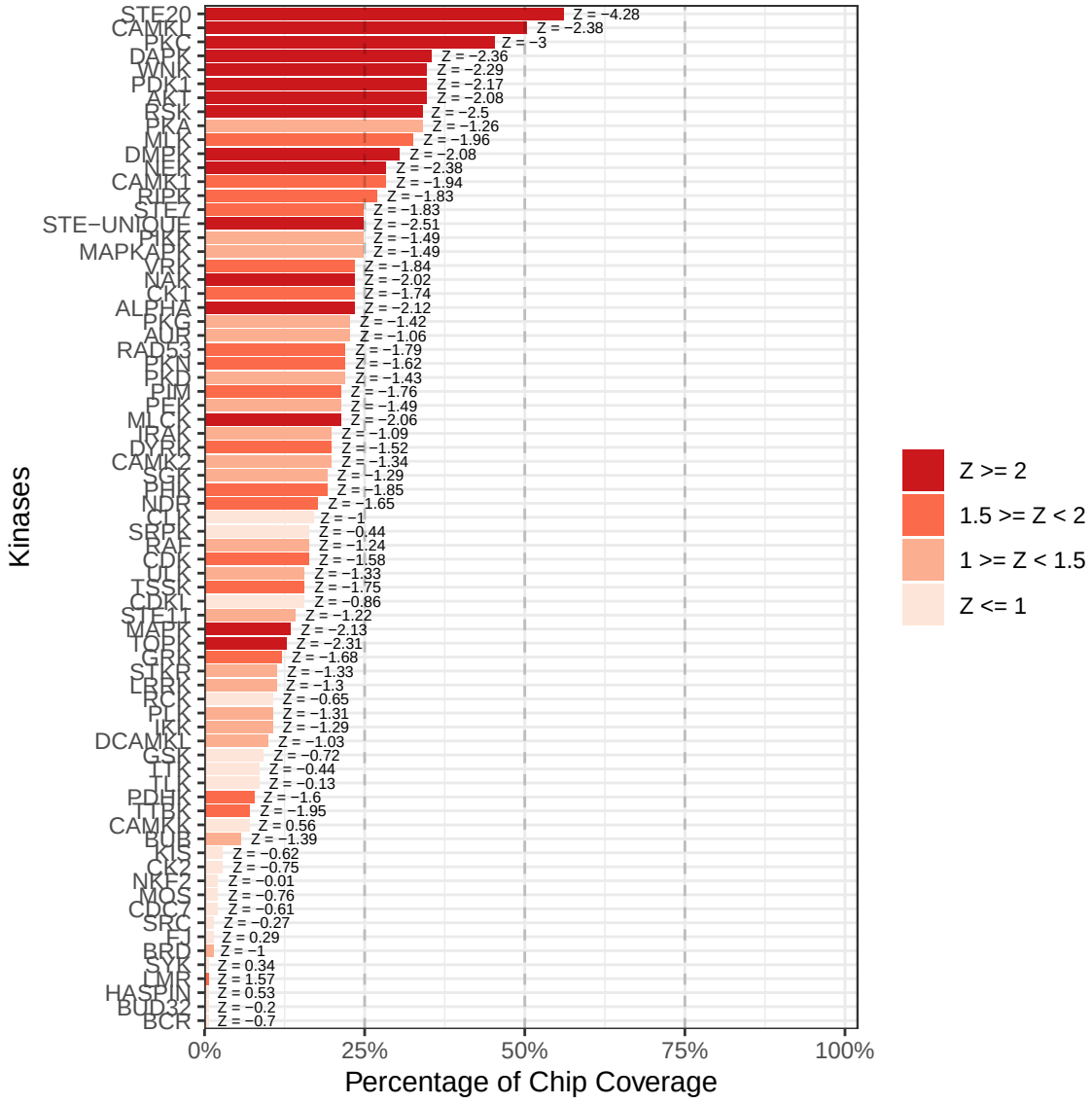
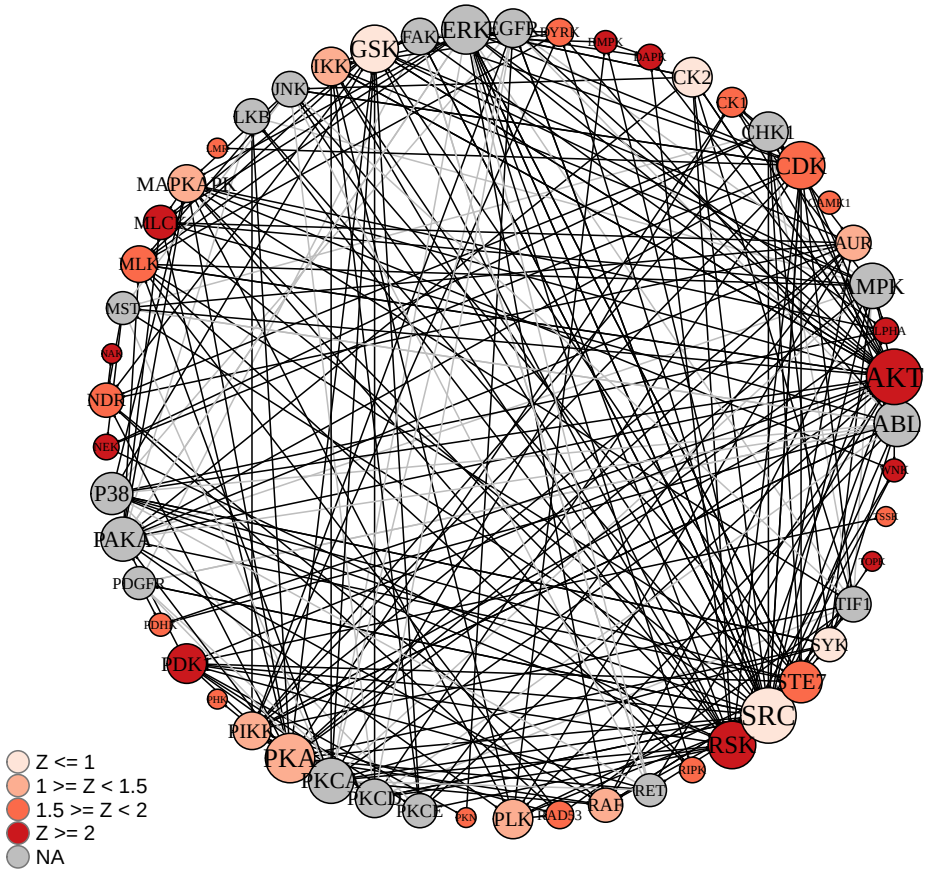
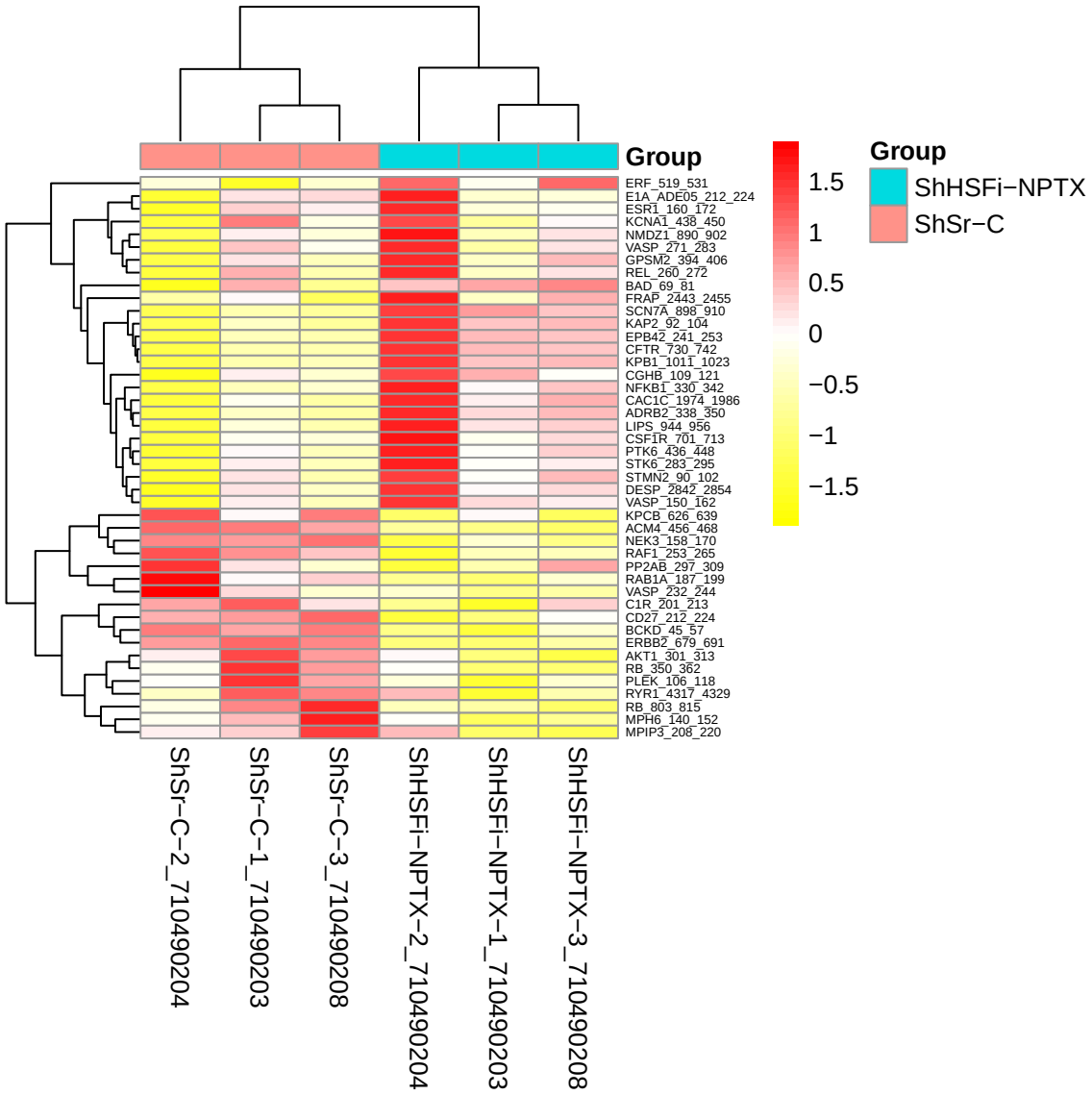


Figure 11.18: Percentage of peptides each kinase family phosphorylates


ERK
EGFR
GSK
FAK
IKK
JNK
LKB
CK2
CHK1
CDK
MAPKAPK
MLCK
MLK
MST
AUR
AMPK
AKT
ABL
NDR
NEK
P38
PAKA
PDGFR
PDK1
PIKK
PKA
PKCA
PKCD
PKCE
PLK
RAF
RET
RSK
SRC
STE7
SYK
TIF1
Z <= 1
1 >= Z < 1.5
1.5 >= Z < 2
Z >= 2
NA



ShHSFi-NPTX vs ShSr-C

11.4.0.17 Heatmap

After applying the *Filtering Parameters* for this group comparison, only *44*/141 peptides carried forward in the analysis (i.e. *44 hits*). Below are some figures to visualize the differences between these samples for considering these *hits*.

11.4.0.18 Violin Plot

Below, the violin plot visualizes the distribution of selected peptides for the analysis.

11.4.0.19 Waterfall Plot

This waterfall represents the log2 fold changes between the two groups at each peptide.

11.4.0.20 Upstream Kinase Analysis

The lab carefully curated and mapped the kinases that can act and phosphorylate each peptide present on the chip. This was achieved by using multiple sources including GPS 3.0, Kinexus Phosphonet, PhosphoELM and PhosphoSite Plus. Based on that association between peptides and kinases, a random sampling analysis is performed for these hits. The basic idea of *KRSA* is: For each iteration (*2000* iterations performed in this analysis), the same number of hits are randomly selected from the total 141/or 193 peptides present on the chip. Predicted kinases are then mapped to this sample list of peptides and number of kinases are determined. The kinase count from the actual hits and random sampling is then compared to determine the significance.

Kinase	AvgZ
LMR	1.5083333
HASPIN	0.8383333
FJ	0.5891667
SYK	0.5400000
BUD32	0.3550000
NKF2	0.1941667
MLK	-2.2150000
PDK1	-2.3316667
DAPK	-2.3583333
RIPK	-2.4866667
ALPHA	-2.5166667
CAMKL	-2.7816667
WNK	-2.7908333
NEK	-2.8341667
PKC	-3.1291667
STE20	-4.6366667

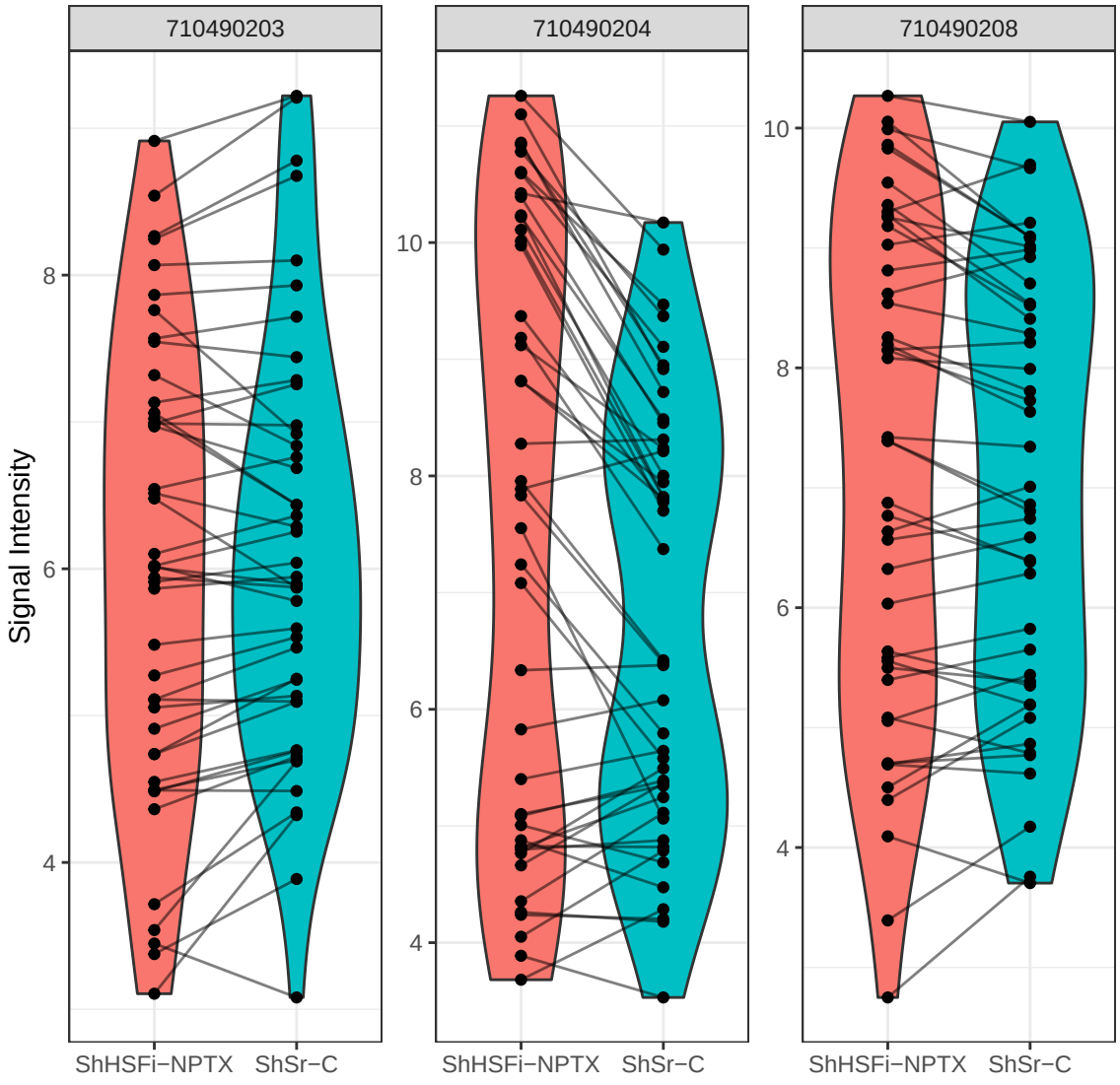


Figure 11.19: Violin plot of two groups

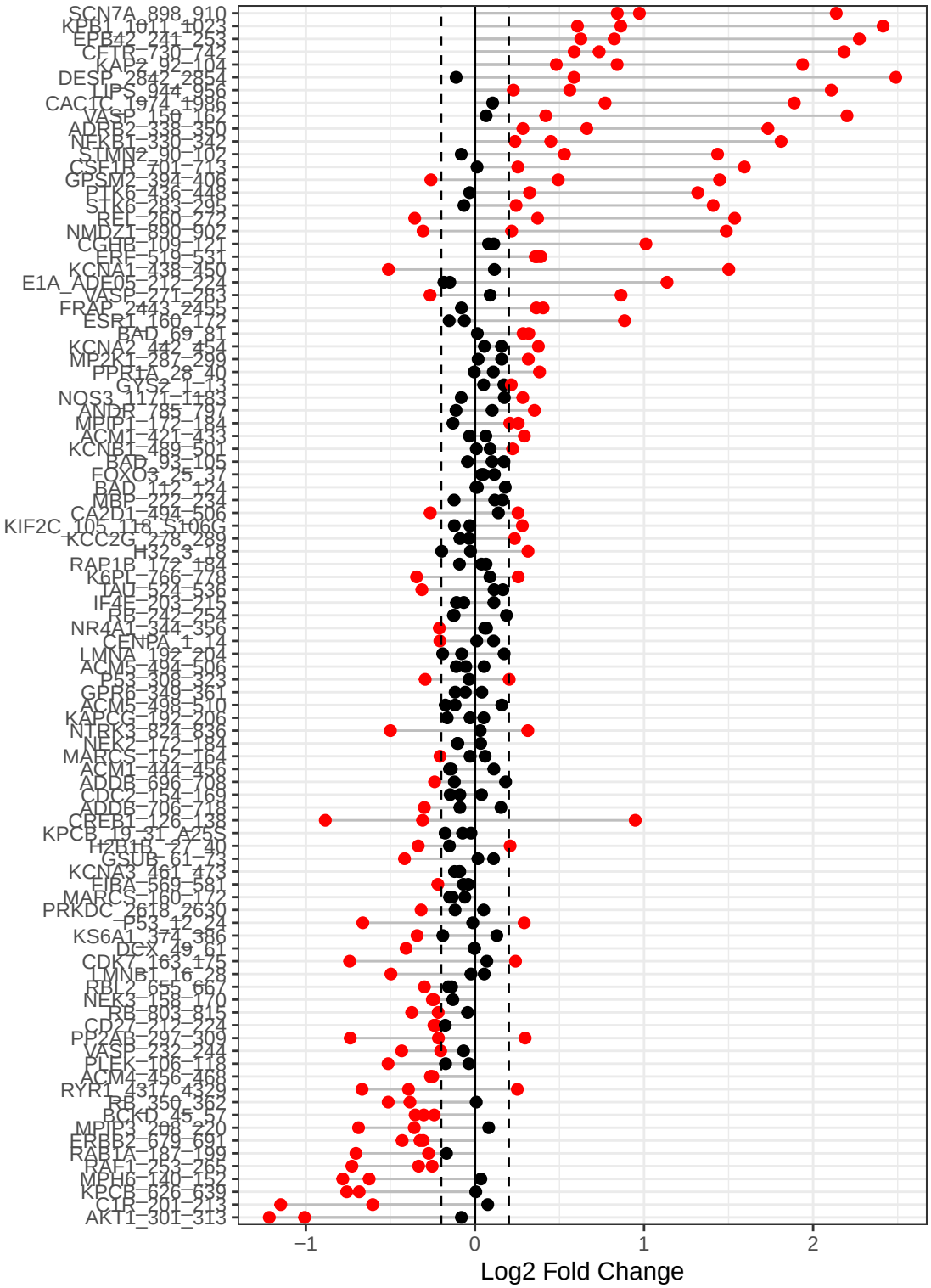


Figure 11.20: Waterfall Plot to show the distribution of change in peptide phosphorylation

Method	NumberOfPeptides
meanLFC.0.15	49
meanLFC.0.2	44
meanLFC.0.3	29
meanLFC.0.4	24
710490203.0.15	56
710490203.0.2	49
710490203.0.3	28
710490203.0.4	19
710490204.0.15	62
710490204.0.2	54
710490204.0.3	41
710490204.0.4	31
710490208.0.15	54
710490208.0.2	42
710490208.0.3	29
710490208.0.4	18

11.4.0.21 Z Scores Plot

We will plot the individual and averaged Z scores using both the across and within chip analyses.

11.4.0.22 Reverse KRSA Plot

We will use the reverse KRSA plot function, to plot the log2 fold change values for all peptides mapped to kinase hits. This will help us examine the activity of the kinase

11.4.0.23 Coverage Plot

To view the coverage of kinases across the full list of peptides on the chip, we will use the coverage plot function

11.4.0.24 Ball Model Network

We will view the ball model network function, to generate a model representing the protein-protein interactions between kinases

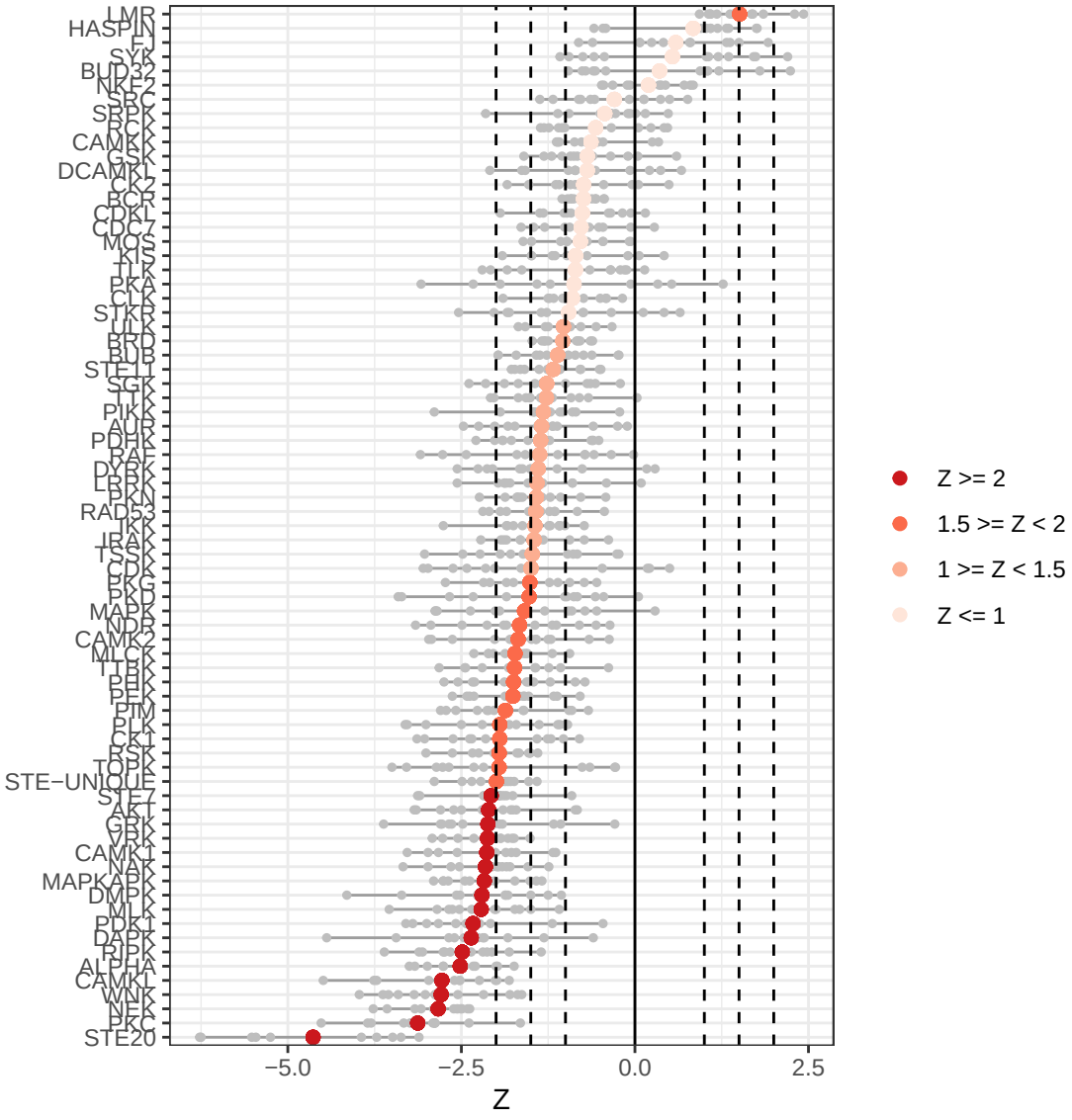


Figure 11.21: Waterfall plot of the Z Scores of each kinase family

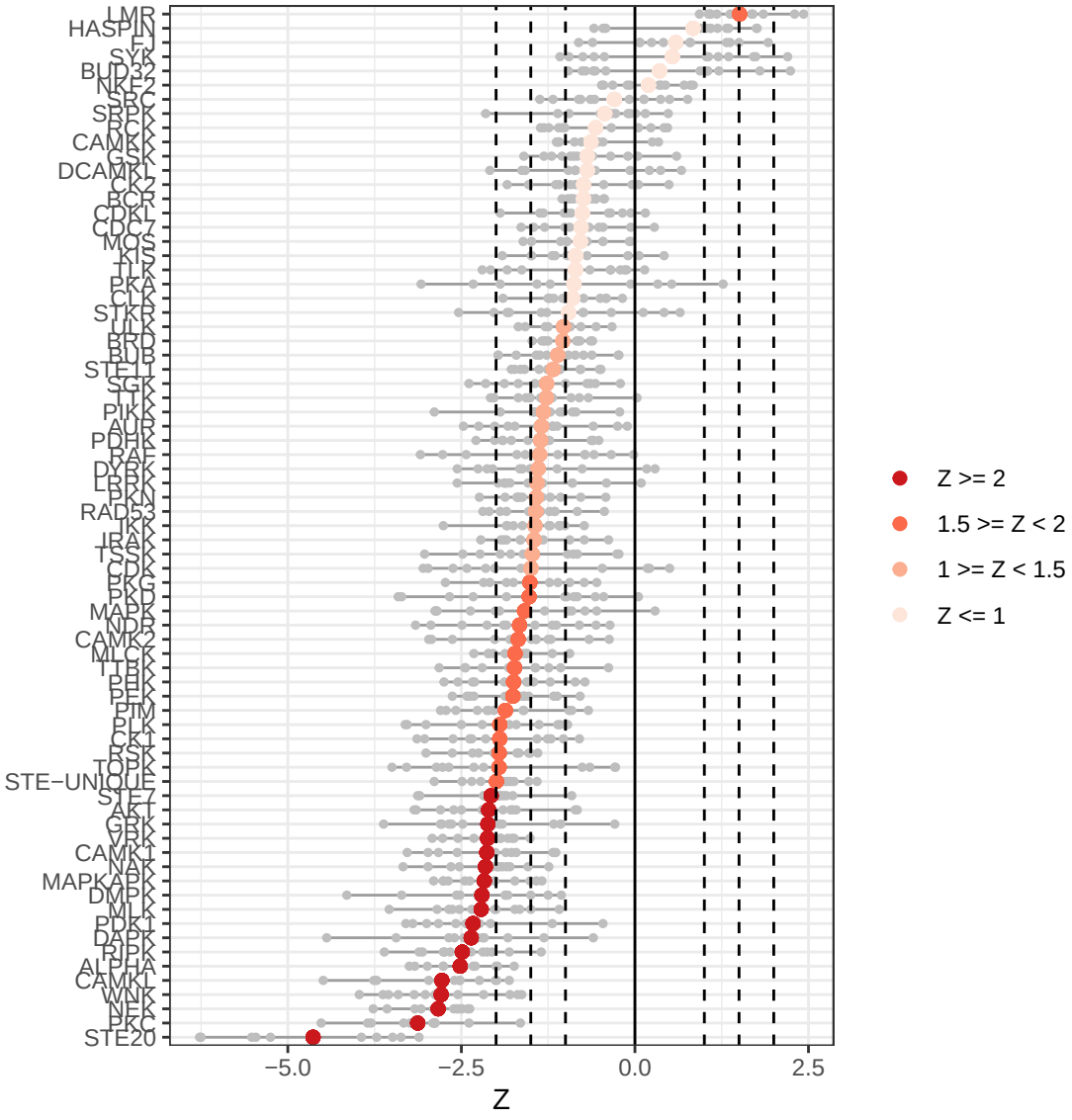


Figure 11.22: Waterfall plot of the Z Scores of each kinase family

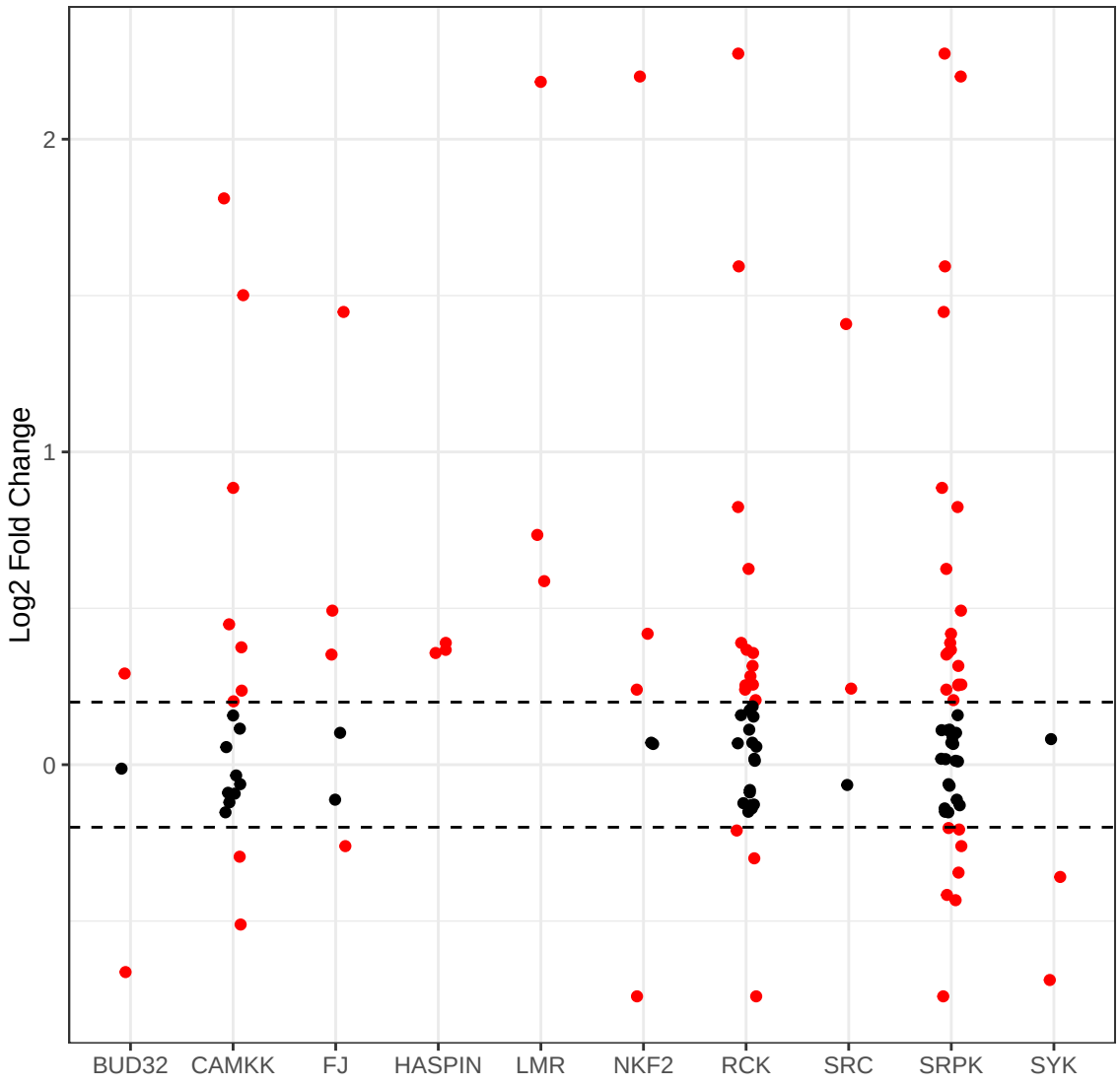


Figure 11.23: Kinase Activity summary for each kinase family based on peptide phosphorylation

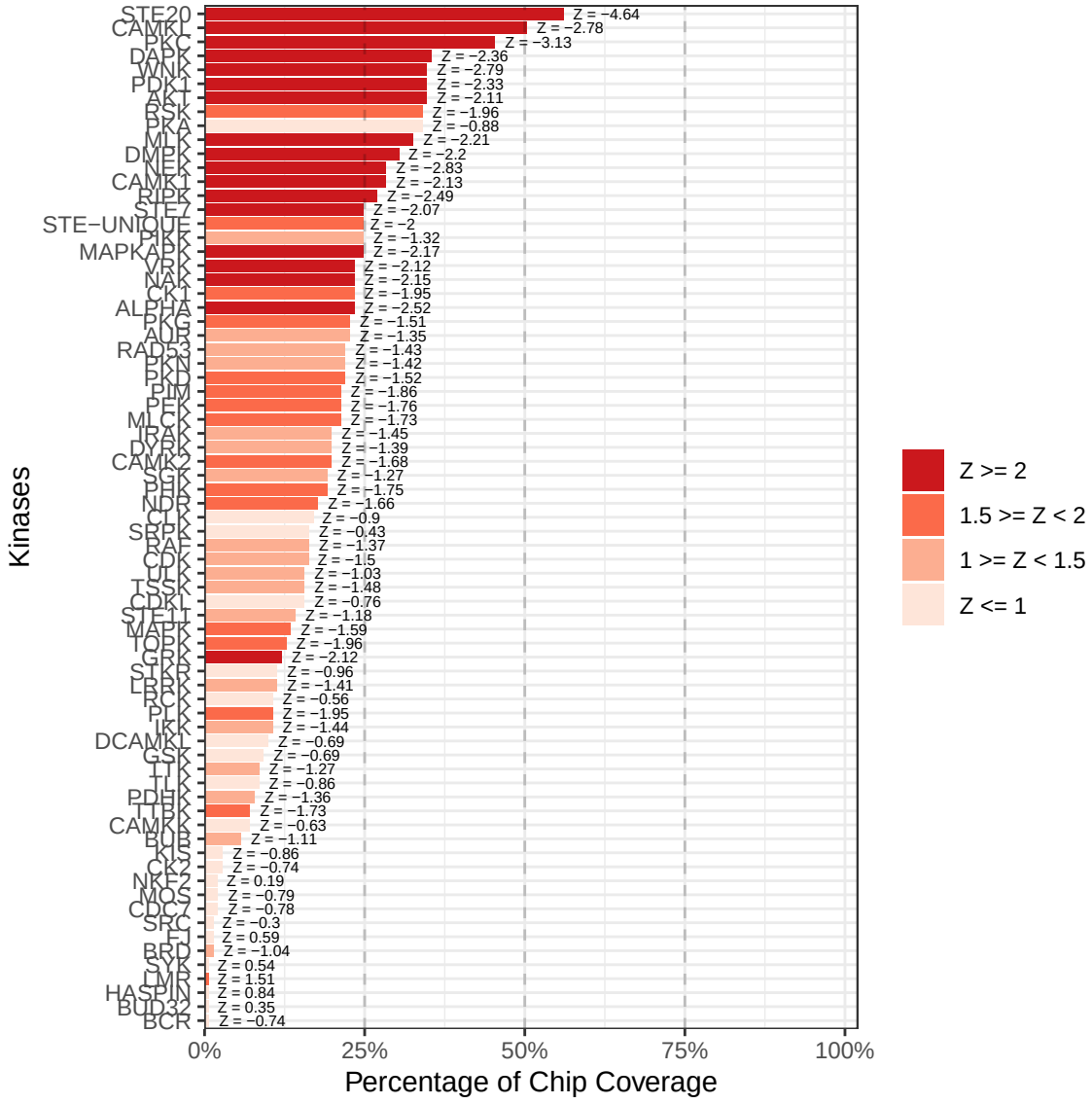


Figure 11.24: Percentage of peptides each kinase family phosphorylates

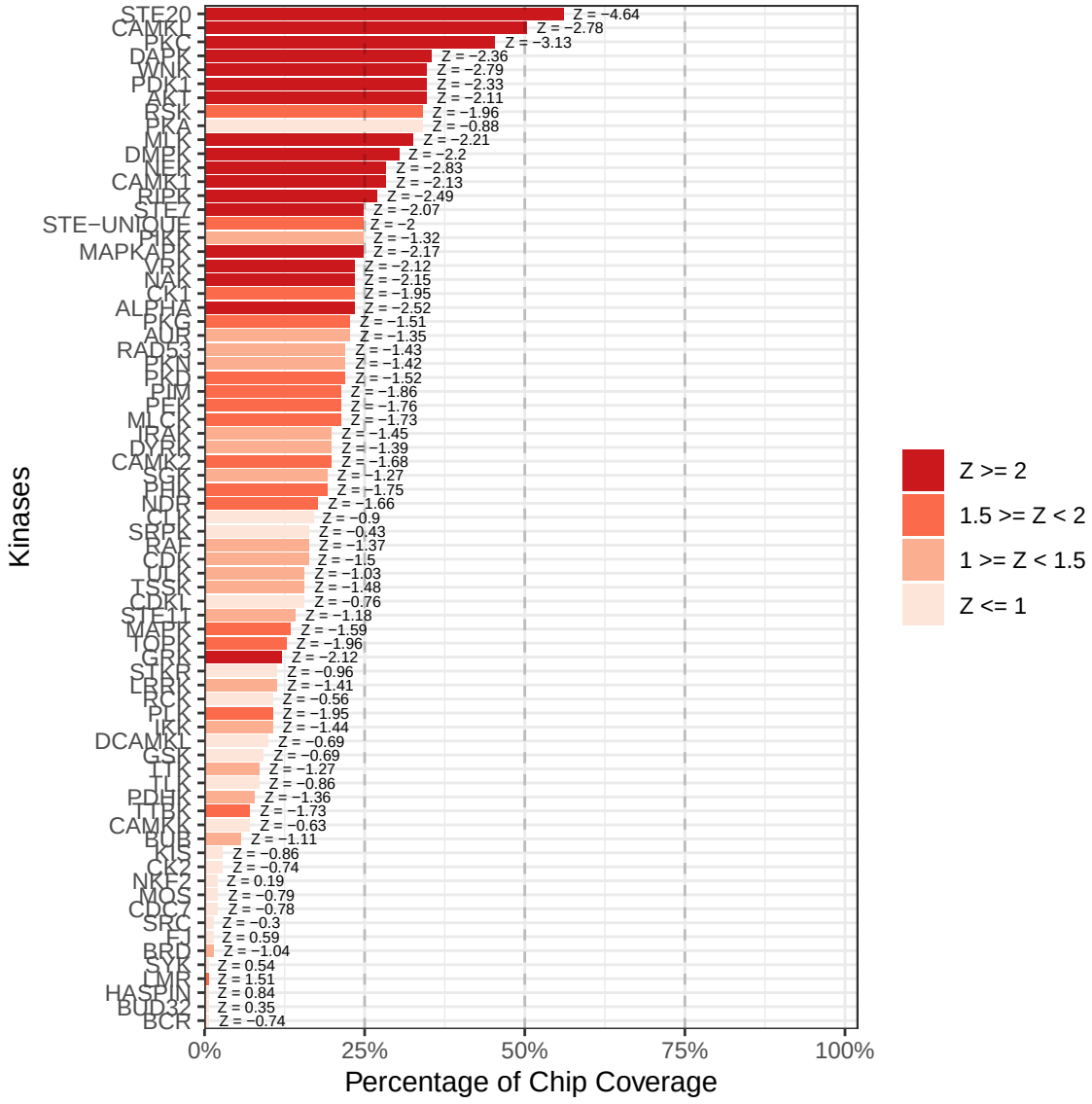
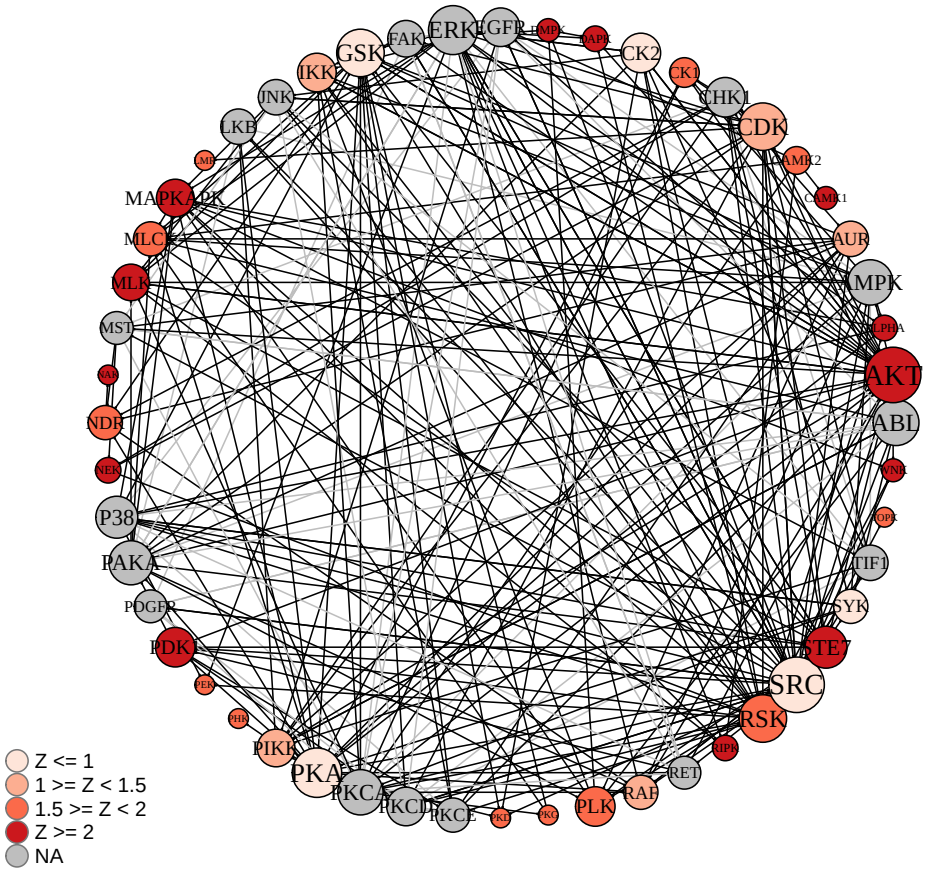
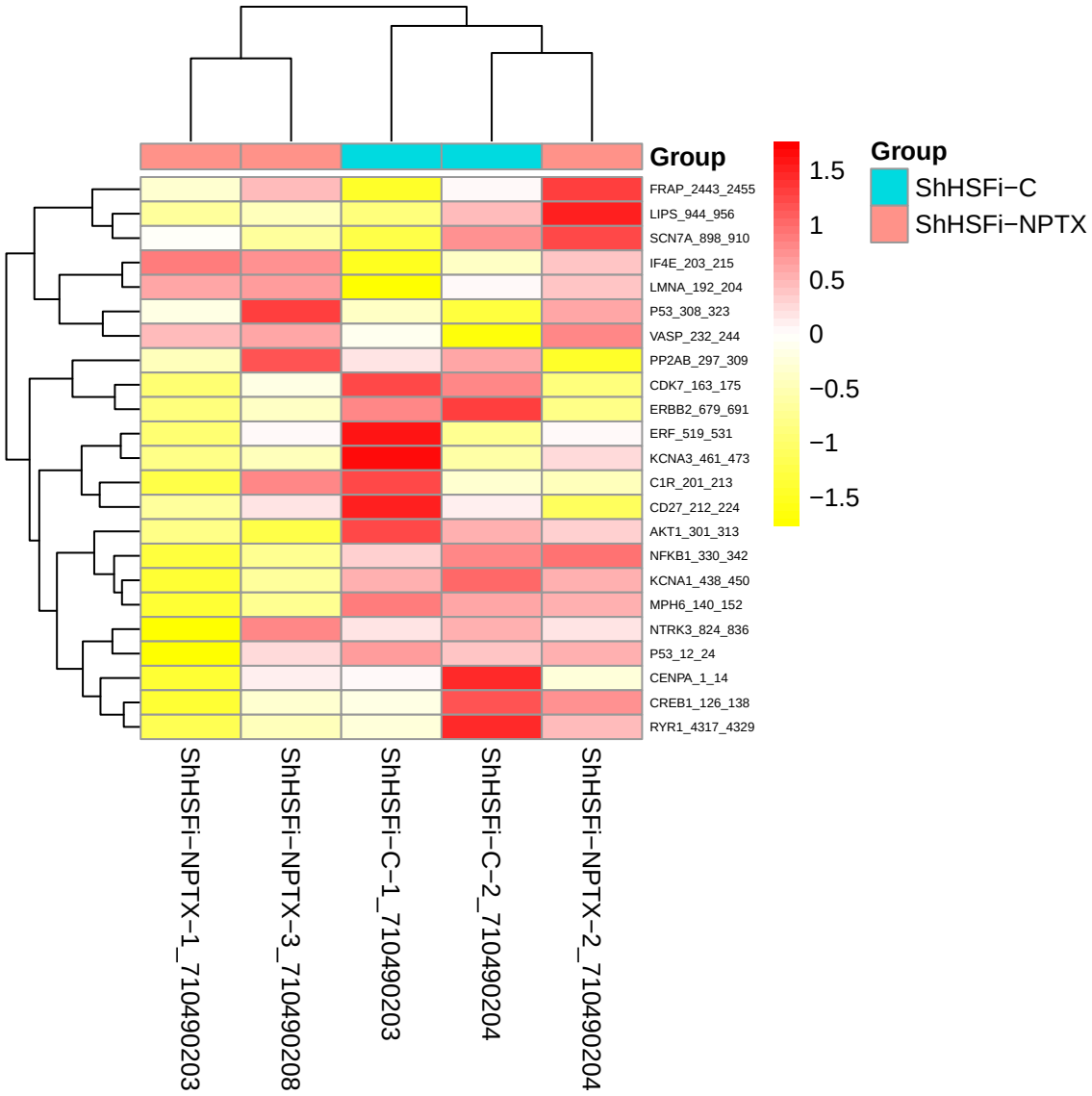


Figure 11.25: Percentage of peptides each kinase family phosphorylates


ERK
EGFR
HMPK
MAPK
CK2
GSK
FAK
IKK
CK1
CHK1
JNK
CDK
LKB
AMPK2
MP
CAMK1
MAPKAPK
AUR
MLCK
MLK
AMPK
MST
ALPHA
AKT
NDR
ABL
WNK
P38
GPK
PAKA
TIF1
PDGFR
SYK
PDK1
STE7
SRC
RSK
PHK
PIKK
MPB
PKA
RET
PKCA
RAF
PKCD
PKCE
PKD
PKC
PLK
Z <= 1
1 >= Z < 1.5
1.5 >= Z < 2
Z >= 2
NA



ShHSFi-NPTX vs ShHSFi-C

11.4.0.25 Heatmap

After applying the *Filtering Parameters* for this group comparison, only *23*/141 peptides carried forward in the analysis (i.e. *23 hits*). Below are some figures to visualize the differences between these samples for considering these *hits*.

11.4.0.26 Violin Plot

Below, the violin plot visualizes the distribution of selected peptides for the analysis.

11.4.0.27 Waterfall Plot

This waterfall represents the log2 fold changes between the two groups at each peptide.

11.4.0.28 Upstream Kinase Analysis

The lab carefully curated and mapped the kinases that can act and phosphorylate each peptide present on the chip. This was achieved by using multiple sources including GPS 3.0, Kinexus Phosphonet, PhosphoELM and PhosphoSite Plus. Based on that association between peptides and kinases, a random sampling analysis is performed for these hits. The basic idea of *KRSA* is: For each iteration (*2000* iterations performed in this analysis), the same number of hits are randomly selected from the total 141/or 193 peptides present on the chip. Predicted kinases are then mapped to this sample list of peptides and number of kinases are determined. The kinase count from the actual hits and random sampling is then compared to determine the significance.

Kinase	AvgZ
SYK	1.38750
HASPIN	1.27125
BUD32	0.69250
NKF2	0.61250
CAMKK	0.54625
KIS	0.41250
TTK	0.25375
CDK	0.02375
CK1	-1.94500
TSSK	-1.97750
MLCK	-2.00125
WNK	-2.02625
PKA	-2.13250
CAMK1	-2.14250
PIM	-2.16625
AKT	-2.17875
PKC	-2.18125
STE20	-2.81250

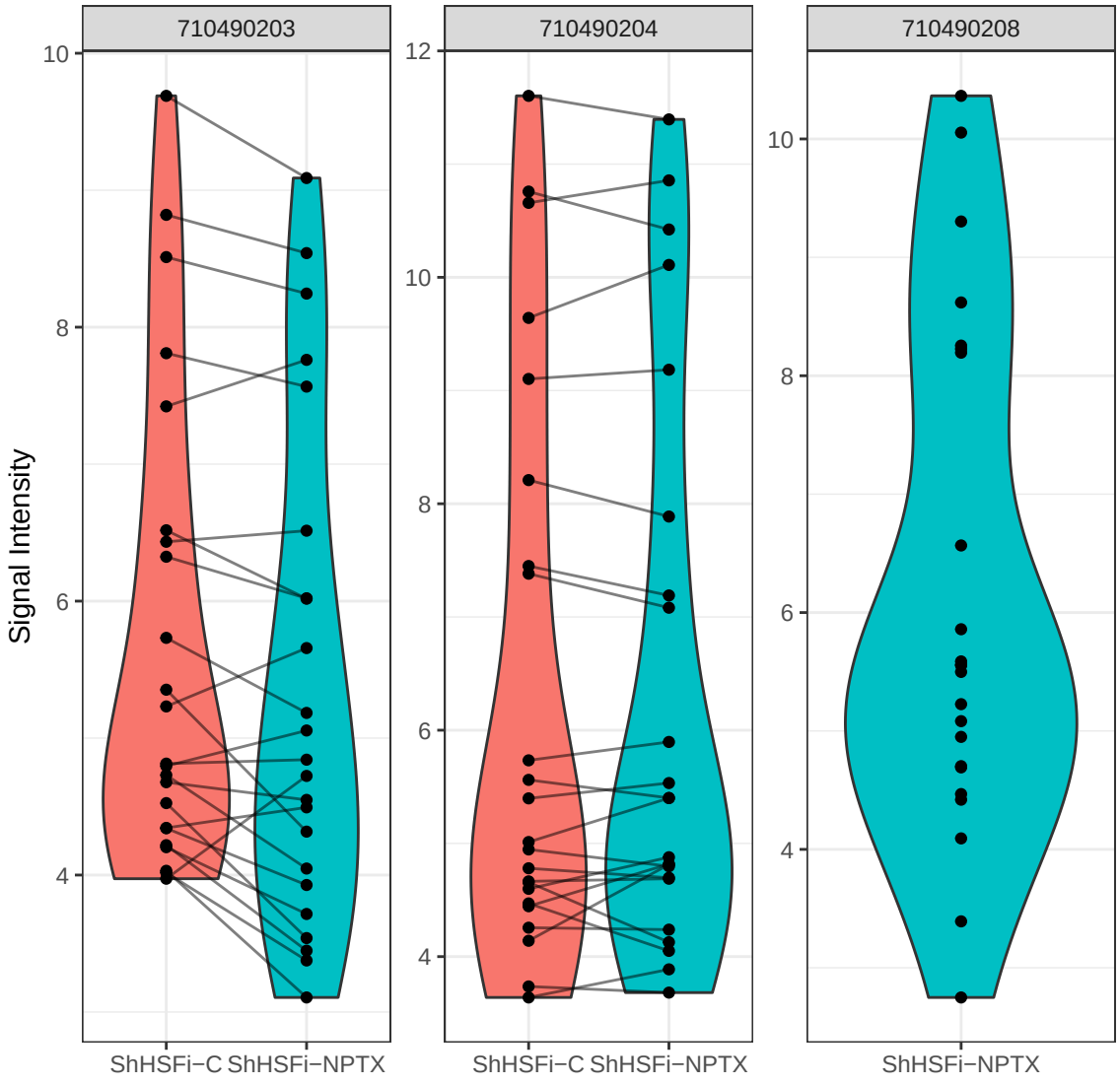


Figure 11.26: Violin plot of two groups

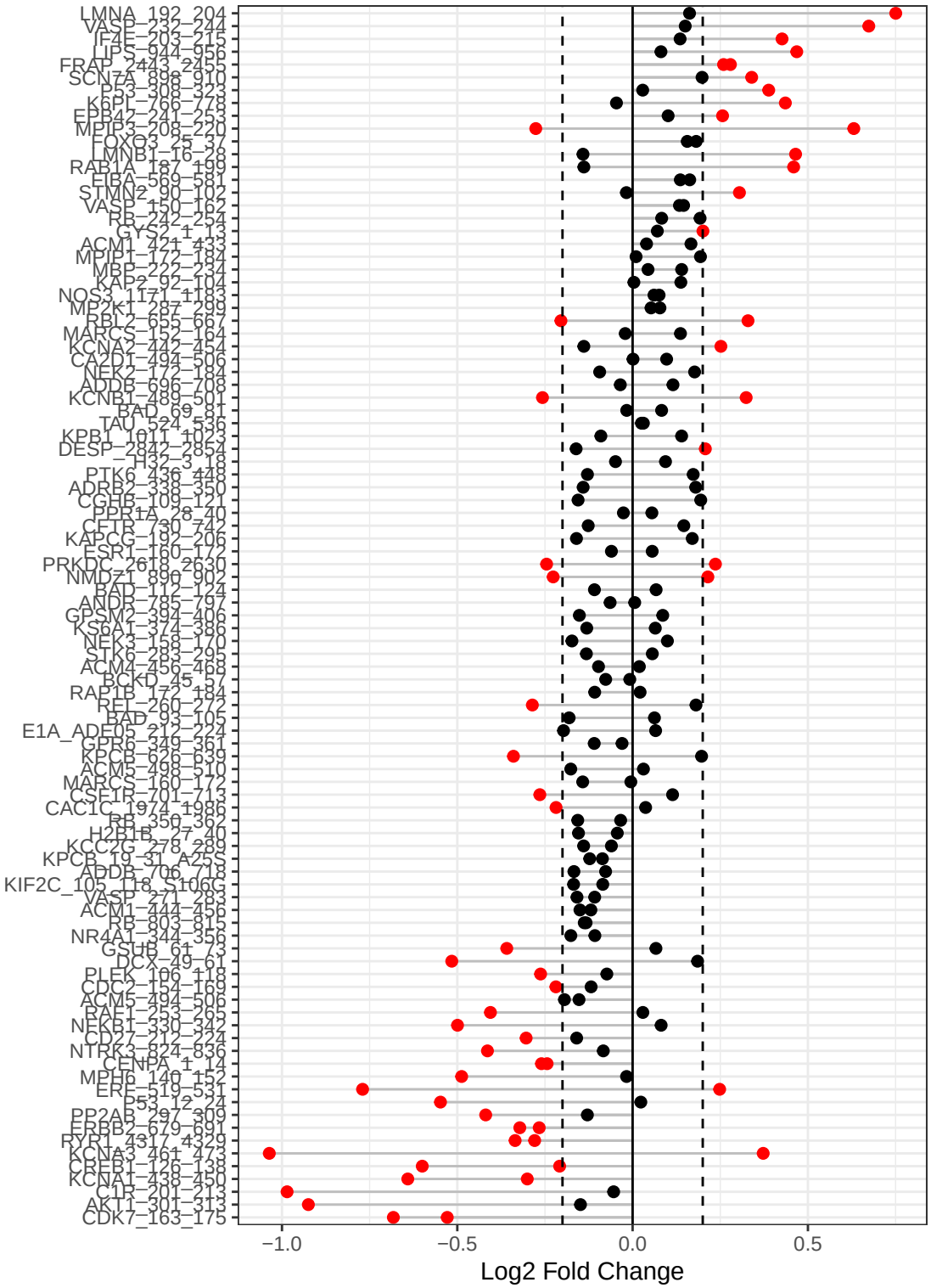


Figure 11.27: Waterfall Plot to show the distribution of change in peptide phosphorylation

Method	NumberOfPeptides
meanLFC.0.15	34
meanLFC.0.2	23
meanLFC.0.3	9
meanLFC.0.4	7
710490203.0.15	50
710490203.0.2	31
710490203.0.3	20
710490203.0.4	15
710490204.0.15	44
710490204.0.2	28
710490204.0.3	15
710490204.0.4	8

11.4.0.29 Z Scores Plot

We will plot the individual and averaged Z scores using both the across and within chip analyses.

11.4.0.30 Reverse KRSA Plot

We will use the reverse KRSA plot function, to plot the log2 fold change values for all peptides mapped to kinase hits. This will help us examine the activity of the kinase

11.4.0.31 Coverage Plot

To view the coverage of kinases across the full list of peptides on the chip, we will use the coverage plot function

11.4.0.32 Ball Model Network

We will view the ball model network function, to generate a model representing the protein-protein interactions between kinases

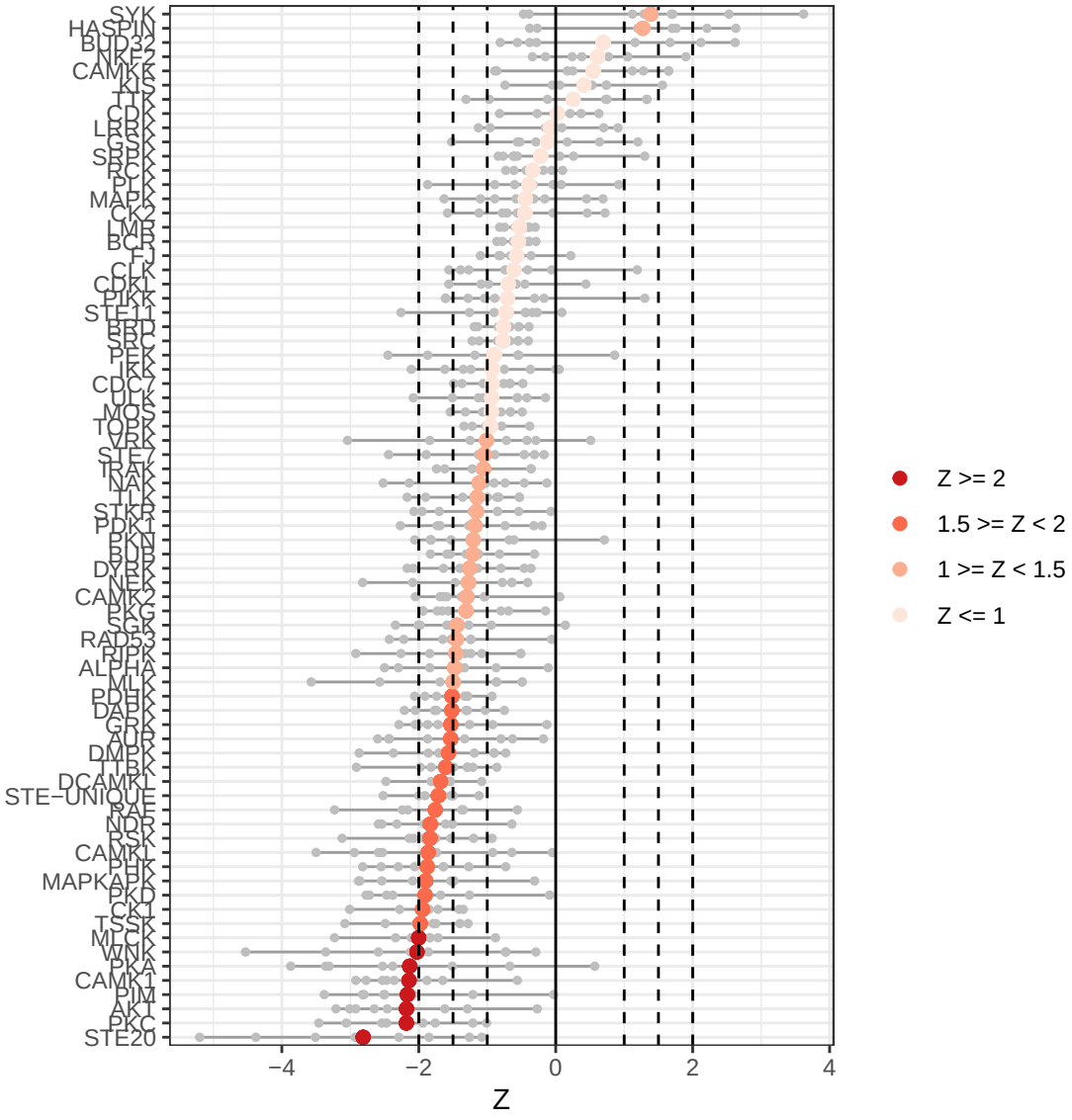


Figure 11.28: Waterfall plot of the Z Scores of each kinase family

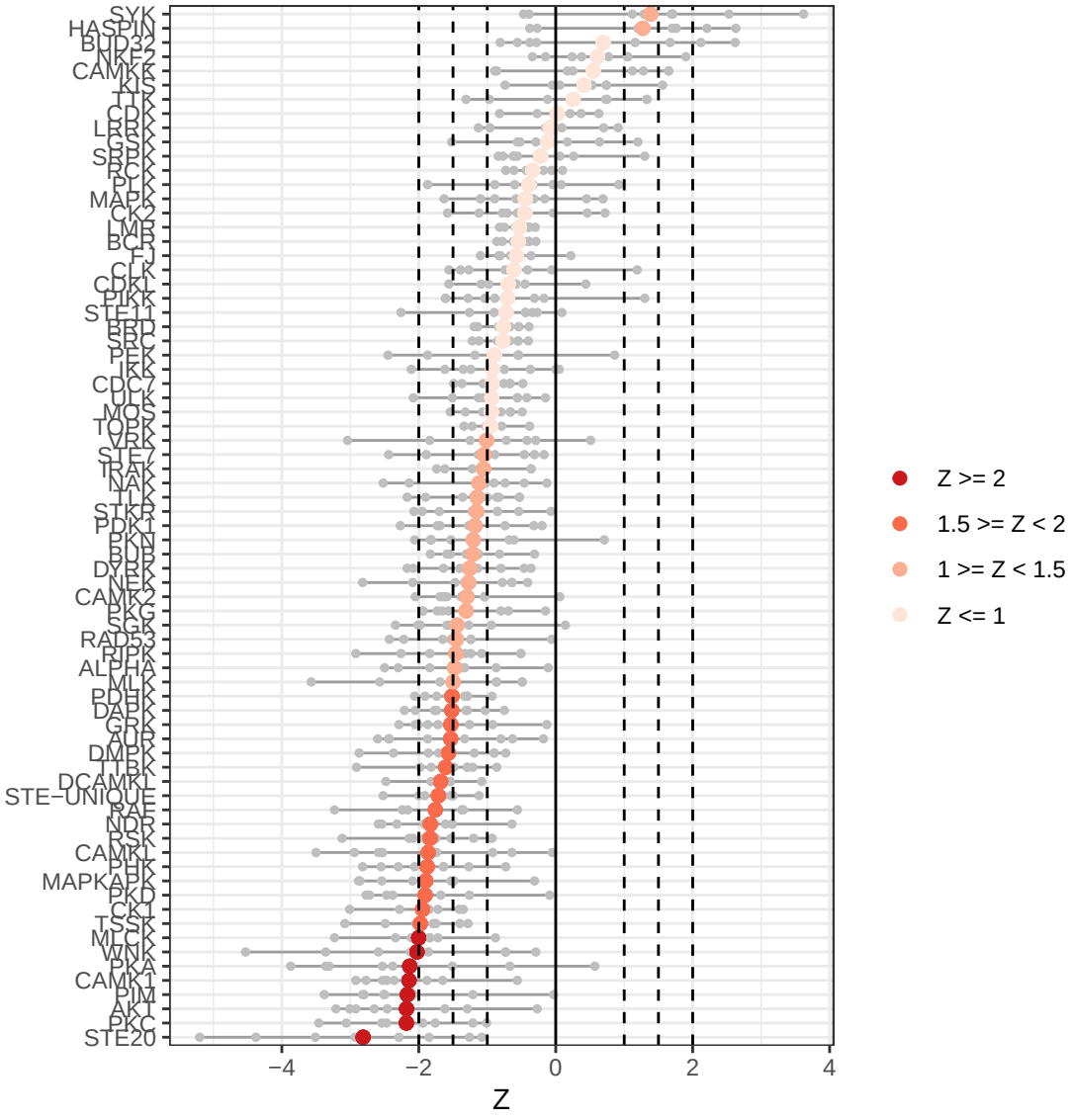


Figure 11.29: Waterfall plot of the Z Scores of each kinase family

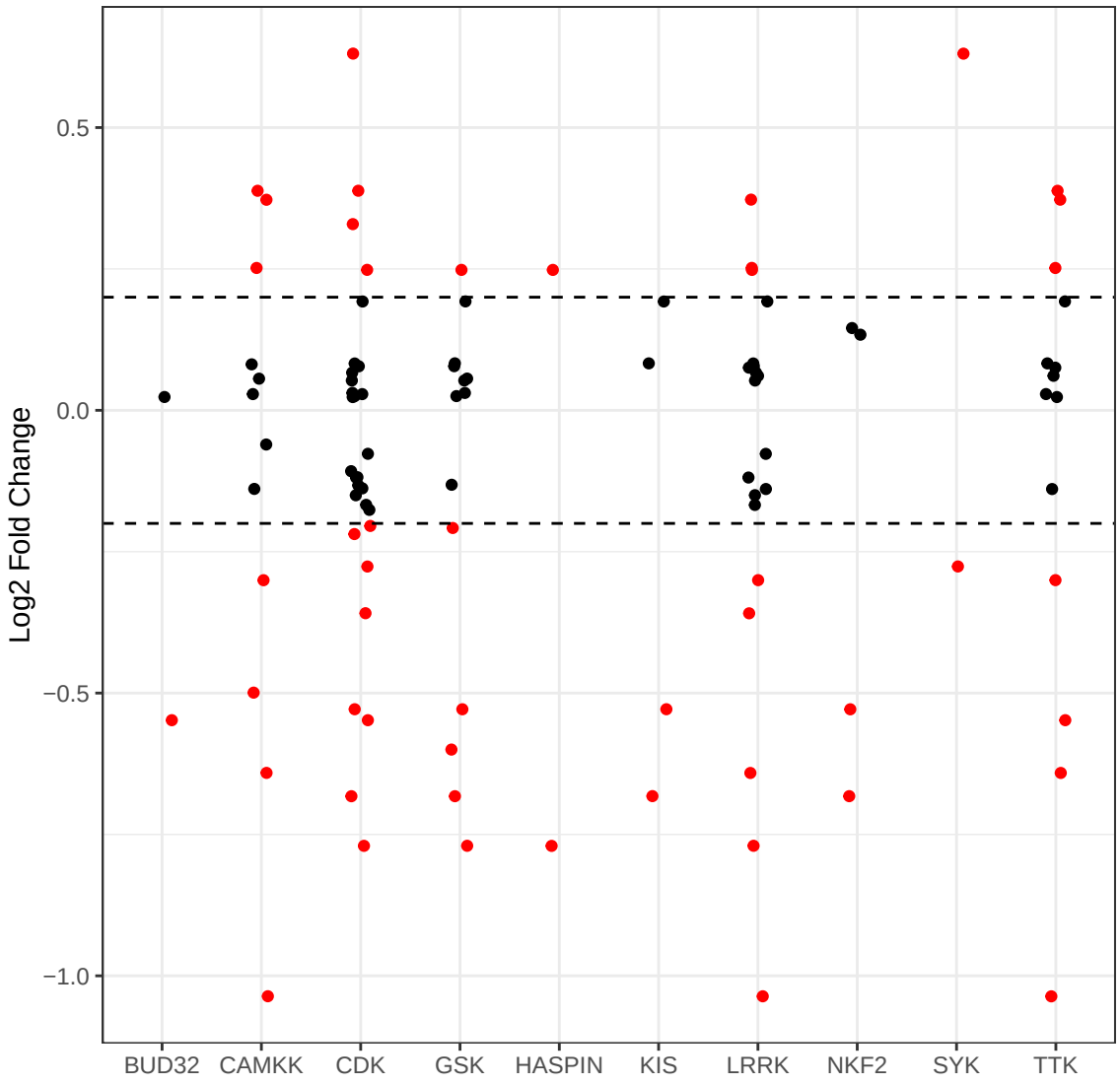


Figure 11.30: Kinase Activity summary for each kinase family based on peptide phosphorylation

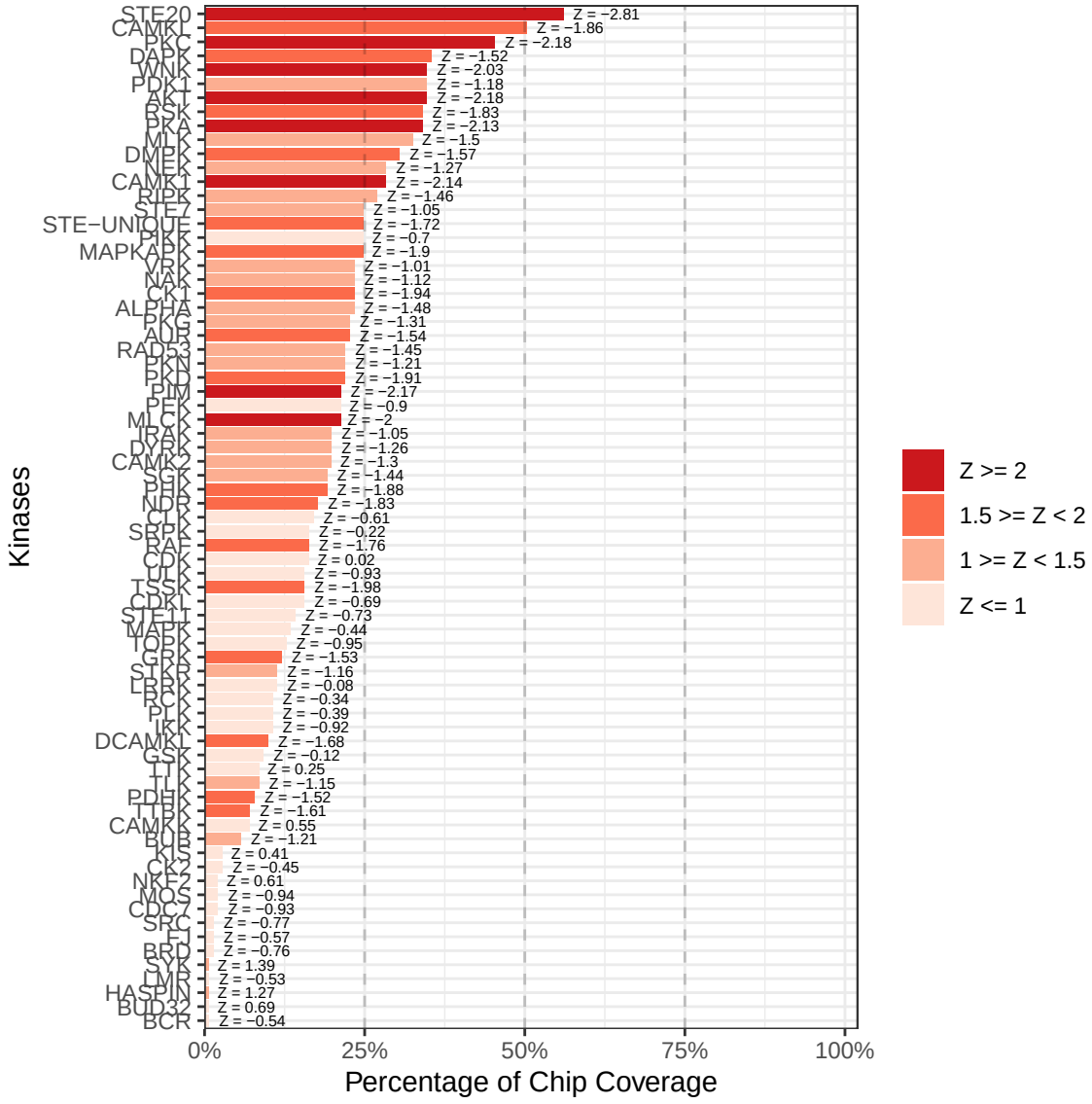


Figure 11.31: Percentage of peptides each kinase family phosphorylates

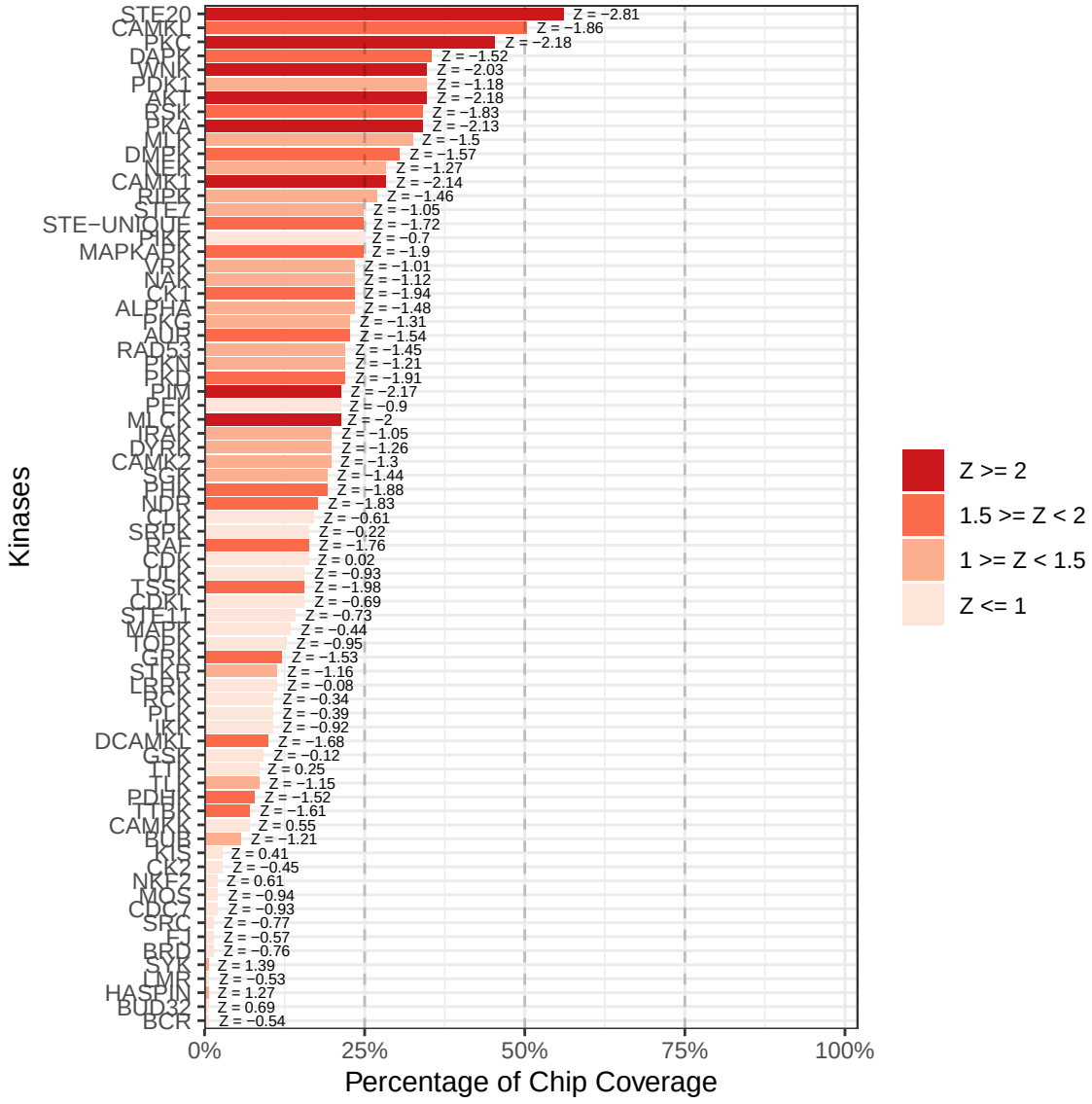
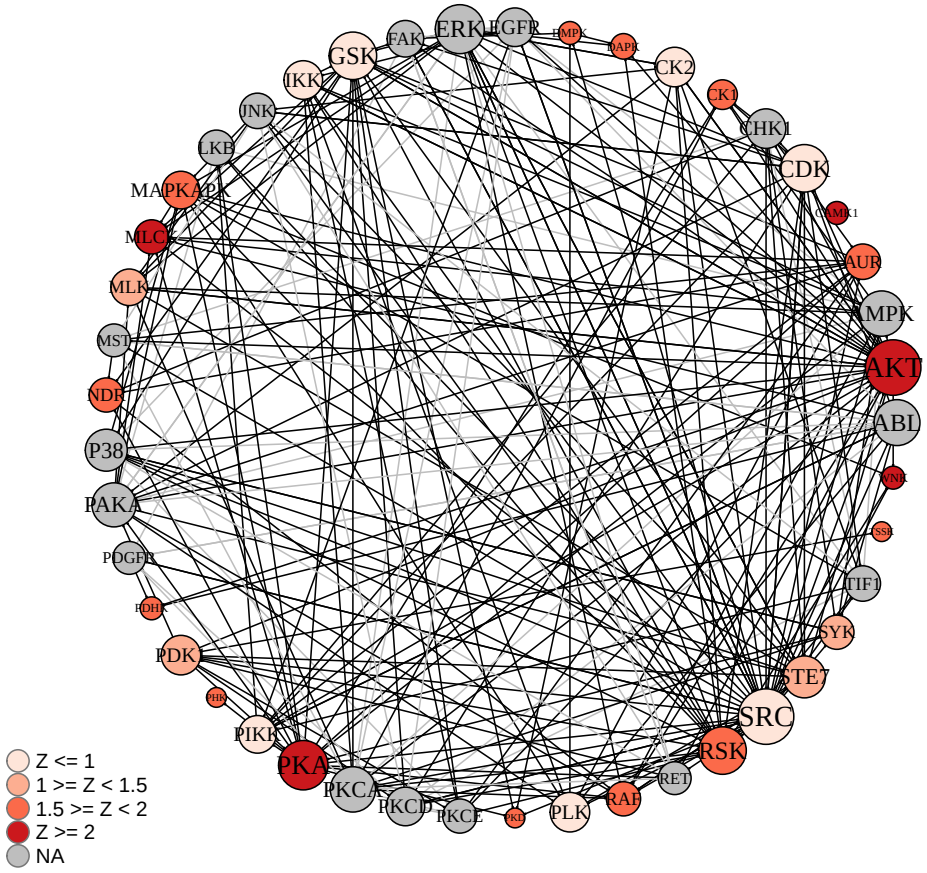
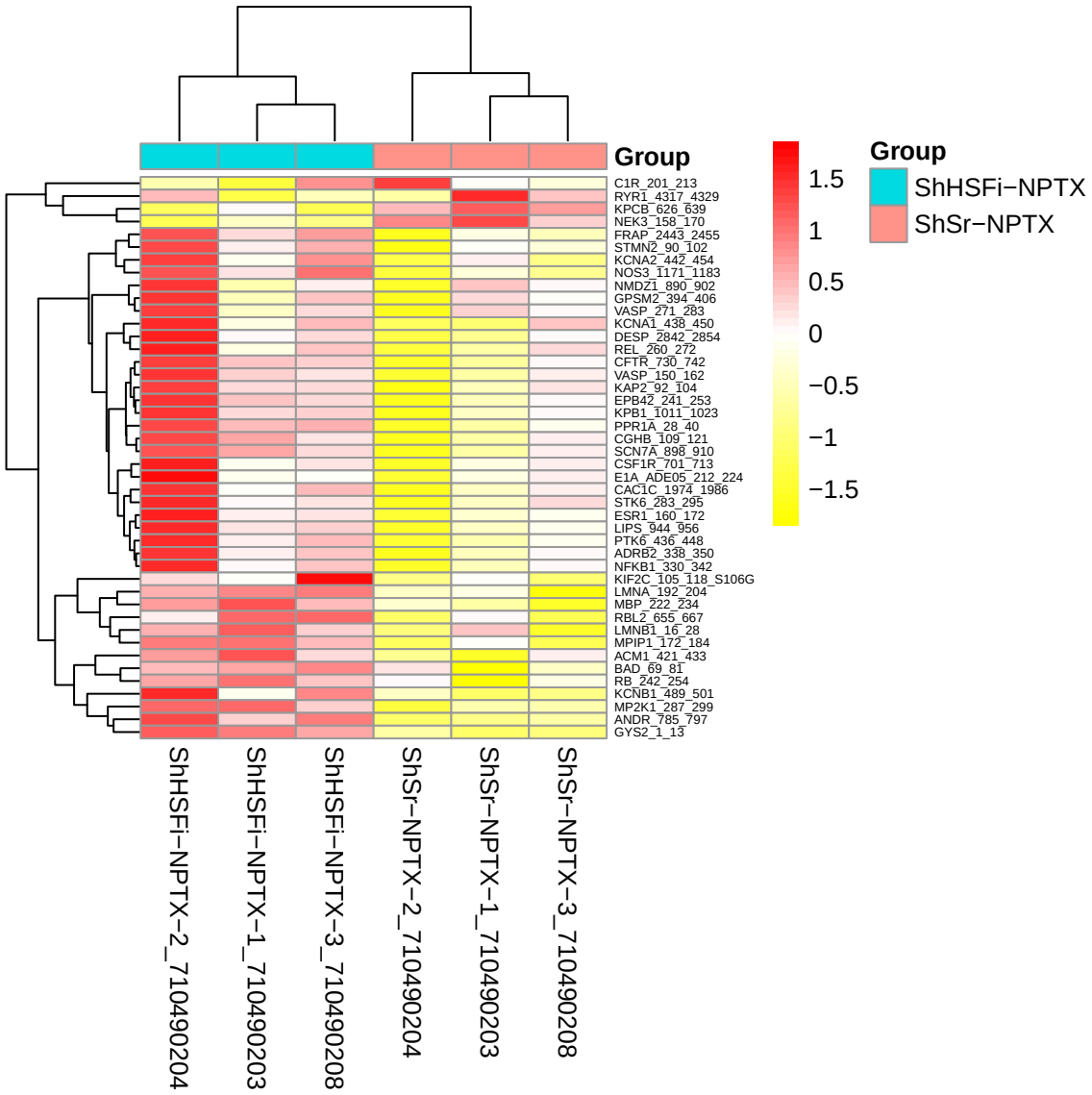


Figure 11.32: Percentage of peptides each kinase family phosphorylates


ERK
EGFR
BMPR
MAPK
FAK
GSK
CK2
IKK
CK1
JNK
CHK1
LKB
CDK
MAPKAPK
CAMK1
MLCK
AUR
MLK
AMPK
MST
AKT
NDR
ABL
P38
ZAK
PAKA
SSK
PDGFR
TIF1
PDHK
SYK
PDK
STE7
PHK
SRC
PIKK
RSK
PKA
RET
PKCA
PKCD
RAF
PKCE
PKG
PLK
Z <= 1
1 >= Z < 1.5
1.5 >= Z < 2
Z >= 2
NA



ShHSFi-NPTX vs ShSr-NPTX

11.4.0.33 Heatmap

After applying the *Filtering Parameters* for this group comparison, only *44*/141 peptides carried forward in the analysis (i.e. *44 hits*). Below are some figures to visualize the differences between these samples for considering these *hits*.

11.4.0.34 Violin Plot

Below, the violin plot visualizes the distribution of selected peptides for the analysis.

11.4.0.35 Waterfall Plot

This waterfall represents the log2 fold changes between the two groups at each peptide.

11.4.0.36 Upstream Kinase Analysis

The lab carefully curated and mapped the kinases that can act and phosphorylate each peptide present on the chip. This was achieved by using multiple sources including GPS 3.0, Kinexus Phosphonet, PhosphoELM and PhosphoSite Plus. Based on that association between peptides and kinases, a random sampling analysis is performed for these hits. The basic idea of *KRSA* is: For each iteration (*2000* iterations performed in this analysis), the same number of hits are randomly selected from the total 141/or 193 peptides present on the chip. Predicted kinases are then mapped to this sample list of peptides and number of kinases are determined. The kinase count from the actual hits and random sampling is then compared to determine the significance.

Kinase	AvgZ
FJ	1.2133333
LMR	1.0883333
NKF2	0.4233333
BUD32	0.0308333
KIS	0.0141667
CK1	-2.2116667
RSK	-2.2425000
CAMK1	-2.2683333
AKT	-2.2766667
DMPK	-2.2900000
DAPK	-2.3250000
WNK	-2.6383333
CAMKL	-2.9783333
PKC	-3.3791667
STE20	-3.9416667

Method	NumberOfPeptides
meanLFC.0.15	56

Method	NumberOfPeptides
meanLFC.0.2	44
meanLFC.0.3	32
meanLFC.0.4	24
710490203.0.15	65
710490203.0.2	49
710490203.0.3	29
710490203.0.4	17
710490204.0.15	63
710490204.0.2	54
710490204.0.3	41
710490204.0.4	33
710490208.0.15	50
710490208.0.2	38
710490208.0.3	22
710490208.0.4	12

11.4.0.37 Z Scores Plot

We will plot the individual and averaged Z scores using both the across and within chip analyses.

11.4.0.38 Reverse KRSA Plot

We will use the reverse KRSA plot function, to plot the log2 fold change values for all peptides mapped to kinase hits. This will help us examine the activity of the kinase

11.4.0.39 Coverage Plot

To view the coverage of kinases across the full list of peptides on the chip, we will use the coverage plot function

11.4.0.40 Ball Model Network

We will view the ball model network function, to generate a model representing the protein-protein interactions between kinases

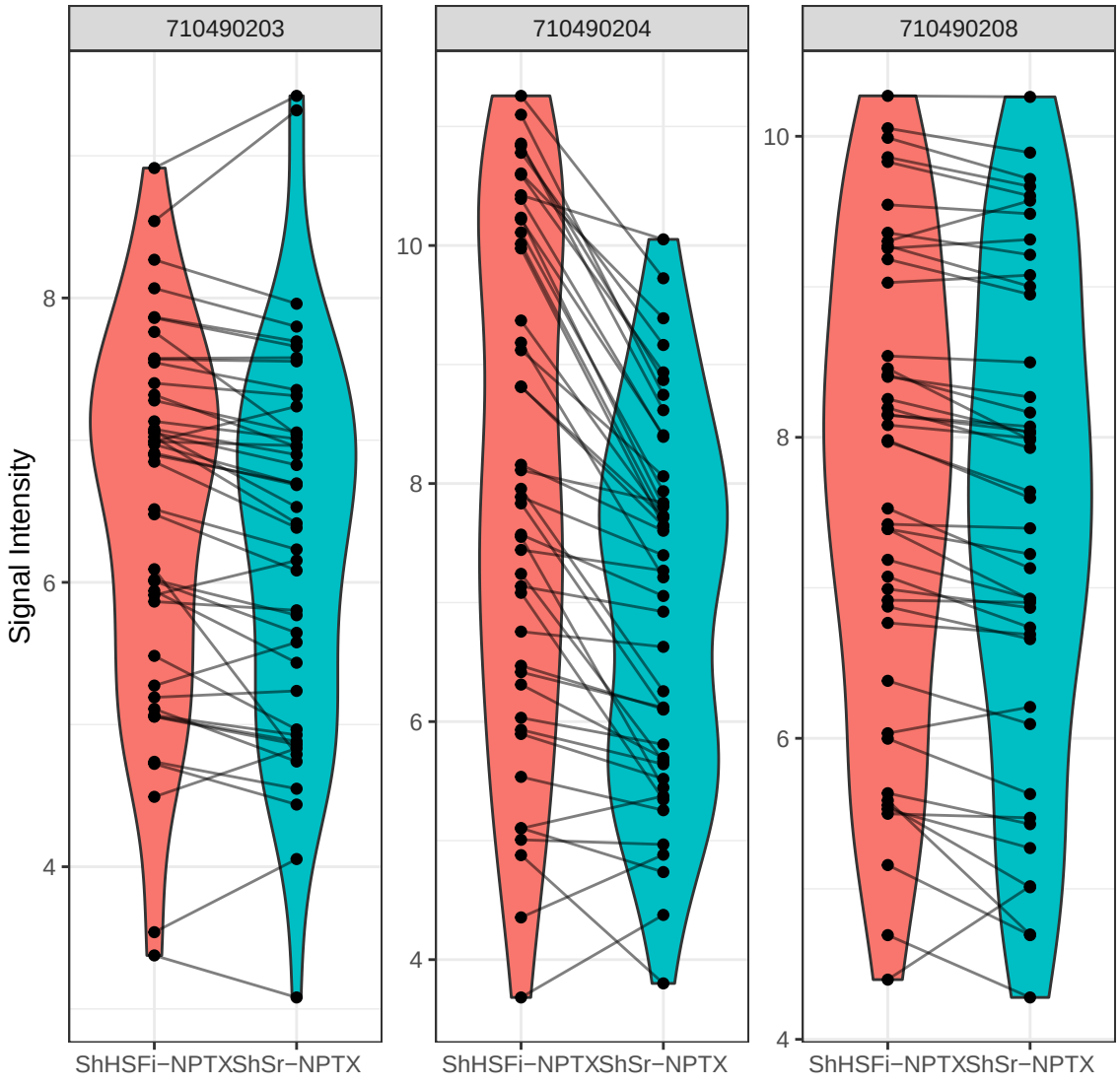


Figure 11.33: Violin plot of two groups

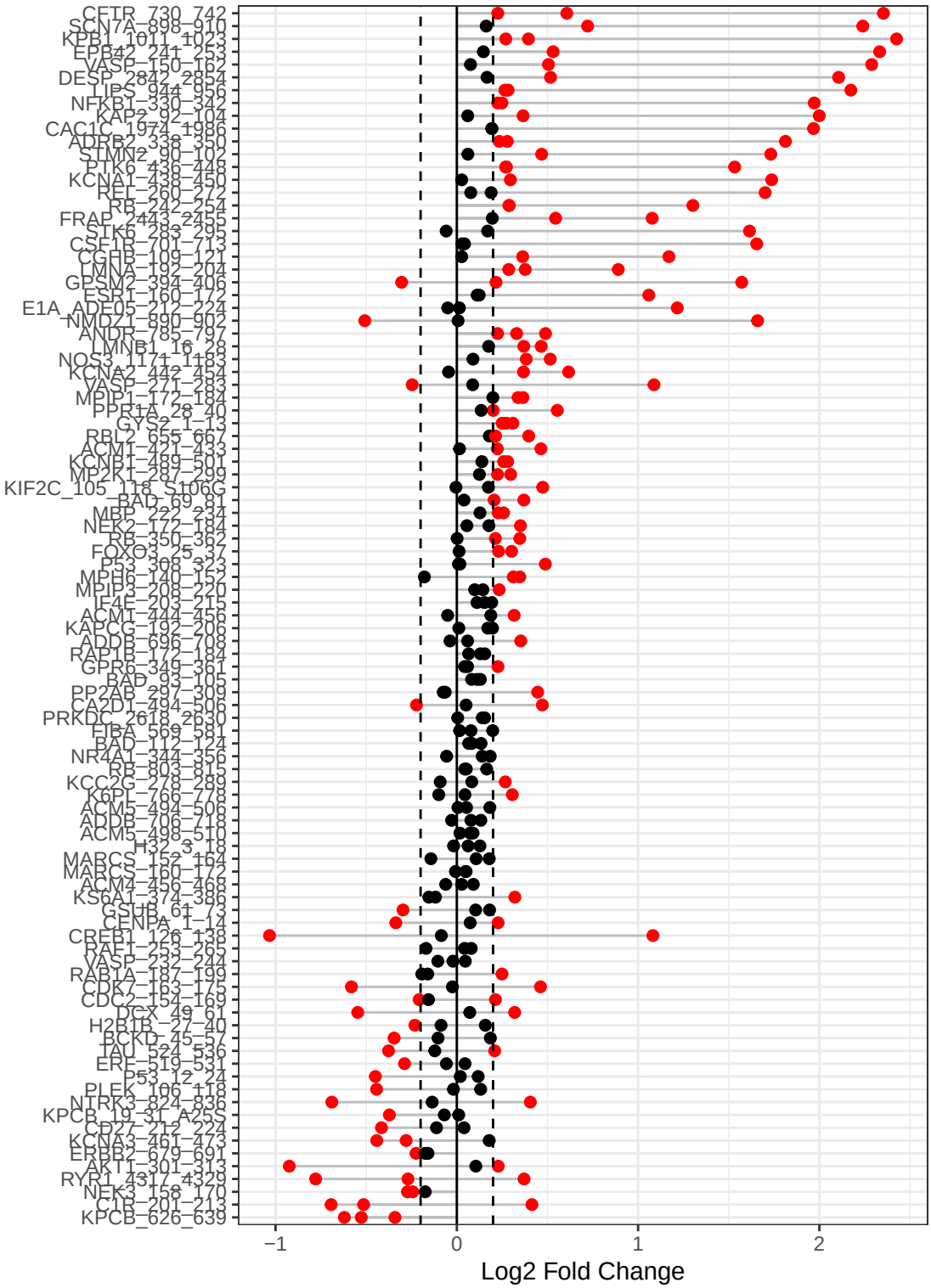


Figure 11.34: Waterfall Plot to show the distribution of change in peptide phosphorylation

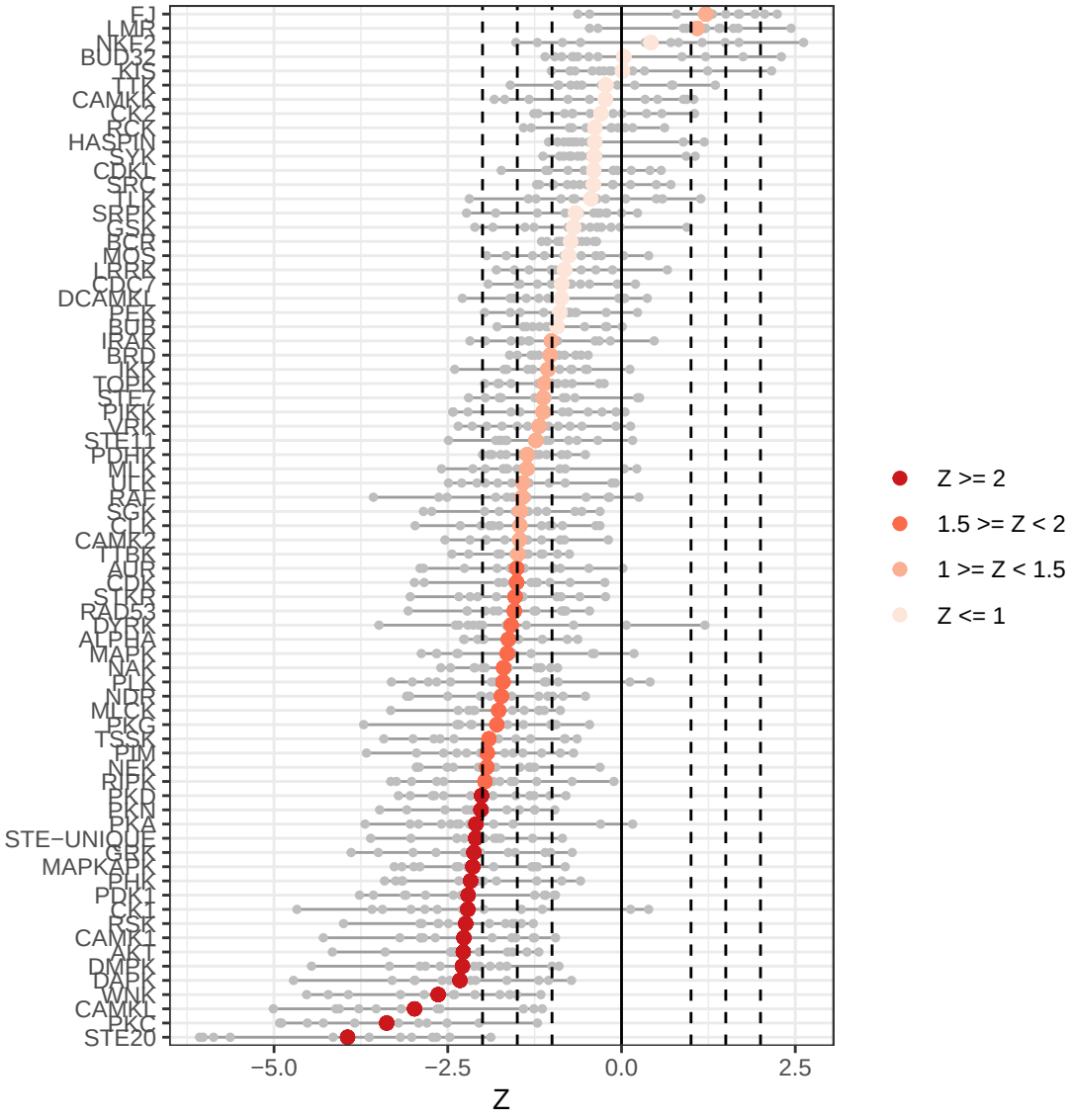


Figure 11.35: Waterfall plot of the Z Scores of each kinase family

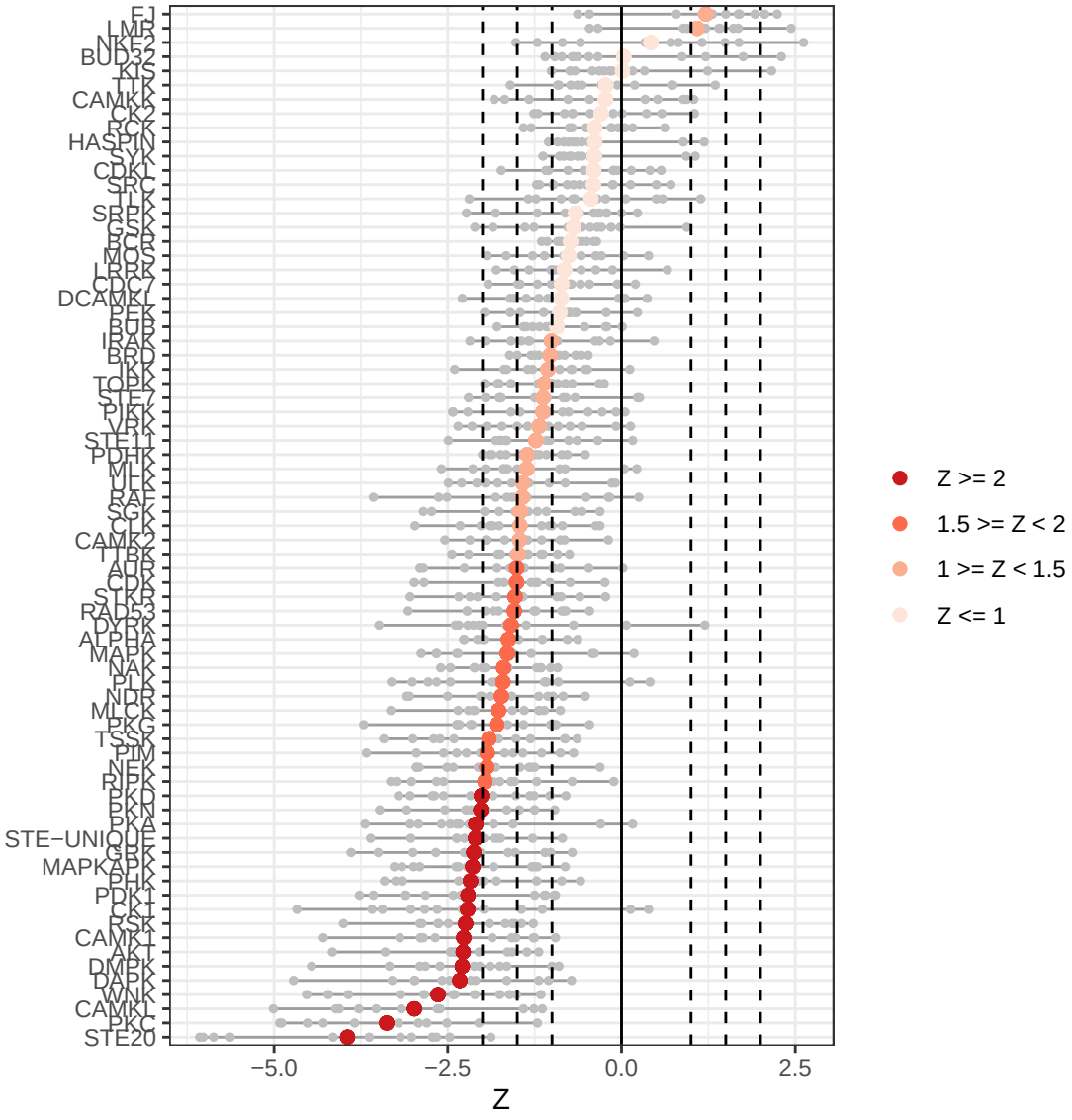


Figure 11.36: Waterfall plot of the Z Scores of each kinase family

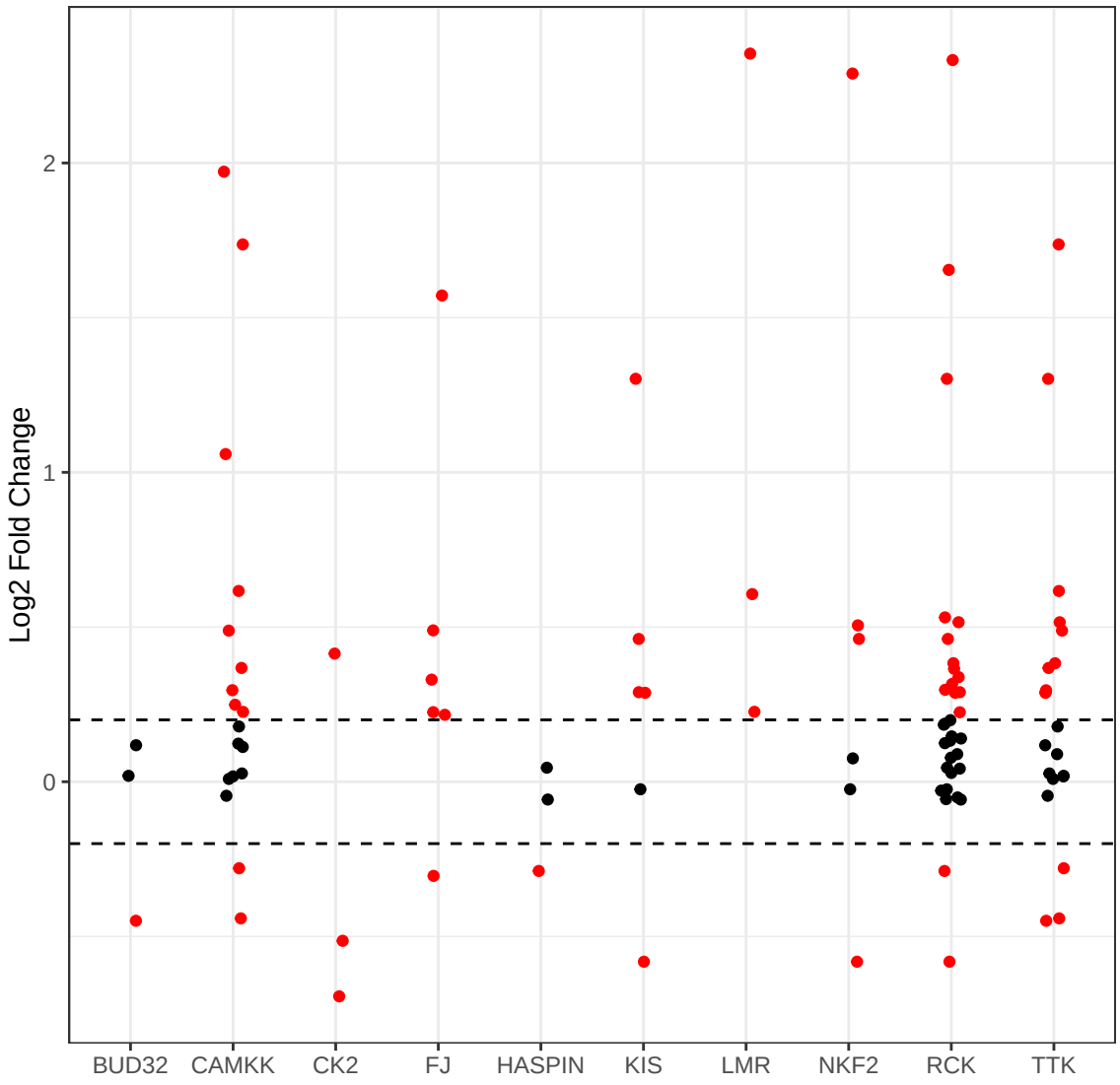


Figure 11.37: Kinase Activity summary for each kinase family based on peptide phosphorylation

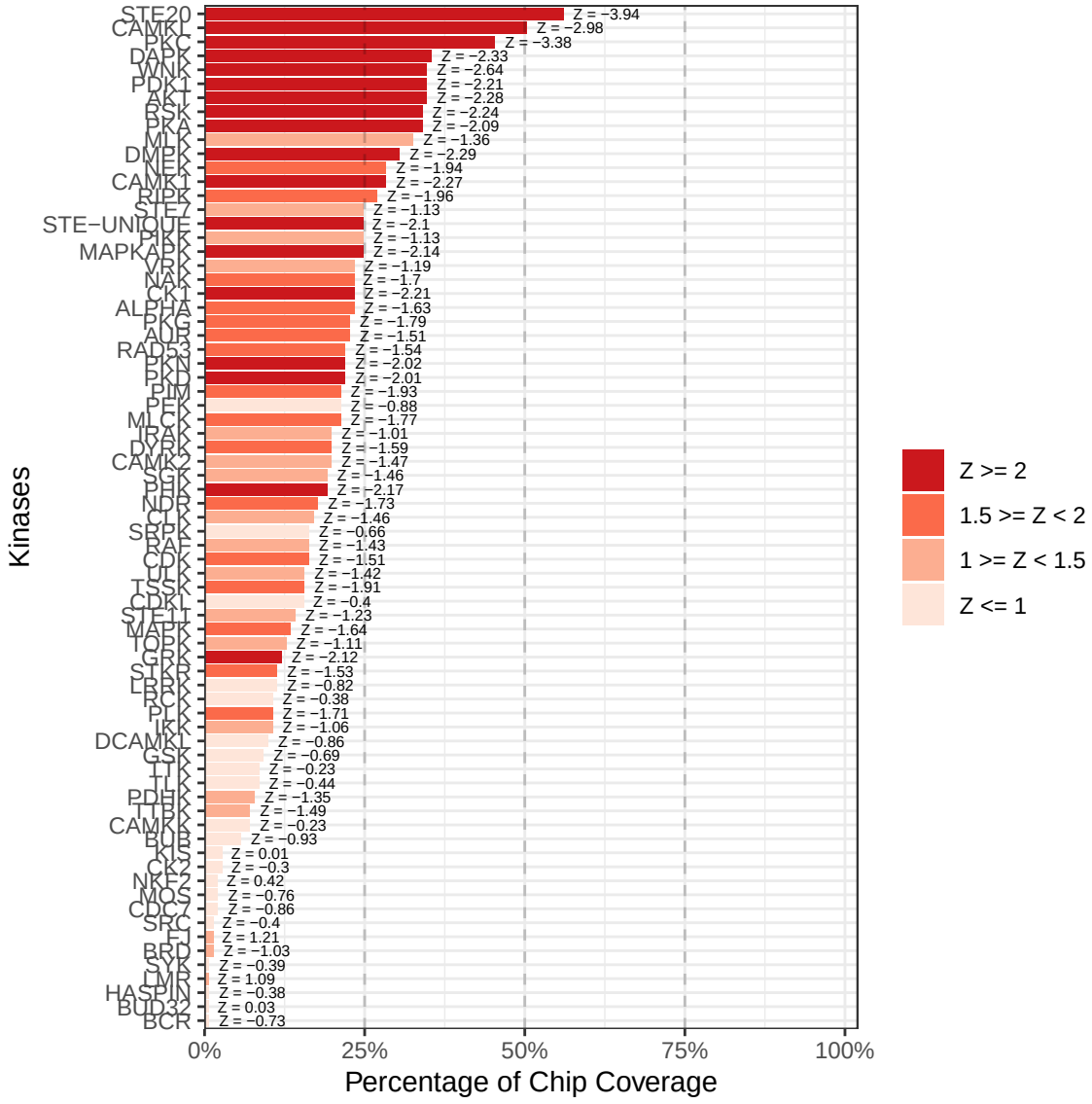


Figure 11.38: Percentage of peptides each kinase family phosphorylates

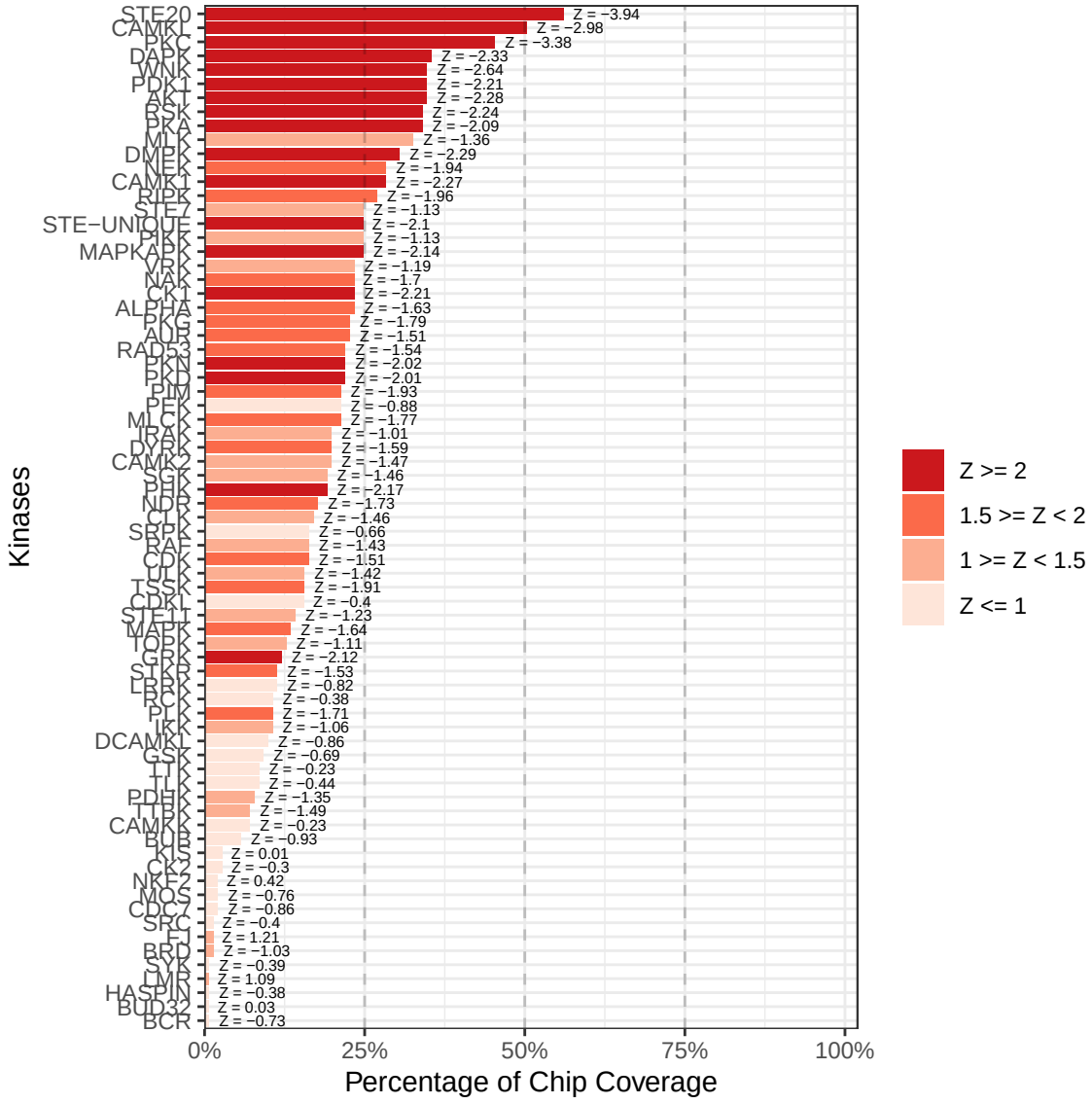
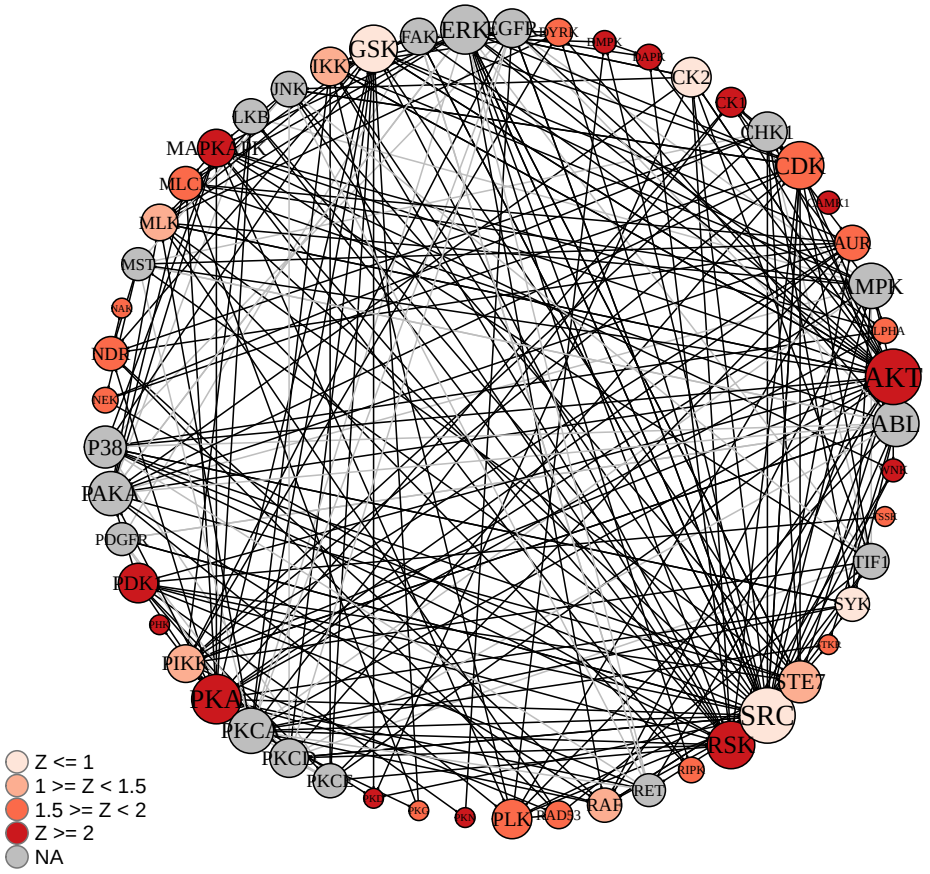


Figure 11.39: Percentage of peptides each kinase family phosphorylates


ERK
EGFR
FAK
GSK
IKK
JNK
LKB
MAPKAPK
MLCK
MLK
MST
NDR
NEK
P38
PAKA
PDGFR
PDK
PIKK
PKA
PKCA
PKCB
PKCE
PKG
PLK
HADES
RAF
RET
RSK
SRC
STE7
SYK
TIF1
VNK
CK2
CHK1
CDK
AUR
AMPK
ALPHA
AKT
ABL
Z <= 1
1 >= Z < 1.5
1.5 >= Z < 2
Z >= 2
NA

12 Session Info

```
#> - Session info ---------------------------------------------------------------
#>  setting  value
#>  version  R version 4.5.3 (2026-03-11)
#>  os       macOS Tahoe 26.2
#>  system   aarch64, darwin20
#>  ui       X11
#>  language (EN)
#>  collate  en_US.UTF-8
#>  ctype    en_US.UTF-8
#>  tz       America/New_York
#>  date     2026-10-24
#>  pandoc   3.6.4 @ /opt/homebrew/bin/ (via rmarkdown)
#>  quarto   1.6.43 @ /usr/local/bin/quarto
#>
#> - Packages -------------------------------------------------------------------
#>  ! package      * version  date (UTC) lib source
#>  P backports      1.5.1    2026-04-03 [?] CRAN (R 4.5.2)
#>  P BiocManager    1.30.27  2025-11-14 [?] CRAN (R 4.5.2)
#>  P bit            4.6.0    2025-03-06 [?] CRAN (R 4.5.0)
#>  P bit64          4.8.6    2026-09-01 [?] CRAN (R 4.5.2)
#>  P broom          1.0.13   2026-05-14 [?] CRAN (R 4.5.2)
#>  P cachem         1.1.0    2024-05-16 [?] CRAN (R 4.5.0)
#>  P cli            3.6.6    2026-04-09 [?] CRAN (R 4.5.2)
#>  P codetools      0.2-20   2024-03-31 [?] CRAN (R 4.5.3)
#>  P crayon         1.5.3    2024-06-20 [?] CRAN (R 4.5.0)
#>  P devtools       2.5.2    2026-04-30 [?] CRAN (R 4.5.2)
#>  P digest         0.6.39   2025-11-19 [?] CRAN (R 4.5.2)
#>  P dplyr        * 1.2.1    2026-04-03 [?] CRAN (R 4.5.2)
#>  P ellipsis       0.3.3    2026-04-04 [?] CRAN (R 4.5.2)
#>  P EnvStats       3.1.0    2025-04-24 [?] CRAN (R 4.5.0)
#>  P evaluate       1.0.5    2025-08-27 [?] CRAN (R 4.5.0)
#>  P farver         2.1.2    2024-05-13 [?] CRAN (R 4.5.0)
#>  P fastmap        1.2.0    2024-05-15 [?] CRAN (R 4.5.0)
#>  P forcats      * 1.0.1    2025-09-25 [?] CRAN (R 4.5.0)
#>  P fs             2.1.0    2026-04-18 [?] CRAN (R 4.5.2)
```

```
#>  P furrr        * 0.4.0    2026-03-31 [?] CRAN (R 4.5.2)
#>  P future       * 1.76.0   2026-09-24 [?] CRAN (R 4.5.2)
#>  P generics       0.1.4    2025-05-09 [?] CRAN (R 4.5.0)
#>  P ggplot2      * 4.0.3    2026-04-22 [?] CRAN (R 4.5.2)
#>  P globals        0.20.0   2026-10-11 [?] CRAN (R 4.5.2)
#>  P glue           1.8.1    2026-04-17 [?] CRAN (R 4.5.2)
#>  P gt           * 1.3.0    2026-01-22 [?] CRAN (R 4.5.2)
#>  P gtable         0.3.6    2024-10-25 [?] CRAN (R 4.5.0)
#>  P hms            1.1.4    2025-10-17 [?] CRAN (R 4.5.0)
#>  P htmltools      0.5.9    2025-12-04 [?] CRAN (R 4.5.2)
#>  P igraph         2.3.4    2026-09-30 [?] CRAN (R 4.5.2)
#>  P jsonlite       2.0.0    2025-03-27 [?] CRAN (R 4.5.0)
#>  P knitr        * 1.52     2026-09-06 [?] CRAN (R 4.5.2)
#>  P KRSA         * 1.99.354 2026-08-07 [?] repository (https://github.com/CogDisResLab/KRSA
#>  P labeling       0.4.3    2023-08-29 [?] CRAN (R 4.5.0)
#>  P lifecycle      1.0.5    2026-01-08 [?] CRAN (R 4.5.2)
#>  P listenv        1.1.0    2026-10-04 [?] CRAN (R 4.5.2)
#>  P lubridate    * 1.9.5    2026-02-04 [?] CRAN (R 4.5.2)
#>  P magrittr       2.0.5    2026-04-04 [?] CRAN (R 4.5.2)
#>  P memoise        2.0.1    2021-11-26 [?] CRAN (R 4.5.0)
#>  P otel           0.2.0    2025-08-29 [?] CRAN (R 4.5.0)
#>  P parallelly     1.48.0   2026-06-29 [?] CRAN (R 4.5.2)
#>  P pheatmap       1.0.13   2025-06-05 [?] CRAN (R 4.5.0)
#>  P pillar         1.11.1   2025-09-17 [?] CRAN (R 4.5.0)
#>  P pkgbuild       1.4.8    2025-05-26 [?] CRAN (R 4.5.0)
#>  P pkgconfig      2.0.3    2019-09-22 [?] CRAN (R 4.5.0)
#>  P pkgload        1.5.3    2026-06-15 [?] CRAN (R 4.5.2)
#>  P purrr        * 1.2.2    2026-04-10 [?] CRAN (R 4.5.2)
#>  P R6             2.6.1    2025-02-15 [?] CRAN (R 4.5.0)
#>  P RColorBrewer   1.1-3    2022-04-03 [?] CRAN (R 4.5.0)
#>  P readr        * 2.2.0    2026-02-19 [?] CRAN (R 4.5.2)
#>    renv           1.1.5    2025-07-24 [1] CRAN (R 4.5.3)
#>  P rlang          1.3.0    2026-07-05 [?] CRAN (R 4.5.2)
#>  P rmarkdown      2.32     2026-09-01 [?] CRAN (R 4.5.2)
#>  P S7             0.2.2    2026-04-22 [?] CRAN (R 4.5.2)
#>  P scales         1.4.0    2025-04-24 [?] CRAN (R 4.5.0)
#>  P sessioninfo    1.2.4    2026-06-04 [?] CRAN (R 4.5.2)
#>  P stringi        1.8.9    2026-08-04 [?] CRAN (R 4.5.2)
#>  P stringr      * 1.6.0    2025-11-04 [?] CRAN (R 4.5.0)
#>  P tibble       * 3.3.1    2026-01-11 [?] CRAN (R 4.5.2)
#>  P tidyr        * 1.3.2    2025-12-19 [?] CRAN (R 4.5.2)
#>  P tidyselect     1.2.1    2024-03-11 [?] CRAN (R 4.5.0)
#>  P tidyverse    * 2.0.0    2023-02-22 [?] CRAN (R 4.5.0)
```

```
#>  P timechange     0.4.0    2026-01-29 [?] CRAN (R 4.5.2)
#>  P tzdb           0.5.0    2025-03-15 [?] CRAN (R 4.5.0)
#>  P usethis        3.2.2    2026-09-10 [?] CRAN (R 4.5.2)
#>  P utf8           1.2.6    2025-06-08 [?] CRAN (R 4.5.0)
#>  P vctrs          0.7.3    2026-04-11 [?] CRAN (R 4.5.2)
#>  P vroom          1.7.1    2026-03-31 [?] CRAN (R 4.5.2)
#>  P withr          3.0.3    2026-06-19 [?] CRAN (R 4.5.2)
#>  P xfun           0.61     2026-09-16 [?] CRAN (R 4.5.2)
#>  P xml2           1.6.0    2026-06-22 [?] CRAN (R 4.5.2)
#>  P yaml           2.3.12   2025-12-10 [?] CRAN (R 4.5.2)
#>
#>  [1] /Users/jvergis/Documents/work/Su_STK_2/renv/library/macos/R-4.5/aarch64-apple-darwin2
#>  [2] /Users/jvergis/Library/Caches/org.R-project.R/R/renv/sandbox/macos/R-4.5/aarch64-appl
#>
#>  * -- Packages attached to the search path.
#>  P -- Loaded and on-disk path mismatch.
#>
#> ------------------------------------------------------------------------------
```